Chemical Thermodynamics

Question1

A liquid when kept inside a thermally insulated closed vessel at $25^{\circ}\mathrm{C}$ was mechanically stirred from outside. What will be the correct option for the following thermodynamic parameters ?

JEE Main 2025 (Online) 22nd January Morning Shift

Options:

A. $\Delta \mathrm{U}=0, \mathrm{q}<0, \mathrm{w}>0$

B. $\Delta \mathrm{U}>0, \mathrm{q}=0, \mathrm{w}>0$

C. $\Delta \mathrm{U}=0, \mathrm{q}=0, \mathrm{w}=0$

D. $\Delta \mathrm{U}<0, \mathrm{q}=0, \mathrm{w}>0$

Answer: B

Solution:

Let's analyze the situation step by step:

System: The liquid inside a **thermally insulated** (adiabatic) and **closed** vessel.

Process: The liquid is **mechanically stirred** from outside.

Heat Exchange (q)

Because the vessel is **thermally insulated**, there is **no heat exchange** between the system and surroundings. Hence,

$q = 0$.

Work (w)

The external mechanical stirring does **work on the system** by agitating the liquid. Work done **on** the system is conventionally taken as **positive**:

$w > 0$.

Change in Internal Energy (ΔU)

From the First Law of Thermodynamics,

$\Delta U \ = \ q \ + \ w$.

Since $q = 0$ and $w > 0$, we get

$\Delta U = w > 0$.

Hence:

$\boxed{\Delta U > 0, \quad q = 0, \quad w > 0.}$

Answer: Option B is correct.

Question2

Match List - I with List - II.

	List - I (Partial Derivatives)		List - II (Thermodynamic Quantity)
(A)	$\left(\frac{\partial \mathrm{G}}{\partial \mathrm{T}}\right)_{\mathrm{P}}$	(I)	Cp
(B)	$\left(\frac{\partial \mathrm{H}}{\partial \mathrm{T}}\right)_{\mathrm{P}}$	(II)	$-\mathrm{S}$
(C)	$\left(\frac{\partial \mathrm{G}}{\partial \mathrm{P}}\right)_{\mathrm{T}}$	(III)	Cv
(D)	$\left(\frac{\partial \mathrm{U}}{\partial \mathrm{T}}\right)_{\mathrm{V}}$	(IV)	V

Choose the correct answer from the options given below :

JEE Main 2025 (Online) 22nd January Evening Shift

Options:

A. (A)-(I), (B)-(II), (C)-(IV), (D)-(III)

B. (A)-(II), (B)-(III), (C)-(I), (D)-(IV)

C. (A)-(II), (B)-(I), (C)-(IV), (D)-(III)

D. (A)-(II), (B)-(I), (C)-(III), (D)-(IV)

Answer: C

Solution:

We want to match each partial derivative in List–I to the appropriate thermodynamic quantity in List–II:

---

1. $\left(\frac{\partial G}{\partial T}\right)_P$

A well-known thermodynamic identity is:

$\left(\frac{\partial G}{\partial T}\right)_P = -S.$

Hence,

$(A) \quad \longrightarrow \quad -S \quad$ (II).

---

2. $\left(\frac{\partial H}{\partial T}\right)_P$

By definition of the heat capacity at constant pressure,

$\left(\frac{\partial H}{\partial T}\right)_P = C_P.$

Hence,

$(B) \quad \longrightarrow \quad C_P \quad$ (I).

---

3. $\left(\frac{\partial G}{\partial P}\right)_T$

A standard thermodynamic relation (for a single-component system) is:

$\left(\frac{\partial G}{\partial P}\right)_T = V.$

Hence,

$(C) \quad \longrightarrow \quad V \quad (\text{IV}).$

---

4. $\left(\frac{\partial U}{\partial T}\right)_V$

By definition of the heat capacity at constant volume,

$\left(\frac{\partial U}{\partial T}\right)_V = C_V.$

Hence,

$(D) \quad \longrightarrow \quad C_V \quad (\text{III}).$

Final Matching

$(A) \rightarrow (\text{II}), \quad (B) \rightarrow (\text{I}), \quad (C) \rightarrow (\text{IV}), \quad (D) \rightarrow (\text{III}).$

Which corresponds to:

Option C: (A)-(II), (B)-(I), (C)-(IV), (D)-(III).

Question3

Ice at $-5^{\circ}\mathrm{C}$ is heated to become vapor with temperature of $110^{\circ}\mathrm{C}$ at atmospheric pressure. The entropy change associated with this process can be obtained from

JEE Main 2025 (Online) 23rd January Morning Shift

Options:

A. $\int_{268\ \mathrm{K}}^{273\ \mathrm{K}} \mathrm{C_{p,m}dT} + \frac{\Delta \mathrm{H_{m,\ fusion}}}{\mathrm{T_f}} + \frac{\Delta \mathrm{H_{m,\ vaporisation}}}{\mathrm{T_b}} + \int_{273\ \mathrm{K}}^{373\ \mathrm{K}} \mathrm{C_{p,m}dT} + \int_{373\ \mathrm{K}}^{383\ \mathrm{K}} \mathrm{C_{p,m}dT}$

B. $\int_{268\ \mathrm{K}}^{383\ \mathrm{K}} \mathrm{C_p dT} + \frac{\Delta \mathrm{H_{melting}}}{273} + \frac{\Delta \mathrm{H_{boiling}}}{373}$

C. $\int_{268\ \mathrm{K}}^{383\ \mathrm{K}} \mathrm{C_p dT} + \frac{\mathrm{q_{rev}}}{\mathrm{T}}$

D. $\int_{268\ \mathrm{K}}^{273\ \mathrm{K}} \frac{\mathrm{C_{p,m}}}{\mathrm{T}}\mathrm{dT} + \frac{\Delta \mathrm{H_{m,\ fusion}}}{\mathrm{T_f}} + \frac{\Delta \mathrm{H}^{373\ \mathrm{K}}_{\mathrm{m,\ vaporisation}}}{\mathrm{T_b}} + \int_{273\ \mathrm{K}} \frac{\mathrm{C_{p,m}dT}}{T} + \int_{373\ \mathrm{K}}^{383\ \mathrm{K}} \frac{\mathrm{C_{p,m}dT}}{\mathrm{T}}$

Answer: D

Solution:

$$\underset{268\ \mathrm{K}}{\text{Ice}} \rightarrow \underset{273\ \mathrm{K}}{\text{Ice}} \rightleftharpoons \underset{273\ \mathrm{K}}{\text{Water}} \rightarrow \underset{373\ \mathrm{K}}{\text{Water}} \rightleftharpoons \underset{373\ \mathrm{K}}{\text{Water vapour}} \rightarrow \underset{383\ \mathrm{K}}{\text{Water vapour}}$$

(1) (2) (3) (4) (5)

$$\Delta S_{\text{overall}} = \Delta S_1 + \Delta S_2 + \Delta S_3 + \Delta S_4 + \Delta S_5$$

$$\Delta\ S_2 = \frac{\Delta H_{\text{m fusion}}}{273}\ T_f = 273'K'$$

$$\Delta S_3 = \int_{273}^{373} \frac{C_{p,\ m} dT}{T}$$

$$\Delta\ S_4 = \frac{\Delta H_{\text{m vaporisation}}}{373}\ T_b = 373'K'$$

$$\Delta S_5 = \int_{373}^{383} \frac{C_{p,\ m} dT}{T}$$

$$\text{Answer } = (2)$$

Question4

The effect of temperature on spontaneity of reactions are represented as :

	ΔH	ΔS	Temperature	Spontaneity
(A)	+	−	any T	Non spontaneous
(B)	+	+	low T	spontaneous
(C)	−	−	low T	Non spontaneous
(D)	−	+	any T	spontaneous

The incorrect combinations are :

JEE Main 2025 (Online) 23rd January Evening Shift

Options:

A. (A) and (C) only

B. (B) and (D) only

C. (A) and (D) only

D. (B) and (C) only

Answer: D

Solution:

Let's analyze each combination using the Gibbs free energy equation:

$\Delta G = \Delta H - T\Delta S$

A reaction is spontaneous when $\Delta G < 0$.

Now, consider each case:

(A) $\Delta H > 0$ and $\Delta S < 0$

Here, $\Delta G = +(\text{positive}) - T(-(\text{positive})) = \text{positive} + T(\text{positive})$

This is positive for any temperature.

Thus, the reaction is non-spontaneous for any T, which is correct.

(B) $\Delta H > 0$ and $\Delta S > 0$

Now, $\Delta G = +(\text{positive}) - T(\text{positive})$.

For ΔG to be negative, the temperature has to be **high** enough such that $T\Delta S > \Delta H$.

The table incorrectly states that the reaction is spontaneous at low T.

(C) $\Delta H < 0$ and $\Delta S < 0$

In this case, $\Delta G = -(\text{positive}) - T(-(\text{positive})) = -(\text{positive}) + T(\text{positive})$.

At very low temperatures, the term $T\Delta S$ is small, making ΔG negative (spontaneous).

The table mistakenly indicates that the reaction is non-spontaneous at low T.

(D) $\Delta H < 0$ and $\Delta S > 0$

Then, $\Delta G = -(\text{positive}) - T(\text{positive})$.

This expression is negative at all temperatures, so the reaction is correctly marked as spontaneous.

Thus, the incorrect combinations are:

(B): It should be spontaneous at high T, not low T.

(C): It should be spontaneous at low T, not non-spontaneous.

Therefore, the incorrect combinations are (B) and (C) only.

The correct answer is Option D.

Question5

Let us consider an endothermic reaction which is non-spontaneous at the freezing point of water. However, the reaction is spontaneous at boiling point of water. Choose the correct option.

JEE Main 2025 (Online) 24th January Morning Shift

Options:

A. Both ΔH and ΔS are (-ve)

B. ΔH is $(+\text{ve})$ but ΔS is (-ve)

C. ΔH is $(-\text{ve})$ but ΔS is (+ve)

D. Both ΔH and ΔS are (+ve)

Answer: D

Solution:

Reaction is spontaneous at relatively high temperature and non-spontaneous at low temperature $\Delta G = \Delta H - T\Delta S$It is only possible when ΔH and ΔS both are positive.

Option (1)

Question6

Which of the following mixing of 1 M base and 1 M acid leads to the largest increase in temperature?

JEE Main 2025 (Online) 24th January Evening Shift

Options:

A. 50 mL HCl and 20 mL NaOH

B. 30 mL HCl and 30 mL NaOH

C. 45 mL CH_3COOH and 25 mL NaOH

D. 30 mL CH_3COOH and 30 mL NaOH

Answer: B

Solution:

The rise in temperature of neutralization reaction will be maximum for maximum number of moles of strong acid and strong base neutralized and lower volume of final solution.

(a)	HCl	+ NaOH →	NaCl	+ H_2O
mmol	30		30	
	-		-	

Option (D) and (C) have weak acids and in option (A) only 20 mmol of HCl will be neutralized with 70 mL final volume.

Question7

$$S(g)+\frac{3}{2}O_2(g) \rightarrow SO_3(g)+2x\text{kcal}$$

$$SO_2(g)+\frac{1}{2}O_2(g) \rightarrow SO_3(g)+y\text{kcal}$$

The heat of formation of $SO_2(g)$ is given by :

JEE Main 2025 (Online) 24th January Evening Shift

Options:

A. $2x+y$ kcal

B. $\frac{2x}{y}$ kcal

C. $y-2x$ kcal

D. $x+y$ kcal

Answer: C

Solution:

To determine the heat of formation for $SO_2(g)$, we need to consider the given reactions and their enthalpy changes.

The reactions are:

$S(g)+\frac{3}{2}O_2(g) \rightarrow SO_3(g)+2x\text{kcal}$

$SO_2(g)+\frac{1}{2}O_2(g) \rightarrow SO_3(g)+y\text{kcal}$

For reaction 2, we have:

$\Delta H_r = (\Delta H_f)_{SO_3} - (\Delta H_f)_{SO_2} = -y$

Given that:

$-y = -2x - (\Delta H_f)_{SO_2}$

Rearranging this equation gives:

$(\Delta H_f)_{SO_2} = y - 2x$

Thus, the heat of formation for $SO_2(g)$ is $y - 2x$ kcal.

Question8

Ice and water are placed in a closed container at a pressure of 1 atm and temperature 273.15 K . If pressure of the system is increased 2 times, keeping temperature constant, then identify correct observation from following

JEE Main 2025 (Online) 28th January Morning Shift

Options:

A. Liquid phase disappears completely.

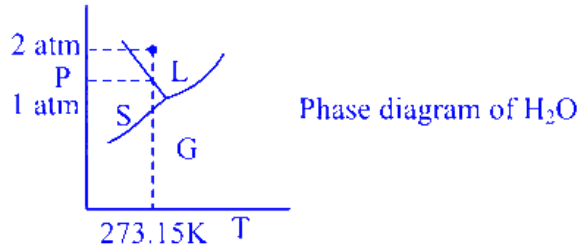
B. The amount of ice decreases.

C. The solid phase (ice) disappears completely.

D. Volume of system increases .

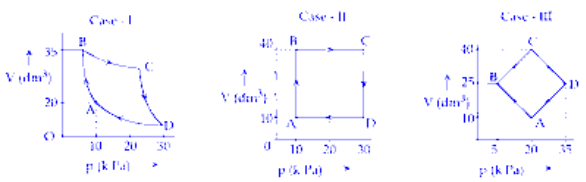
Answer: C

Solution:



If pressure is made two time then mixture of ice and water will completely convert into water (liquid) form.

Question9



An ideal gas undergoes a cyclic transformation starting from the point A and coming back to the same point by tracing the path A→B→C→D→A as shown in the three cases above.

Choose the correct option regarding ΔU :

JEE Main 2025 (Online) 28th January Evening Shift

Options:

A.

$\Delta U(\text{ Case-III }) > \Delta U(\text{ Case-II }) > \Delta U(\text{ Case-I })$

B. $\Delta U(\text{ Case-I }) > \Delta U(\text{ Case-II }) > \Delta U(\text{ Case-III })$

C. $\Delta U(\text{ Case-I }) = \Delta U(\text{ Case-II }) = \Delta U(\text{ Case-III })$

D. $\Delta U(\text{ Case-I }) > \Delta U(\text{ Case-III }) > \Delta U(\text{ Case-II })$

Answer: C

Solution:

As internal energy ' U ' is a state function, its cyclic integral must be zero in a cyclic process

$\therefore \Delta U \text{ case } (I) = \Delta U \text{ case } (II) = \Delta U \text{ case } (\text{ III })$

Question10

500 J of energy is transferred as heat to 0.5 mol of Argon gas at 298 K and 1.00 atm. The final temperature and the change in internal energy respectively are: Given: R = 8.3 J K^{-1} mol^{-1}

JEE Main 2025 (Online) 29th January Morning Shift

Options:

A.

368 K and 500 J

B.

348 K and 300 J

C.

378 K and 300 J

D.

378 K and 500 J

Answer: B

Solution:

Amount of energy transpired as hat, $Q = 500$ J

Number of moles of Argon, $n = 0.5$ mol

Temperature (Initial), $T_i = 298$ K

Pressure, $P = 1.00$ atm

Final temperature, $T_F = ?$

Change in internal energy, $\Delta V = ?$

$R = 8.3 \text{ J K}^{-1} \text{ mol}^{-1}$

For a constant pressure process, the heat added is related to the temperature changes as

$Q = nC_p\Delta T$ ..... (1)

Also, the change in internal energy formula is
$\Delta U = nC_V\Delta T \quad -(2)$ .... (2)

For monoatomic gas (Argon),

$$C_v = \frac{3}{2}R$$

So, $C_p = C_v + R \quad (C_p - C_v = R)$

$$= \frac{3}{2}R + R = \frac{5}{2}R$$

In equation (1) $\Rightarrow$

$Q = nC_p\Delta T$

$Q = nC_p\left(T_F - T_i\right)$

$500 \text{ J} = 0.5 \text{ mol} \times \frac{5}{2}R\left(T_F - T_i\right) \quad R = 8.3 \text{JK}^{-1} \text{ mol}^{-1}$

$500 \text{ J} = 0.5 \text{ mol} \times \frac{5}{2} \times 8.3 \text{ J}K^{-1} \text{ mol}^{-1}\left(T_F - T_i\right)$

$T_F - T_i = \frac{500 \text{ J}}{0.5\text{mol} \times \frac{5}{2} \times 8.3 \text{ J K}^{-1} \text{ mol}^{-1}}$

$$= \frac{500}{0.5 \times \frac{5}{2} \times 8.3}\text{k}$$

$$= 48.2\text{k}$$

$T_F = 48.2\text{k} + 298\text{k}$

$$= 346.2\text{k}$$

$$\approx 348\text{k}$$

In equation (2) $\Rightarrow$

$\Delta U = nC_V\Delta T$

$$= 0.5 \text{ mol} \times \frac{3}{2}R \times 48.2 \text{ K}$$

$$= 0.5 \text{ mol} \times \frac{3}{2} \times 8.3 \text{ J K}^{-1}\text{mol}^{-1} \times 48.2 \text{ K}$$

$$= 300.045 \text{ J} \approx 300 \text{ J}$$

The final temperature is, 348 k

change in internal energy is 300 J

So, correct answer is Option 2) 348 K and 300 J

Question11

If $\quad C$ (diamond) $\rightarrow C$ (graphite) $+X \text{ kJ mol}^{-1}$

C (diamond) $+\mathrm{O}_2(\mathrm{~g}) \rightarrow \mathrm{CO}_2(\mathrm{~g})+\mathrm{YkJmol}^{-1}$

C (graphite) $+\mathrm{O}_2(\mathrm{~g}) \rightarrow \mathrm{CO}_2(\mathrm{~g})+\mathrm{ZkJmol}^{-1}$

at constant temperature. Then

JEE Main 2025 (Online) 29th January Evening Shift

Options:

A.

−X = Y + Z

B.

X = Y − Z

C.

X = −Y + Z

D.

X = Y + Z

Answer: B

Solution:

Given,

$C_{(diamond)} \longrightarrow C_{(graphite)} + X\,kJ\,mol^{-1}$ ..... (1)

$C_{(diamond)} + O_2(g) \longrightarrow CO_2(g) + Y\,kJ\,mol^{-1}$ ..... (2)

$C_{(graphite)} + O_2(g) \longrightarrow CO_2(g) + Z\,kJ\,mol^{-1}$ ..... (3)

Condition for temperature : constant

Hess's law is applied here. (Hess's law of constant heat summation)

The given reactions (2) and (3),

(2) - (3) $\Rightarrow$

$C_{(diamond)} + O_2(g) - \left(C_{(graphite)} + O_2(g)\right) \longrightarrow CO_2(g) + Y - (CO_2(g) + Z)$

$C_{(diamond)} + O_2(g) - C_{(grpahite)} - O_2(g) \longrightarrow CO_2(g) + Y - CO_2(g) - Z$

$C_{(diamond)} - C_{(graphite)} \rightarrow Y - Z$

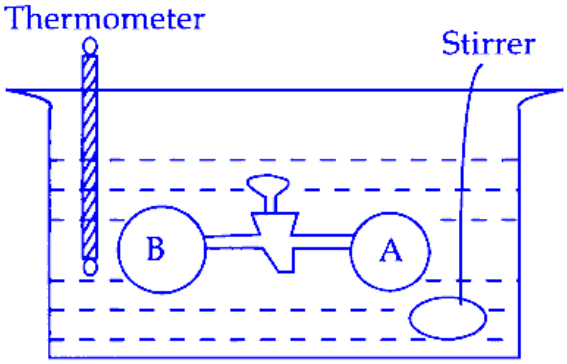
$C_{(diamond)} \rightarrow C_{(graphite)} + (Y - Z)$ ..... (4)

Comparing (1) and (4),

$X = Y - Z$

So, the correct answer is option (2) $X = Y - Z$

Question12



Two vessels A and B are connected via stopcock. The vessel A is filled with a gas at a certain pressure. The entire assembly is immersed in water and is allowed to come to thermal equilibrium with water. After opening the stopcock the gas from vessel A expands into vessel B and no change in temperature is observed in the thermometer. Which of the following statement is true ?

JEE Main 2025 (Online) 2nd April Morning Shift

Options:

A. $dq \neq 0$

B. The pressure in the vessel B before opening the stopcock is zero

C. $dw \neq 0$

D. $dU \neq 0$

Answer: B

Solution:

It is free expansion of gas $\Rightarrow P_{ext} = 0$

Where $w = 0, q = 0$ and $\Delta U = 0$

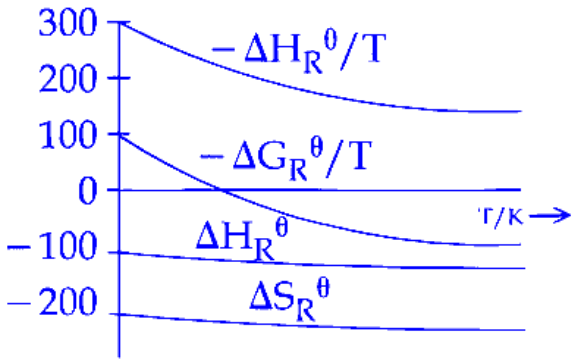
Question13

Which of the following graphs correctly represents the variation of thermodynamic properties of Haber's process?

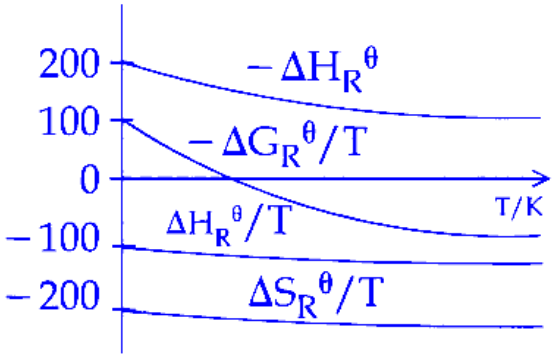
JEE Main 2025 (Online) 2nd April Evening Shift

Options:

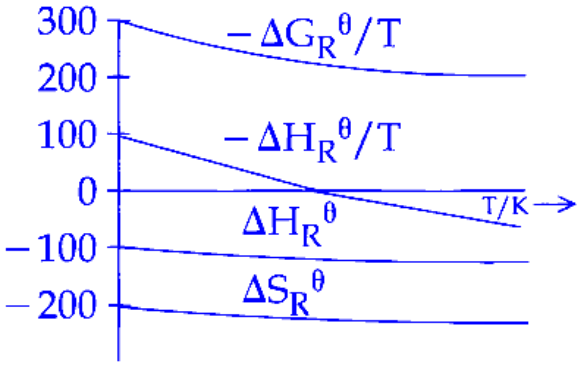
A.



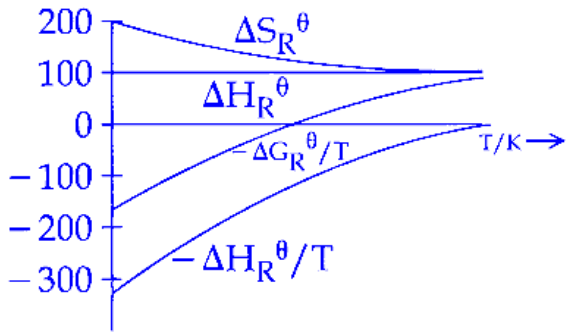
B.



C.



D.



Answer: A

Solution:

In the Haber process, the chemical reaction is as follows:

$N_2(g) + 3H_2(g) \rightarrow 2NH_3$

The change in enthalpy (ΔH°) is negative, indicating that the reaction is exothermic.

The change in entropy (ΔS°) is also negative because the number of moles of gas decreases during the reaction.

Key points to consider:

As temperature increases, the term $-\frac{\Delta H_R^0}{T}$ decreases.

The free energy change is given by:

$\Delta G^\circ = -RT \ln K_{eq}$

Rearranging gives:

$R \ln K_{eq} = -\frac{\Delta G^0}{T}$

For an exothermic reaction, as temperature increases, the equilibrium constant (K_{eq}) decreases.

Both ΔH° and ΔS° are almost constant with respect to temperature.

Question14

Arrange the following in order of magnitude of work done by the system/on the system at constant temperature.

(a) $|w_{reversible}|$ for expansion in infinite stages.

(b) $|w_{irreversible}|$ for expansion in single stage.

(c) $|w_{reversible}|$ for compression in infinite stages.

(d) $|w_{irreversible}|$ for compression in single stage.

Choose the correct answer from the options given below :

JEE Main 2025 (Online) 2nd April Evening Shift

Options:

A.

$d > c = a > b$

B.

$c = a > d > b$

C.

$a > c > b > d$

D.

$a > b > c > d$

Answer: A

Solution:

For isothermal process

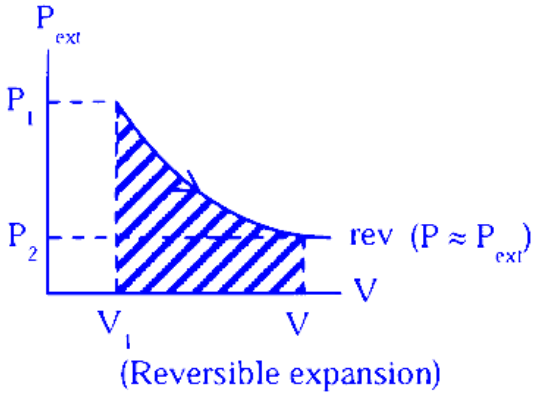
$$|W_{reversible}|_{expansion} = |W_{reversible}|_{compression}$$

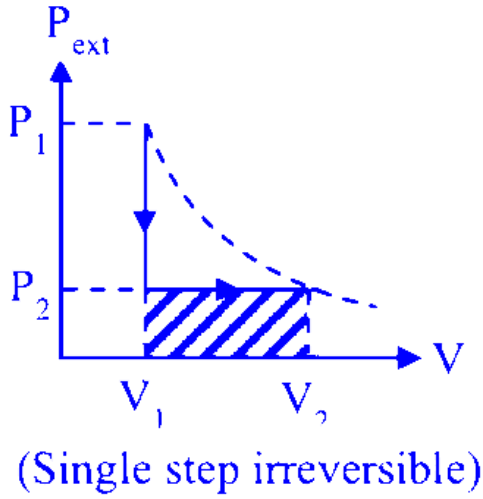
$$= -nRT \ln \frac{V_f}{V_i}$$

$$|W_{irreversible}|_{expansion} < |W_{irreversible}|_{compression}$$

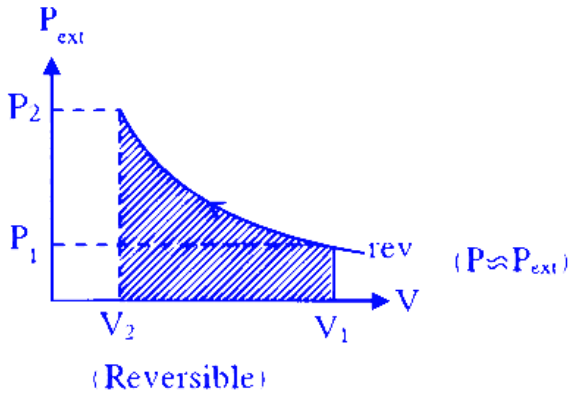
$$d > c = a > b$$

$$|W_{irreversible}|_{expansion} = -P_{ext}\ (V_f - V_i)$$

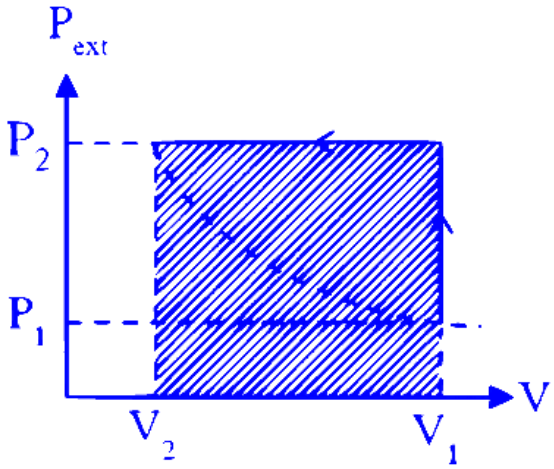




(Single step irreversible)



(Reversible)



(Single step irreversible)

Graphical representation

We can compare work by area of PV graph.

Question15

Given below are two statements:

Statement I : When a system containing ice in equilibrium with water (liquid) is heated, heat is absorbed by the system and there is no change in the temperature of the system until whole ice gets melted.

Statement II : At melting point of ice, there is absorption of heat in order to overcome intermolecular forces of attraction within the molecules of water in ice and kinetic energy of

molecules is not increased at melting point.

In the light of the above statements, choose the correct answer from the options given below

JEE Main 2025 (Online) 3rd April Evening Shift

Options:

A. Both Statement I and Statement II are false

B. Statement I is true but Statement II is false

C. Both Statement I and Statement II are true

D. Statement I is false but Statement II is true

Answer: C

Solution:

The correct choice is Option C: Both Statement I and Statement II are true.

Explanation:

Statement I is true because

During the ice–water phase change at 0 °C, any heat added goes into the latent heat of fusion.

The temperature remains fixed at 0 °C until all the ice has melted.

Statement II is also true because

The heat absorbed at the melting point is used to overcome (break) the hydrogen-bonding forces between water molecules in ice—in other words, it increases the system's potential energy.

Since no heat goes into raising the kinetic energy of the molecules, the temperature (which measures average kinetic energy) does not increase during melting.

Mathematically, for a mass m of ice, the heat needed is

$Q = m\,L_f$

where L_f is the latent heat of fusion of ice. This heat does not change the temperature but is entirely consumed in changing the phase (i.e., increasing potential energy).

Question16

Let us consider a reversible reaction at temperature, T.In this reaction, both ΔH and ΔS were observed to have positive values. If the equilibrium temperature is Te , then the reaction becomes spontaneous at:

JEE Main 2025 (Online) 4th April Morning Shift

Options:

A. $\mathrm{Te} > \mathrm{T}$

B. $\mathrm{T} > \mathrm{Te}$

C. $\mathrm{T} = \mathrm{Te}$

D. $\mathrm{Te} = 5\ \mathrm{T}$

Answer: B

Solution:

For reaction to be spontaneous according to 2nd law:

$$\Delta \mathrm{G} < 0$$
$$\Rightarrow \Delta \mathrm{H} - \mathrm{T}\Delta\ \mathrm{S} < 0$$
$$\Rightarrow\ \mathrm{T} > \left(\frac{\Delta \mathrm{H}}{\Delta\ \mathrm{S}}\right) = \mathrm{T_e}$$
$$\Rightarrow\ \mathrm{T} > \mathrm{T_e}$$

Question17

One mole of an ideal gas expands isothermally and reversibly from $10\mathrm{dm}^3$ to $20\mathrm{dm}^3$ at 300 K . $\Delta \mathrm{U}, \mathrm{q}$ and work done in the process respectively are

Given: $\mathrm{R} = 8.3\ \mathrm{J\ K^{-1}\ mol^{-1}}$

$\ln 10 = 2.3$

$\log 2 = 0.30$

$\log 3 = 0.48$

JEE Main 2025 (Online) 4th April Morning Shift

Options:

A. $0, 21.84\ \mathrm{kJ}, -1.726\ \mathrm{J}$

B. $0, 21.84\ \mathrm{kJ}, 21.84\ \mathrm{kJ}$

C. $0, 1.718\ \mathrm{kJ}, -1.718\ \mathrm{kJ}$

D. $0, -17.18\ \mathrm{kJ}, 1.718\ \mathrm{J}$

Answer: C

Solution:

$$(10\ \mathrm{L}, 300\ \mathrm{K}) \xrightarrow{\mathrm{n=1}} (20\ \mathrm{L}, 300\ \mathrm{K})$$
$$-\mathrm{q} = \mathrm{w} = -\mathrm{nRT}\ln\frac{\mathrm{V_2}}{\mathrm{V_1}}$$
$$= -8.3 \times 300 \times \ln\left(\frac{20}{10}\right)$$
$$= -1.718\ \mathrm{kJ}$$
$$\Rightarrow \mathrm{q} = 1.718\ \mathrm{kJ}$$
$$\mathrm{w} = -1.718\ \mathrm{kJ}$$
$$\Delta \mathrm{U} = 0 (\because \Delta\ \mathrm{T} = 0)$$

Question18

Consider the given data :

(a) $\mathrm{HCl(g) + 10H_2O(l) \rightarrow HCl.10H_2O \Delta H = -69.01\ kJ\ mol^{-1}}$

(b) $\mathrm{HCl(g) + 40H_2O(l) \rightarrow HCl.40H_2O \Delta H = -72.79\ kJ\ mol^{-1}}$

Choose the correct statement :

JEE Main 2025 (Online) 4th April Evening Shift

Options:

A. The heat of dilution for the HCl $(\mathrm{HCl.10H_2O}$ to $\mathrm{HCl.40H_2O})$ is $3.78\ \mathrm{kJ\ mol^{-1}}$.

B. Dissolution of gas in water is an endothermic process.

C. The heat of solution depends on the amount of solvent.

D. The heat of formation of HCl solution is represented by both (a) and (b).

Answer: C

Solution:

From the given information

ΔH is negative so it means dissolution of gas $\mathrm{HCl(g)}$ is exothermic.

$\mathrm{HCl(\ g) + 10H_2O(l) \rightarrow HCl.10H_2O}$ ..... (1)

$\Delta H_1 = -69.01 \frac{\mathrm{kJ}}{\mathrm{mol}}$

$\mathrm{HCl(\ g) + 40H_2O(l) \rightarrow HCl.40H_2O}$ ..... (2)

$\Delta H_2 = -72.79 \frac{\mathrm{kJ}}{\mathrm{mol}}$

Hence heat of solution depends upon amount of solvent

By equation ..... (2) − equation ..... (1)

$\mathrm{HCl.10H_2O + 30H_2O(\ell) \rightarrow HCl.40H_2O}$

So Heat of dilution $= -72.79 - (-69.01)$

$= -3.78 \frac{\mathrm{kJ}}{\mathrm{mol}}$

Hence option (3) is incorrect.

For heat of formation reactant should be in elemental form hence option (4) is incorrect

---

Question19

Total enthalpy change for freezing of 1 mol of water at $10^{\circ}\mathrm{C}$ to ice at $-10^{\circ}\mathrm{C}$ is _____

(Given: $\Delta_{\text{fus}}\ \mathrm{H} = x\ \mathrm{kJ/mol}$

$\mathrm{C_p\,[H_2O(l)] = y\,J\,mol^{-1}\,K^{-1}}$
$\mathrm{C_p\,[H_2O(\,s)] = z\,J\,mol^{-1}\,K^{-1}}$

JEE Main 2025 (Online) 7th April Morning Shift

Options:

A. $-x-10y-10z$

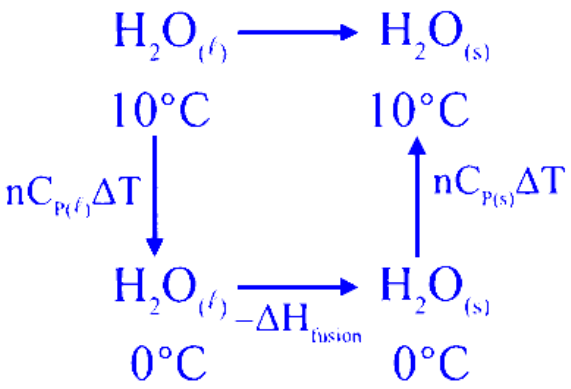
B. $x-10y-10z$

C. $-10(100x+y+z)$

D. $10(100\mathrm{x}+y+z)$

Answer: C

Solution:



$\Delta H=1\times y(0-10)-x\times 1000+1\times z(-10^{\circ}-0^{\circ})$
$\Delta H=-10(100x+y+z)$ Joule.

Question20

The correct statement amongst the following is :

JEE Main 2025 (Online) 7th April Evening Shift

Options:

A.

The standard state of a pure gas is the pure gas at a pressure of 1 bar and temperature 273 K.

B.

$\Delta_f H^{\circ}_{500}$ is zero for $O_2(g)$.

C.

$\Delta_f H^{\circ}_{298}$ is zero for $O(g)$.

D.

The term 'standard state' implies that the temperature is 0°C.

Answer: B

Solution:

Option A:

It claims that the standard state for a pure gas is defined at 1 bar and 273 K.

In modern thermochemistry, the standard state generally means the pure substance at 1 bar, but the standard temperature most commonly used is 298 K (25°C), not 273 K (0°C).

Thus, Option A is incorrect.

Option B:

It states that the standard enthalpy of formation $\Delta_f H^\circ_{500}$ is zero for $O_2(g)$.

By convention, the standard enthalpy of formation for an element in its most stable form is defined as zero. Even though typical tables list $\Delta_f H^\circ_{298}$ values, if one were to evaluate the formation enthalpy at any temperature where the element remains in its standard state, the reaction

$O_2(g) \rightarrow O_2(g)$

has no change in enthalpy.

Therefore, by definition, $\Delta_f H^\circ_{500}(O_2(g)) = 0$.

This makes Option B correct.

Option C:

It claims that $\Delta_f H^\circ_{298}$ is zero for $O(g)$ (atomic oxygen).

However, the standard state of oxygen is diatomic $O_2(g)$, not atomic oxygen. Since $O(g)$ is not the most stable form of oxygen under standard conditions, its formation enthalpy is not zero.

Hence, Option C is incorrect.

Option D:

It asserts that the term "standard state" implies that the temperature is 0°C (273 K).

In fact, "standard state" typically specifies a standard pressure (1 bar) and a chosen temperature—commonly 298 K for tabulated thermodynamic data—not necessarily 0°C.

Thus, Option D is also incorrect.

To summarize:

The only correct statement is Option B.

Therefore, the correct answer is: Option B.

Question21

The hydration energies of K^+ and Cl^- are $-x$ and $-y$ kJ/mol respectively. If the lattice energy of KCl is $-z$ kJ/mol, then the heat of solution of KCl is :

JEE Main 2025 (Online) 7th April Evening Shift

Options:

A.

$x + y + z$

B.

$z - (x + y)$

C.

$-z - (x + y)$

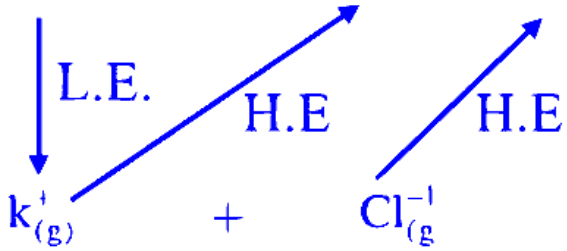
D.

$x - y - z$

Answer: B

Solution:

$$KCl_{(s)} + H_2O \xrightarrow{\Delta H \text{ sol.}} K^+_{(aq)} + Cl^{-1}_{(aq)}$$



$$\begin{aligned}\Delta H_{Sol^n.} &= L \cdot E \cdot + (H \cdot E)_{K^+_{(g)}} + (HE)_{Cl^{-1}_{(g)}} \\ &= Z - x - y \\ &= z - (x + y)\end{aligned}$$

Question22

Consider the following cases of standard enthalpy of reaction $\left(\Delta H_r^{\circ} \text{ in } kJmol^{-1}\right)$

$$C_2H_6(g) + \frac{7}{2}O_2(g) \rightarrow 2CO_2(g) + 3H_2O(l) \Delta H_1^{\circ} = -1550$$
$$C(\text{ graphite }) + O_2(g) \rightarrow CO_2(g) \Delta H_2^{\circ} = -393.5$$
$$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(l) \Delta H_3^{\circ} = -286$$

The magnitude of $\Delta H^{\circ}_{fC_2H_6(g)}$ is _____ $kJmol^{-1}$ (Nearest integer).

JEE Main 2025 (Online) 22nd January Evening Shift

Answer: 95

Solution:

$$C_2H_6(g) + \frac{7}{2}O_2(g) \rightarrow 2CO_2(g) + 3H_2O(I) \Delta H_1^{\circ} = -1550 \ldots \text{ (i)}$$

$$C \text{ (graphite) } + O_2(g) \rightarrow CO_2(g) \Delta H_2^{\circ} = -393.5 \ldots \text{ (ii)}$$

$$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(I) \Delta H_3^{\circ} = -286 \ldots\ldots (iii)$$

From $2 \times eq^n(ii) + 3 \times eq^n(iii) - eq^n(i)$

$$2C_{(graphite)} + 3H_2(g) \rightarrow C_2H_6(g) : \Delta H^{\circ}_{fC_2H_6}$$

$$\left(\Delta H_f^{\circ}\right)_{C_2H_6} = 2 \times (-393.5) + 3 \times (-286) - (-1550)$$

$$= -95 \text{ kJ/mol}$$

Question23

The standard enthalpy and standard entropy of decomposition of N_2O_4 to NO_2 are $55.0\ kJ\ mol^{-1}$ and $175.0\ J/K/mol$ respectively. The standard free energy change for this reaction at $25^{\circ}C$ in J mol^{-1} is _____ (Nearest integer)

JEE Main 2025 (Online) 23rd January Morning Shift

Answer: 2850

Solution:

$\Delta H^{\circ}_{rxn} = 55\ kJ/mol, \quad T = 298\ K$

$\Delta S^{o}_{rxn} = 175\ J/mol$
$\Delta G^{o}_{rxn} = \Delta H^{o}_{rxn} - T\Delta\ S^{o}_{rxn}$
$\Rightarrow \Delta G^{o}_{rxn} = 55000\ J/mol - 298 \times 175\ J/mol$
$\Rightarrow \Delta G^{o}_{rxn} = 55000 - 52150$
$\Rightarrow \Delta G^{o}_{rxn} = 2850\ J/mol$

Question24

The bond dissociation enthalpy of $X_2 \Delta H^{\circ}_{bond}$ calculated from the given data is _____ $kJmol^{-1}$. (Nearest integer)

$M^+X^-(s) \rightarrow M^+(g) + X^-(g) \Delta H^{\circ}_{lattice} = 800\ kJ\ mol^{-1}$
$M(\ s) \rightarrow M(\ g) \Delta H^{\circ}_{sub} = 100\ kJ\ mol^{-1}$

$M(\ g) \rightarrow M^+(g) + e^-(g) \Delta H^{\circ}_{i} = 500\ kJ\ mol^{-1}$

$X(\ g) + e^-(g) \rightarrow X^-(g) \Delta H^{\circ}_{eg} = -300\ kJ\ mol^{-1}$

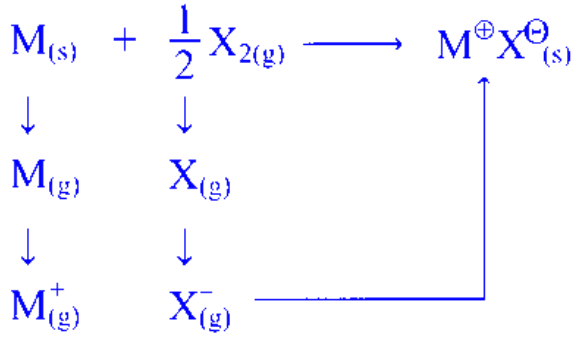
$M(\ s) + \frac{1}{2}X_2(\ g) \rightarrow M^+X^-(s) \Delta H^{\circ}_{f} = -400\ kJ\ mol^{-1}$

[Given : M^+X^- is a pure ionic compound and X forms a diatomic molecule X_2 in gaseous state]

JEE Main 2025 (Online) 23rd January Evening Shift

Answer: 200

Solution:



$$\therefore \Delta H_f(MX) = \Delta H_{sub} \ (M) + \text{ I.E. } (M) + \frac{1}{2}[\text{ B.E. } (X-X)] + EG(X) + \text{ L.E. } (MX)$$

$$-400 = (100) + (500) + \frac{1}{2}(\text{ B.E. }) + (-300) + (-800)$$

$$\therefore \text{ B.E. } = 200 \text{ kJ mole}^{-1}$$

Question25

Standard entropies of X_2, Y_2 and XY_5 are 70, 50 and $110 \text{ J K}^{-1} \text{ mol}^{-1}$ respectively. The temperature in Kelvin at which the reaction

$$\frac{1}{2}X_2 + \frac{5}{2}Y_2 \rightleftharpoons XY_5 \Delta H^{\ominus} = -35 \text{ kJ mol}^{-1}$$

will be at equilibrium is _____ (Nearest integer)

JEE Main 2025 (Online) 24th January Morning Shift

Answer: 700

Solution:

To determine the temperature at which the given reaction is at equilibrium, we need to apply the concept of Gibbs free energy change (ΔG^0) at equilibrium, where ΔG^0 is equal to zero.

The given reaction is:

$\frac{1}{2}X_2 + \frac{5}{2}Y_2 \rightleftharpoons XY_5$

We are given the standard entropies and enthalpy change ($\Delta H^{\ominus}$):

Standard entropies:

$X_2 = 70 \, \text{J K}^{-1} \, \text{mol}^{-1}$

$Y_2 = 50 \, \text{J K}^{-1} \, \text{mol}^{-1}$

$XY_5 = 110 \, \text{J K}^{-1} \, \text{mol}^{-1}$

Standard enthalpy change:

$\Delta H^{\ominus} = -35 \, \text{kJ/mol}$

First, calculate the standard entropy change (ΔS^0_{Rxn}) of the reaction:

$\Delta S^0_{Rxn} = 110 - \left(\frac{1}{2} \times 70 + \frac{5}{2} \times 50\right)$

$= 110 - (35 + 125)$

$= 110 - 160$

$= -50\,\text{J K}^{-1}\,\text{mol}^{-1}$

At equilibrium, $\Delta G^0 = 0$, and the relation $\Delta G^0 = \Delta H^0 - T\Delta S^0$ applies. Thus:

$0 = -35000\,\text{J/mol} - T(-50\,\text{J K}^{-1}\,\text{mol}^{-1})$

Solving for T:

$T \cdot 50 = 35000$

$T = \frac{35000}{50}$

$T = 700\,\text{K}$

Therefore, the temperature at which the reaction is at equilibrium is 700 Kelvin.

Question26

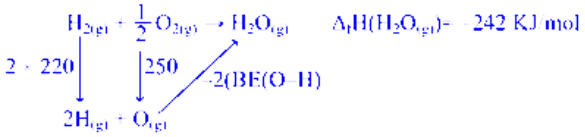
The formation enthalpies, $\Delta H_f^{\ominus}$ for $H_{(g)}$ and $O_{(g)}$ are 220.0 and $250.0\ \text{kJ mol}^{-1}$, respectively, at 298.15 K , and $\Delta H_f^{\ominus}$ for $H_2O_{(g)}$ is $-242.0\ \text{kJ mol}^{-1}$ at the same temperature. The average bond enthalpy of the $O-H$ bond in water at 298.15 K is _____ kJ mol^{-1} (nearest integer).

JEE Main 2025 (Online) 28th January Morning Shift

Answer: 466

Solution:

$\frac{1}{2}H_{2(g)} \rightarrow H(g) \quad ; \Delta_f H\left(H_{(g)}\right) = 220\text{KJ/mol}$
$\frac{1}{2}O_{2(g)} \rightarrow O(g) \quad ; \Delta_f H\left(O_{(g)}\right) = 250\text{KJ/mol}$



$\Delta H_f\left(H_2O_{(l)}\right) = -242 = 440 + 250 - 2(\text{B. E.}(O-H))$
$\text{BE}(O-H) = 466\text{KJ/mol}$

Question27

Consider the following data :

Heat of formation of $CO_2(g) = -393.5\ \text{kJ mol}^{-1}$

Heat of formation of $H_2O(l) = -286.0\ \text{kJ mol}^{-1}$

Heat of combustion of benzene $= -3267.0\ \text{kJ mol}^{-1}$

The heat of formation of benzene is _____ kJmol^{-1}. (Nearest integer)

JEE Main 2025 (Online) 28th January Evening Shift

Answer: 48

Solution:

To determine the heat of formation of benzene ($\Delta H_f[C_6H_6]$), we use the given data:

Heat of formation of $CO_2(g) = -393.5$ kJ/mol

Heat of formation of $H_2O(l) = -286.0$ kJ/mol

Heat of combustion of benzene = -3267.0 kJ/mol

The reaction for the combustion of benzene is:

$$C_6H_6 + \frac{15}{2}O_2(g) \rightarrow 6CO_2(g) + 3H_2O(l)$$

Using the formula for the enthalpy change of the reaction (ΔH_R):

$$\Delta H_R = \Delta H_C = \Sigma \Delta H_f(\text{Products}) - \Sigma \Delta H_f(\text{Reactants})$$

Substitute values into the equation:

$$-3267 = 6 \times (-393.5) + 3 \times (-286) - \Delta H_f[C_6H_6]$$

Solving this equation, we find:

$$\Delta H_f[C_6H_6] = 48 \text{ kJ/mol}$$

Thus, the heat of formation of benzene is 48 kJ/mol.

Question28

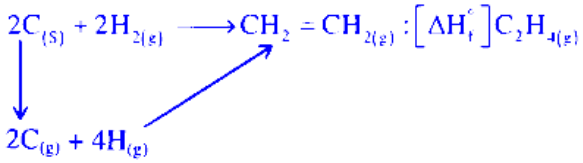
$\Delta H^{\ominus}_{\text{sub}}\ [\text{C (graphite)}] = 710\,\text{kJ mol}^{-1}$
$\Delta_{\text{C-H}}H^{\ominus} = 414\,\text{kJ mol}^{-1}$
$\Delta_{\text{H-H}}H^{\ominus} = 436\,\text{kJ mol}^{-1}$
$\Delta_{\text{C}} = \text{CH}^{\ominus} = 611\,\text{kJ mol}^{-1}$

The $\Delta H_f \ominus$ for $CH_2 = CH_2$ is_____ kJmol^{-1} (nearest integer value)

JEE Main 2025 (Online) 3rd April Morning Shift

Answer: 25

Solution:



$$
\begin{aligned}
& [\Delta H_f^o]_{C_2H_{4(fl)}} = 2 \times [\Delta H_{sub}^o]_{C_{(s)}} + 2 \times \Delta H_{H-H}^o - 1 \times \Delta H_{C=C}^o - 4 \times \Delta H_{C-H}^\circ \\
\Rightarrow \quad & [\Delta H_f^o]_{C_2H_{4(\ g)}} = (2 \times 710) + (2 \times 436) - 611 - 4 \times 414 \\
\Rightarrow \quad & [\Delta H_f^o]_{C_2H_{4(\ g)}} = 25 \text{ kJ/mol}
\end{aligned}
$$

Question29

A sample of n -octane (1.14 g) was completely burnt in excess of oxygen in a bomb calorimeter, whose heat capacity is 5 kJ K^{-1}. As a result of combustion reaction, the temperature of the calorimeter is increased by 5 K . The magnitude of the heat of combustion of octane at constant volume is_____ kJmol^{-1} (nearest integer).

JEE Main 2025 (Online) 3rd April Evening Shift

Answer: 2500

Solution:

Calculate the moles of octane:

Moles of octane $= \frac{1.14 \text{ g}}{114 \text{ g/mol}} = 0.01$ moles

Determine the heat evolved during combustion:

The heat capacity of the calorimeter is 5 kJ/K and the temperature increase is 5 K.

Heat evolved $= C \times \Delta T = 5 \text{ kJ/K} \times 5 \text{ K} = 25 \text{ kJ}$

Calculate the magnitude of the heat of combustion:

The heat of combustion per mole of octane is found by dividing the total heat evolved by the moles of octane combusted:

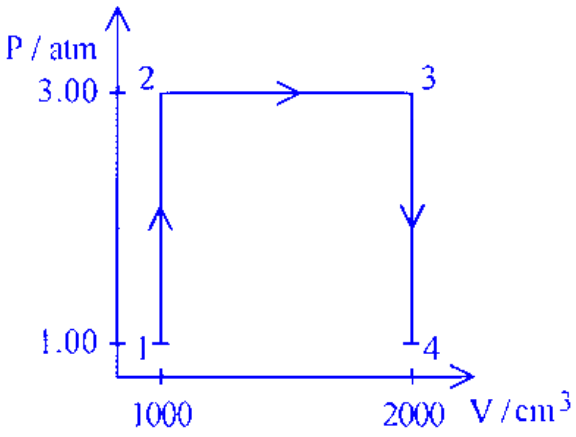
Magnitude of Heat of combustion $= \frac{25 \text{ kJ}}{0.01 \text{ moles}} = 2500 \text{ kJ/mol}$

Therefore, the magnitude of the heat of combustion of octane at constant volume is 2500 kJ/mol.

Question30

A perfect gas (0.1 mol) having $\overline{C}_v = 1.50\text{R}$ (independent of temperature) undergoes the above transformation from point 1 to point 4. If each step is reversible, the total work done (w) while going from point 1 to point 4 is $(-)$_____J (nearest integer)

[Given: $\text{R} = 0.082 \text{ L atm K}^{-1} \text{ mol}^{-1}$ **]**



JEE Main 2025 (Online) 3rd April Evening Shift

Answer: 304

Solution:

To determine the total work done when a perfect gas undergoes a transformation from point 1 to point 4, we can analyze each step of the process:

Given Data

Moles of gas, $n = 0.1\,\text{mol}$

Specific heat at constant volume, $\overline{\mathrm{C}}_v = 1.50\mathrm{R}$

Gas constant, $\mathrm{R} = 0.082\,\mathrm{L\,atm\,K^{-1}\,mol^{-1}}$

Work Done Calculation

Step 1: From Point 1 to Point 2 ($W_{1\to2}$)

Since this is an isochoric process (constant volume), the work done, $W_{1\to2}$, is zero.

$\mathrm{W}_{1\to2} = 0$

Step 2: From Point 2 to Point 3 ($W_{2\to3}$)

This step is isobaric (constant pressure), and the work done can be calculated as:

$\mathrm{W}_{2\to3} = -P\Delta V = -P(V_3 - V_2)$

Given:

$P = 3\,\text{atm}$

Change in volume, $\Delta V = V_3 - V_2 = 2 - 1 = 1\,\text{L}$

$$\mathrm{W}_{2\to3} = -3\,\text{atm} \times 1\,\text{L} = -3\,\text{L atm}$$

Converting to Joules (using $1\,\text{L atm} = 101.3\,\text{J}$):

$$\mathrm{W}_{2\to3} = -3 \times 101.3\,\text{J} = -304\,\text{J}$$

Step 3: From Point 3 to Point 4 ($W_{3\to4}$)

This is another isochoric process, so the work done, $W_{3\to4}$, is zero.

$$\mathrm{W}_{3\to4} = 0$$

Total Work Done

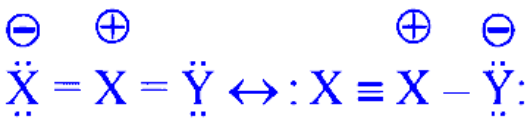
The total work done from point 1 to point 4 is the sum of work done in each step:

Total work done $= W_{1\to2} + W_{2\to3} + W_{3\to4} = 0 + (-304) + 0 = -304\,\text{J}$

Thus, the total work done during this transformation is $-304\,\text{J}$.

Question31

Resonance in $\mathrm{X_2Y}$ can be represented as



The enthalpy of formation of X_2Y $\left(X = X(g) + \frac{1}{2}Y = Y(g) \to X_2Y(g)\right)$ is 80 kJ mol^{-1}. The magnitude of resonance energy of X_2Y is __ kJ mol^{-1} (nearest integer value).

Given: Bond energies of $X \equiv X$, $X = X$, $Y = Y$ and $X = Y$ are 940, 410, 500, and 602 kJ mol^{-1} respectively.
valence X: 3, Y: 2

JEE Main 2025 (Online) 8th April Evening Shift

Answer: 98

Solution:

$\Delta \mathrm{H_{R.E}} = \Delta \mathrm{H_{f(exp)}} - \Delta \mathrm{H_{f(\ Theo\ \ )}}$

$\Delta \mathrm{H_{f(exp)}}$ for $\mathrm{X_2Y_{(g)}} = 80\ \text{kJ/mole}$

for $\Delta \mathrm{H_{f\ (Theo)}}$

$\mathrm{X_{2(\ g)}} + \frac{1}{2}\mathrm{Y_{2(\ g)}} \to \mathrm{X_2Y_{(g)}} \Delta \mathrm{H_f} = ?$

$\Delta H_{f(\text{ Theo }\ )} = \left(BE_{X=X} + \frac{1}{2}BE_{Y=Y}\right) - (BE_{X=X} + BE_{X=Y})$

$= \left(940 + \frac{1}{2} \times 500\right) - (410 + 602)$

$= 178\ \text{kJ/mole}$

$\Delta H_{R.E}\ \ = 80 - 178$

$\ \ = -98\ \text{kJ/mol}$

$|\Delta H_{R.E}\ \ | = 98$

Question32

If three moles of an ideal gas at 300K expand isotherrnally from 30dm^3 to 45dm^3 against a constant opposing pressure of 80kPa, then the amount of heat transferred is _____J.

[27-Jan-2024 Shift 1]

Answer: 1200

Solution:

Using, first law of thermodynamics,

$\Delta U = Q + W,$

$\Delta U = 0:$ Process is isothermal

$Q = -W$

$W = -P_{ext}\,\Delta V:$ Irreversible

$= -80 \times 10^3 (45 - 30) \times 10^{-3}$

$= -1200J$

Question33

For a certain thermochemical reaction M → N at T = 400K, $\Delta H^{\ominus}$ = 77.2kJmol^{-1}, ΔS = 122JK^{-1}, log equilibrium constant (logK) is _____× 10^{-1}.

[27-Jan-2024 Shift 2]

Answer: 37

Solution:

$\Delta G^\circ = \Delta H^\circ - T\,\Delta S^\circ$

$= 77.2 \times 10^3 - 400 \times 122 = 28400J$

$\Delta G^\circ = -2.303\,RT \log K$

$\Rightarrow 28400 = -2.303 \times 8.314 \times 400 \log K$

$\Rightarrow \log K = -3.708 = -37.08 \times 10^{-1}$

Question34

Which of the following is not correct?

[29-Jan-2024 Shift 1]

Options:

A.

ΔG is negative for a spontaneous reaction

B.

ΔG is positive for a spontaneous reaction

C.

ΔG is zero for a reversible reaction

D.

ΔG is positive for a non-spontaneous reaction

Answer: B

Solution:

$(\Delta G)_{P,\ T}$ = (+)ve for non-spontaneous process

Question35

Standard enthalpy of vapourisation for CCl_4 is 30.5kJmol^{-1}. Heat required for vapourisation of 284g of CCl_4 at constant temperature is____ kJ. (Given molar mass in gmol^{-1} ; C = 12, Cl = 35.5)

[29-Jan-2024 Shift 2]

Answer: 56

Solution:

$\Delta H^{\circ}_{vap}\ CCl_4 = 30.5\ kJ/mol$

Mass of $CCl_4 = 284$ gm

Molar mass of $CCl_4 = 154 g/mol$

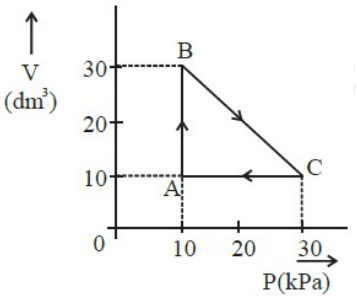
Moles of $CCl_4 = \frac{284}{154} = 1.844$ mol

ΔH°_{vap} for 1 mole = 30.5 kJ/mol

ΔH°_{vap} for 1.844 mol = 30.5 × 1.844

= 56.242 kJ

Question36



An ideal gas undergoes a cyclic transformation starting from the point A and coming back to the same point by tracing the path A → B → C → A as shown in the diagram. The total work done in the process is _____J.

[30-Jan-2024 Shift 1]

Answer: 200

Solution:

Work done is given by area enclosed in the P vs V cyclic graph or V vs P cyclic graph.

Sign of work is positive for clockwise cyclic process for V vs P graph.

$$W = \frac{1}{2} \times (30-10) \times (30-10) = 200\,\text{kPa} - \text{dm}^3$$

$$= 200 \times 1000\,\text{Pa} - \text{L} = 2\text{L} - \text{bar} = 200\text{J}$$

Question37

Two reactions are given below:

$$2\text{Fe}_{(s)} + \frac{3}{2}\text{O}_{2(g)} \rightarrow \text{Fe}_2\text{O}_{3(s)},\ \Delta\text{H}^\circ = -822\,\text{kJ/mol}$$

$$\text{C}_{(s)} + \frac{1}{2}\text{O}_{2(g)} \rightarrow \text{CO}_{(g)},\ \Delta\text{H}^\circ = -110\,\text{kJ/mol}$$

Then enthalpy change for following reaction

$$3\text{C}_{(s)} + \text{Fe}_2\text{O}_{3(s)} \rightarrow 2\text{Fe}_{(s)} + 3\text{CO}_{(g)}$$

[30-Jan-2024 Shift 2]

Answer: 492

Solution:

$2Fe_{(s)} + \frac{3}{2}O_{2(g)} \rightarrow Fe_2O_{3(s)},\ \Delta H^\circ = -822\ kJ/mol$ .....(1)

$C_{(s)} + \frac{1}{2}O_{2(g)} \rightarrow CO_{(g)},\ \Delta H^\circ = -110\ kJ/mol$ .....(2)

$3C_{(s)} + Fe_2O_{3(s)} \rightarrow 2Fe_{(s)} + 3CO_{(g)},\ \Delta H_3 = ?$

$(3) = 3 \times (2) - (1)$

$\Delta H_3 = 3 \times \Delta H_2 - \Delta H_1$

$= 3(-110) + 822$

$= 492\ kJ/mole$

Question38

Consider the following reaction at 298K.

$\frac{3}{2}O_{2(g)} \rightleftharpoons O_{3(g)} \cdot K_p = 2.47 \times 10^{-29}$.

$\Delta_r G^\ominus$ for the reaction is _____kJ. (Given R $= 8.314 JK^{-1}mol^{-1}$)

[31-Jan-2024 Shift 1]

Answer: 163

Solution:

$\frac{3}{2}O_{2(g)} \rightleftharpoons O_{3(g)} \cdot K_p = 2.47 \times 10^{-29}$.

$\Delta_r G^\ominus = -RT \ln K_p$

$= -8.314 \times 10^{-3} \times 298 \times \ln(2.47 \times 10^{-29})$

$= -8.314 \times 10^{-3} \times 298 \times (-65.87)$

$= 163.19\ kJ$

Question39

If 5 moles of an ideal gas expands from 10L to a volume of 100L at 300K under isothermal and reversible condition then work, w, is −xJ. The value of x is_____

(Given R = $8.314 JK^{-1}mol^{-1}$)

[31-Jan-2024 Shift 2]

Answer: None

Solution:

It is isothermal reversible expansion, so work done negative

$$W = -2.303nRT \log\left(\frac{V_2}{V_1}\right)$$

$$= -2.303 \times 5 \times 8.314 \times 300 \log\left(\frac{100}{10}\right)$$

$$= -28720.713J$$

$$\equiv -28721J$$

Question40

Choose the correct option for free expansion of an ideal gas under adiabatic condition from the following :

[1-Feb-2024 Shift 1]

Options:

A.

$q = 0, \Delta T \neq 0, w = 0$

B.

$q = 0, \Delta T < 0, w \neq 0$

C.

$q \neq 0, \Delta T = 0, w = 0$

D.

$q = 0, \Delta T = 0, w = 0$

Answer: D

Solution:

During free expansion of an ideal gas under adiabatic condition $q = 0, \Delta T = 0, w = 0$.

Question41

For a certain reaction at 300K, K = 10, then ΔG° for the same reaction is_____ $\times 10^{-1}$ $kJmol^{-1}$. (Given R = 8.314$JK^{-1}mol^{-1}$)

[1-Feb-2024 Shift 2]

Answer: 57

Solution:

$\Delta G^{\circ} = -RT\ell nK$

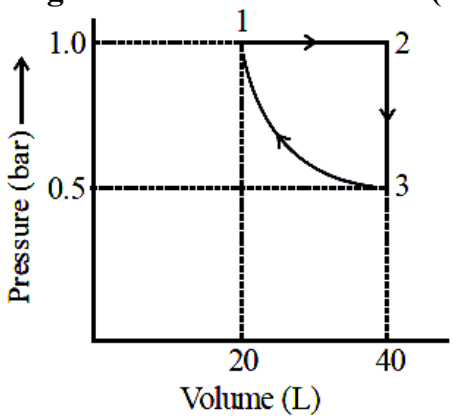
$= -8.314 \times 300 \ln(10)$

$= 5744.14 J/mole$

$= 57.44 \times 10^{-1} kJ/mole$

Question42

One mole of an ideal monoatomic gas is subjected to changes as shown in the graph. The magnitude of the work done (by the system or on the system) is J (nearest integer).



Given : log 2 = 0.3, ln 10 = 2.3
[24-Jan-2023 Shift 2]

Answer: 620

Solution:

Solution:
$1 \rightarrow 2 \Rightarrow$ Isobaric process
$2 \rightarrow 3 \Rightarrow$ Isochoric process
$3 \rightarrow 1 \Rightarrow$ Isothermal process

$W = W_{1\rightarrow 2} + W_{2\rightarrow 3} + W_{3\rightarrow 1}$

$= \left(-P(V_2 - V_1) + 0\left[-P_1V_1 \ln\left(\frac{V_2}{V_1}\right)\right]\right)$

$= \left[-1 \times (40-20) + 0 + \left[-1 \times 20 \ln\left(\frac{20}{40}\right)\right]\right]$

$= -20 + 20 \ln 2$

$= -20 + 20 \times 2.3 \times 0.3$

$= -6.2$ bar L

$|W| = 6.2$ bar $1 = 620J$

Question43

An athlete is given 100g of glucose ($C_6H_{12}O_6$) for energy. This is equivalent to 1800 kJ of energy. The 50% of this energy gained is utilized by the athlete for sports activities at the event. In order to avoid storage of energy, the weight of extra water he would need to perspire is _____ g (Nearest integer)
Assume that there is no other way of consuming stored energy.

Given : The enthalpy of evaporation of water is 45 kJ mol^{-1}
Molar mass of C, H&O are 12.1 and 16gmol^{-1}.
[25-Jan-2023 Shift 1]

Answer: 360

Solution:

Solution:
$C_6H_{12}O_6(s) + 6O_2 \rightarrow 6CO_2(g) + 6H_2O(l)$
Extra energy used to convert $H_2O(l)$ into $H_2O(l)$ into $H_2O(g)$

$= \frac{1800}{2} = 900\,kJ$

$\Rightarrow \quad 900 = n_{H_2O} \times 45$

$n_{H_2O} = \frac{900}{45} = 20\,mole$

$W_{H_2O} = 20 \times 18 = 360g$

Question44

28.0L of CO_2 is produced on complete combustion of 16.8L gaseous mixture of ethene and methane at 25°C and 1 atm. Heat evolved during the combustion process is _____ kJ.
Given : $\Delta H_C(CH_4) = -900\,kJ\,mol^{-1}$
$\Delta H_C(C_2H_4) = -1400\,kJ\,mol^{-1}$
[25-Jan-2023 Shift 2]

Answer: 847

Solution:

Solution:
Let, Volume of C_2H_4 is x litre
Total volume of $CO_2 = 2x + 16.8 - x$

$\Rightarrow \quad 28 = 16.8 + x$

$x = 11.2L$

$n_{CH_4} = \frac{PV}{RT} = \frac{1 \times 5.6}{0.082 \times 298} = 0.229 \text{ mole}$

$n_{C_2H_2} = \frac{11.2}{0.082 \times 298} = 0.458 \text{ mole}$

$\therefore \quad \text{Heat evolved} = 0.229 \times 900 + 0.458 \times 1400$

$= 206.1 + 641.2$

$= 847.3\,kJ$

Question45

Which of the following relations are correct?
(A) $\Delta U = q + p\,\Delta V$
(B) $\Delta G = \Delta H - T\,\Delta S$

(C) $\Delta S = \frac{q_{rev}}{T}$

(D) $\Delta H = \Delta U - \Delta nRT$

Choose the most appropriate answer from the options given below :
[29-Jan-2023 Shift 2]

Options:

A. C and D only

B. B and C only

C. A and B only

D. B and D only

Answer: B

Solution:

Solution:
Only (B) and (C) are correct.
(B) $G = H - TS$
At constant T
$\Delta G = \Delta H - T\Delta S$
(A) First law is given by
$\Delta U = Q + W$
If we apply constant P and reversible work.
$\Delta U = Q - P\Delta V$
(C)By definition of entropy change
$dS = \frac{dq\, q_{rev}}{T}$
At constant T
$\Delta S = \frac{q_{rev}}{T}$
(D) $H = U + PV$
For ideal gas
$H = U + nRT$
At constant T
$\Delta H = \Delta U + \Delta nRT$

Question46

When 2 litre of ideal gas expands isothermally into vacuum to a total volume of 6 litre, the change in internal energy is _____ J. (Nearest integer)
[30-Jan-2023 Shift 1]

Answer: 0

Solution:

Solution:
For ideal gas $U = f(T)$
and for isothermal process, $\Delta U = 0$

Question47

1 mole of ideal gas is allowed to expand reversibly and adiabatically from a temperature of 27°C. The work done is 3 kJ mol^{-1}. The final temperature of the gas is _____ K (Nearest integer). Given $C_v = 20 J mol^{-1} K^{-1}$.
[30-Jan-2023 Shift 2]

Answer: 150

Solution:

Solution:
$q = 0$
$\Delta U = W$
$1 \times 20 \times [T_2 - 300] = -3000$
$T_2 - 300 = -150$
$T_2 = 150K$

Question48

The enthalpy change for the conversion of
$\frac{1}{2}Cl_2(g)$ to $Cl^-(aq)$ is (−) _____
kJ mol^{-1} (Nearest integer).
Given : $\Delta_{dis} H_{Cl_{2(g)}}^{\circ} = 240 kJ mol^{-1}$
$\Delta_{eg} H_{Cl_{(g)}}^{\circ} = -350 kJ mol^{-1}$
$\Delta_{hyd} H_{Cl_{(g)}}^{\circ} = -380 kJ mol^{-1}$
[31-Jan-2023 Shift 1]

Answer: 610

Solution:

Solution:
$\frac{1}{2}Cl_{2(g)} \rightarrow Cl_{(g)} \rightarrow Cl_{(g)}^- \rightarrow Cl_{(aq.)}^-$
$\Delta H^{\circ} = \frac{1}{2} \times 240 + (-350) + (-380)$
$= -610$ ans.

Question49

Enthalpies of formation of $CCl_4(g)$, $H_2O(g)$, $CO_2(g)$ and
HCl(g) are −105, −242, −394 and −92 kJ mol^{-1}
respectively. The magnitude of enthalpy of the reaction given below is _____ kJ mol^{-1} (nearest integer)

$CCl_4(g) + 2H_2O(g) \rightarrow CO_2(g) + 4\,HCl(g)$
[31-Jan-2023 Shift 2]

Answer: 173

Solution:

Solution:
$\Delta_r H = \sum H_p - \sum H_R$
$= (-394 + 4\times -92) - (-105 + (2\times -242))$
$= -173\,kJ/mol$

Question50

0.3g of ethane undergoes combustion at 27°C in a bomb calorimeter. The temperature of calorimeter system (including the water) is found to rise by 0.5°C. The heat evolved during combustion of ethane at constant pressure is _____ kJ mol^{-1}.
(Nearest integer)
[Given: The heat capacity of the calorimeter system is 20 kJ K^{-1}, R = 8.3J$K^{-1}mol^{-1}$.
Assume ideal gas behaviour.
Atomic mass of C and H are 12 and 1gmol^{-1} respectively]
[1-Feb-2023 Shift 2]

Answer: 1006

Solution:

Solution:
(Bomb calorimeter → const volume
Heat released
By combustion of 1 mole
$C_2H_6(\Delta U) = -\frac{20\times 0.5}{0.3}\times 30 = -1000\,kJ$
$C_2H_6(g) + 7/2O_2(g) \rightarrow 2CO_2(g) + 3H_2O(l)$
$\Delta ng = 2 - (2 + 7/2) = -(7/2)$
$\Delta H = \Delta U + \Delta nRT$
$= -1000 - 7/2 \times 8.3 \times 300\,kJ$
$= -1000 - 6.225$
$= -1006\,kJ$
So heat released $= 1006\,kJ\,mol^{-1}$

Question51

The value of log K for the reaction A ⇋ B at 298K is___
(Nearest integer)
Given: $\Delta H^0 = -54.07$ kJ mol^{-1}
$\Delta S^\circ = 10JK^{-1}mol^{-1}$

(Take 2.303 × 8.314 × 298 = 5705)
[6-Apr-2023 shift 1]

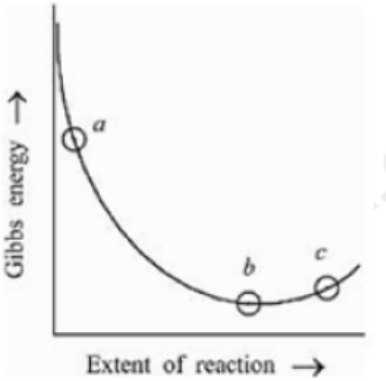
Answer: 10

Solution:

Solution:
$\Delta G^0 = \Delta H^0 - T\,\Delta S$
$\Rightarrow \Delta G^0 = (-54070 - 10 \times 298)$
Also, $\Delta G^0 = (-2.303\,RT \log K)$
$\Rightarrow (-54070 - 10 \times 298)$
$= (-2.303 \times 8.134 \times 298 \log K)$
$\Rightarrow \log K = 10$ Ans: 10

Question52

Consider the graph of Gibbs free energy G vs Extent of reaction. The number of statement/s from the following which are true with respect to points (a), (b) and (c) is____



A. Reaction is spontaneous at (a) and (b)
B. Reaction is at equilibrium at point (b) and nonspontaneous at point (c)
C. Reaction is spontaneous at (a) and nonspontaneous at (c)
D. Reaction is non-spontaneous at (a) and (b)
[6-Apr-2023 shift 1]

Answer: 2

Solution:

Solution:
For, Spontaneous process $dG < 0$
For, Equilibrium $dG = 0$
For, Nonspontaneous process $dG > 0$
$\therefore$ A Wrong
B Correct
C Correct
D Wrong

Question53

Consider the following date
Heat of combustion of $H_2(g) = -241.8\ kJ\ mol^{-1}$
Heat of combustion of $C(s) = -393.5\ kJ\ mol^{-1}$
Heat of combustion of $C_2H_5OH(1) = -1234.7\ kJ\ mol^{-1}$.
The heat of formation of $C_2H_5OH(1)$ is (-)_____ $kJ\ mol^{-1}$ (Nearest integer)
[6-Apr-2023 shift 2]

Answer: 278

Solution:

Solution:

$$2C_{(s)} + 3H_{2(g)} + \frac{1}{2}O_{2(g)} \rightarrow C_2H_5OH_{(l)}$$

$$(\Delta H_f)_{C_2H_5OH_{(l)}} = \Sigma(\Delta H_{comb})_{reactant} - \sum(\Delta H_{comb})_{product}$$

$$= 2 \times (-393.5) + 3(-241.8) - (-1234.7)$$

$$= -277.7\ kJ/mol$$

Question54

When a 60W electric heater is immersed in a gas for 100 s in a constant volume container with adiabatic walls, the temperature of the gas rises by 5°C. The heat capacity of the given gas is _____ JK^{-1} (Nearest integer)
[8-Apr-2023 shift 1]

Answer: 1200

Solution:

Solution:
Adiabatic wall \{no heat exchange between system and surrounding\}

$$C_v \times \Delta T = P \times t/\ sec$$

$$C_v \times 5 = 60 \times 100$$

$$C_v = 1200$$

Question55

For complete combustion of ethane,
$C_2H_4(g) + 3O_2(g) \rightarrow 2CO_2(g) + 2H_2O(l)$
The amount of heat produced as measured in bomb calorimeter is 1406 $KJ\ mol^{-1}$ at 300K. The minimum value of $T\Delta S$ needed to reach equilibrium is _____ (−) KJ (Nearest integer)
Given : $R = 8.3JK^{-1}mol^{-1}$
[8-Apr-2023 shift 2]

Answer: 1411

Solution:

Solution:
$\Delta G = \Delta H - T\,\Delta S$ at equilibrium:-
$\Delta G = 0$
$T\,\Delta S = \Delta H = \Delta U + \Delta ngRT = -1406 + (-2) \times 8.3 \times 300 \times 10^{-3} = -1410.98 \approx 1411$

Question56

The number of incorrect statement from the following is _____ [8-Apr-2023 shift 2]

Options:

A. The electrical work that a reaction can perform at constant pressure and temperature is equal to the reaction Gibbs energy

B. E_{cell}^{0} is dependent on the pressure

C. $\frac{dE_{cell}^{o}}{dT} = \frac{\Delta_r S^{\circ}}{nF}$

D. A cell is operating reversibly if the cell potential is exactly balanced by an opposing source of potential difference

Answer: A

Solution:

Solution:
$dG = vdp - sd\,T$
$dG = -sd\,T$
$\frac{dG}{dT} = -S \Rightarrow \frac{d\,\Delta\,G}{dT} = -\Delta\,S$
$\frac{dE^{\circ}}{dT} = \frac{-\Delta\,S}{-nF}$

Question57

The enthalpy change for the adsorption process and micelle formation respectively are [10-Apr-2023 shift 1]

Options:

A. $\Delta H_{ads} < 0$ and $\Delta H_{mic} < 0$

B. $\Delta H_{ads} > 0$ and $\Delta H_{mic} < 0$

C. $\Delta H_{ads} < 0$ and $\Delta H_{mic} > 0$

D. $\Delta H_{ads} > 0$ and $\Delta H_{mic} > 0$

Answer: C

Solution:

Solution:
Adsorption → Exothermic ($\Delta H_{ads} = -ve$)
Micelle formation → Endothermic ($\Delta H_{mic} = +ve$)
$\Delta H_{ads} < O$ and $\Delta H_{mic} > O$

Question58

Given

(A) $2\,CO(g) + O_2(g) \rightarrow 2CO_2(g)\ \Delta H_1^{o} = -x\ kJ\ mol^{-1}$

(B) $C\ (graphite) + O_2(g) \rightarrow CO_2(g)\ \ \Delta H_2^{\circ} = -y\ KJ\ mol^{-1}$

The ΔH° for the reaction $C(graphite) + \frac{1}{2}O_2(g) \rightarrow CO(g)$ is
[10-Apr-2023 shift 1]

Options:

A. $\frac{x-2y}{2}$

B. $\frac{x+2y}{2}$

C. $\frac{2x-y}{2}$

D. $2y-x$

Answer: A

Solution:

Solution:
$2\,CO(g) + O_2(g) \rightarrow 2CO_2(g)\ \ \Delta H_1^{\circ} = y\,kJ/mol$ .....(2)
$C(\ graphite\ ) + O_2(g) \rightarrow CO_2(g)\ \ \Delta H_2^{\circ} = -y\ \ kJ/mol$..... (2)

$C(\ graphite\ ) + {}^{1/2}O_2(g) \rightarrow CO(g)\ \ \Delta H_3^{\circ} = ?$

$\Delta H_3^{\circ} = H_2^{\circ} - \frac{H_1^{\circ}}{2} = -y - \frac{-x}{2}$

$\Delta H_3^{\circ} = \frac{x}{2} - y = \frac{x-2y}{2}$

Question59

$FeO_4^{2-} \xrightarrow{+2.2V} Fe^{3+} \xrightarrow{+0.70V} Fe^{2+} \xrightarrow{-0.45V} Fe^{0}$

$E_{FeO_4^{2-}/Fe^{2+}}^{\circ}$ is $x \times 10^{-3}V$.

The value of x is _____
[10-Apr-2023 shift 1]

Answer: 1825

Solution:

Solution:

$$FeO_4^{2-} + 3e^{\ominus} \rightarrow Fe^{+3} \quad \Delta G_1$$

$$Fe^{+3} + e^{\ominus} \rightarrow Fe^{+2} \quad \Delta G_2$$

$$FeO_4^{-2} + 4e^{\Theta} \rightarrow Fo^{+2} \quad \Delta G_3$$

$$\Delta G_3 = \Delta G_1 + \Delta G_2$$

$$(-)4E_3^{\circ}F = (-)3 \times 2.2 \times F + (-)1 \times 0.7 \times f$$

$$4E_3^{\circ} = 6.6 + 0.7 = 7.3$$

$$E_3^{\circ} = \frac{7.3}{4} = 1.825 = 1825 \times 10^{-3}$$

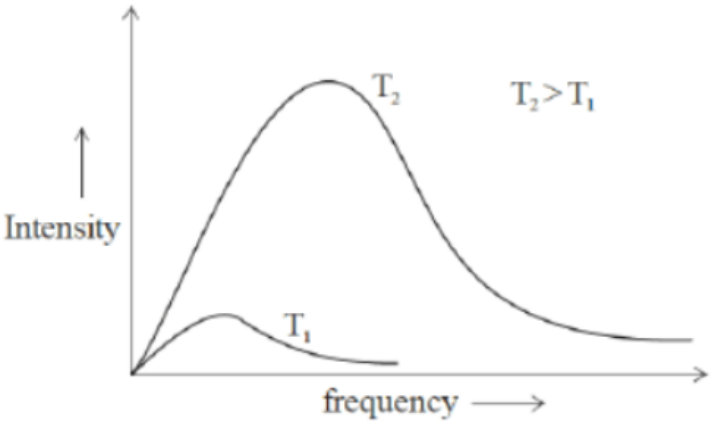
Question60

The number of incorrect statement/s about the black body from the following is _____
(A) Emit or absorb energy in the form of electromagnetic radiation
(B) Frequency distribution of the emitted radiation depends on temperature
(C) At a given temperature, intensity vs frequency curve passes through a maximum value
(D) The maximum of the intensity vs frequency curve is at a higher frequency at higher temperature compared to that at lower temperature
[10-Apr-2023 shift 1]

Answer: 0

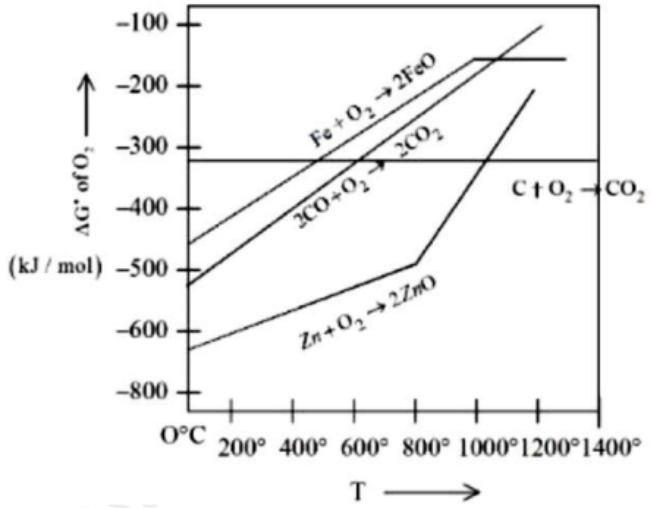
Solution:

Solution:



Question61

Gibbs energy vs T plot for the formation of oxides is given below.



For the given diagram, the correct statement is -
[10-Apr-2023 shift 2]

Options:

A. At 600°C, C can reduce ZnO

B. At 600°C, C can reduce FeO

C. At 600°C, CO cannot reduce FeO

D. At 600°C, CO can reduce ZnO

Answer: B

Solution:

Solution:
$FeO + C \rightarrow Fe + CO_2$
At 600°C ΔG of Reaction is $-Ve$

Question62

$A(g) \rightleftharpoons 2B(g) + C(g)$
For the given reaction, if the initial pressure is 450 mmHg and the pressure at time t is 720 mmHg at a constant temperature T and constant volume V. The fraction of A(g) decomposed under these conditions is $x \times 10^{-1}$. The value of x is _____ (nearest integer)
[10-Apr-2023 shift 2]

Answer: 3

Solution:

Solution:
$A(g) \rightleftharpoons 2B(g) + C(g)$
$t = 0 \quad 450$
time $\quad 450 - x \quad 2x$
$P_T = P_A + P_B + P_C$

$720 = 450 - x + 2x + x$
$2x = 270$
$x = 135$
Fraction of A decomposed $= \frac{135}{450} = 0.3 = 3 \times 10^{-1}$
So, $x = 3$

Question63

The number of endothermic process/es from the following is
A. $I_2(g) \rightarrow 2I(g)$
B. $HCl(g) \rightarrow H(g) + Cl(g)$
C. $H_2O(l) \rightarrow H_2O(g)$
D. $C(s) + O_2(g) \rightarrow CO_2(g)$
E. Dissolution of ammonium chloride in water
[10-Apr-2023 shift 2]

Answer: 4

Solution:

Solution:
A→ Endothermic (Atomisation) B→ Endothermic (Atomisation)
C→ Endothermic (Vapourisation) D→ Exothermic (Combustion)
E→ Endothermic (Dissolution)

Question64

Solid fuel used in rocket is a mixture of Fe_2O_3 and Al (in ratio 1 : 2) the heat evolved (KJ) per gram of the mixture is _____ (Nearest integer)
Givne $\Delta H_f^{\ominus}(Al_2O_3) = -1700\ KJ\ mol^{-1}$
$\Delta H_f^{\ominus}(Fe_2O_3) = -840\ KJ\ mol^{-1}$
[11-Apr-2023 shift 1]

Answer: 4

Solution:

Solution:
$Fe_2O_3 + 2\,Al \rightarrow Al_2O_3 + 2\,Fe$
$\Delta H_r = (\Delta H_f)Al_2O_3 - \Delta H_f^{\circ}(Fe_2O_3)$
$= -1700 - (-840)$
$= -860\ kJ$
Fe_2O_3 & $Al \rightarrow 1 : 2$
$Fe_2O_3 = 1\ mole = (2 \times 25 + 48)$
$= 112 + 48 = 160\ gm$
$Al = 2\ mole = 2 \times 27 = 54\ gm$

Total mass $= 160 + 54 = 214\,\text{gm}$

Heat evolved per gm $= \frac{-860}{214}\,\text{kJ} = -4.01 \approx 4\,\text{kJ}$

Question65

The total number of intensive properties from the following is _____ new line volume, Molar heat capacity, Molarity, E^{θ} cell, Gibbs free energy change, Molar mass, Mole
[11-Apr-2023 shift 2]

Answer: 4

Solution:

Solution:
Extensive ⇒ Mole, Volume, Gibbs free energy.
Intensive ⇒ Molar mass, Molar heat capacity, Molarity, E^{θ} cell.

Question66

One mole of an ideal gas at 350K is in a 2.0L vessel of thermally conducting walls, which are in contact with the surroundings. It undergoes isothermal reversible expansion from 2.0L to 3.0L against a constant pressure of 4 atm. The change in entropy of the surroundings (ΔS) is _____ JK^{-1} (Nearest integer)
Given: R = 8.314$JK^{-1}mol^{-1}$
[12-Apr-2023 shift 1]

Answer: 3

Solution:

Solution:

$\Delta S_{System} = nR\ell\left(\frac{V_2}{V_1}\right) = 1 \times 8.314 \ln\left(\frac{3}{2}\right)$

$\Delta S_{System} = 3.37$

$\Delta S_{Sur.} = 3.37$

Question67

$A_2 + B_2 \rightarrow 2\,AB \cdot \Delta H_f^0 = -200\,\text{kJ mol}^{-1}$ new line AB, A_2 and B_2 are diatomic molecules. If the bond enthalpies of A_2, B_2 and AB are in the ratio 1 : 0.5 : 1, then the bond enthalpy of A_2 is _____ kJ mol^{-1}. (Nearest integer)
[13-Apr-2023 shift 1]

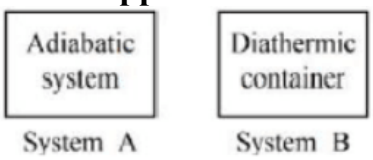
Answer: 800

Solution:

Solution:
$A_2 + B_2 \rightarrow 2\,AB \quad \Delta H_f^\circ = -200\,kJ/mol$
Bond enthalpy of $A_2 = x$
Bond enthalpy of $B_2 = 0.5x$
Bond enthalpy of $AB = x$
$\Delta H_f^\circ = x + 0.5x - 2x = -2(200)$
$-0.5x = -400$
$x = \frac{400}{0.5} = 800\,kJ/mol$
Bond enthalpy of $A_2 = x = 800\,kJ/mol$

Question68

What happens when methane undergoes combustion in systems A and B respectively?



[13-Apr-2023 shift 2]

Options:

A.

System A	System B
Temperature remains same	Temperature rises

B.

System A	System B
Temperature falls	Temperature rises

C.

System A	System B
Temperature falls	Temperature remains same

D.

System A	System B
Temperature rises	Temperature remains same

Answer: D

Solution:

Solution:
Adiabatic boundary does not allow heat exchange thus heat generated in container can't escape out thereby increasing the temperature.
In case of Diathermic container, heat flow can occur to maintain the constant temperature.

Question69

30.4 kJ of heat is required to melt one mole of sodium chloride and the entropy change at the melting point is _____ $28.4JK^{-1}mol^{-1}$ at 1 atm. The melting point of sodium chloride is K. (Nearest Integer)
[15-Apr-2023 shift 1]

Answer: 1070

Solution:

Solution:

$$\Delta S = \frac{\Delta H}{T_{mp}}$$

$$28.4 = \frac{30.4 \times 1000}{T_{mp}}$$

$$T_{mp} = 1070.422K.$$

Question70

$2O_3(g) \rightleftharpoons 3O_2(g)$
At 300K, ozone is fifty percent dissociated. The standard free energy change at this temperature and 1 atm pressure is (−)_____$Jmol^{-1}$. (Nearest integer)
[Given : In 1.35 = 0.3 and .R = $8.3JK^{-1}mol^{-1}$]
[24-Jun-2022-Shift-1]

Answer: 747

Solution:

Solution:

$$2O_3(g) \rightleftharpoons 3O_2(g)$$
$$1-x \qquad \frac{3x}{2}$$

Given, $x = 0.5$

$$\therefore k_p = \frac{[3(0.5)]^3 \times 1}{[2]^3 \times (0.5)^2 \times 1.25}$$

$$\therefore k_p = \frac{27}{8} \times \frac{0.5}{1.25} = 1.35$$

$$\Delta G^\circ = -2.303\, RT \log k_p$$
$$= -2.303 \times 8.3 \times 300 \log 1.35$$
$$= -8.3 \times 300 \ln(1.35)$$
$$= -747 Jmol^{-1}$$

Question71

The standard entropy change for the reaction
$4\,Fe(s) + 3O_2(g) \rightarrow 2Fe_2O_3(s)$ is $-550 JK^{-1}$ at 298K.

[Given : The standard enthalpy change for the reaction is $-165\,kJ\,mol^{-1}$].
The temperature in K at which the reaction attains equilibrium is____(Nearest Integer)
[25-Jun-2022-Shift-1]

Answer: 300

Solution:

Solution:
$\Delta G = \Delta H - T\,\Delta S = 0$ at equilibrium
$\Rightarrow -165 \times 10^3 - T \times (-505) = 0$
$\Rightarrow T = 300K$

Question72

At 25°C and 1 atm pressure, the enthalpy of combustion of benzene (I) and acetylene (.g) are $-3268\,kJ\,mol^{-1}$ and $-1300\,kJ\,mol^{-1}$, respectively. The change in enthalpy for the reaction $3C_2H_2(g) \rightarrow C_6H_6(I)$, is
[25-Jun-2022-Shift-2]

Options:

A. $+324\,kJ\,mol^{-1}$

B. $+632\,kJ\,mol^{-1}$

C. $-632\,kJ\,mol^{-1}$

D. $-732\,kJ\,mol^{-1}$

Answer: C

Solution:

Solution:
$\Delta H = \Sigma\,\Delta H_{Combustion}$ (Reactant) $-\Sigma\,\Delta H_{Combustion}$ (Product)
$= 3 \times (-1300) - [-3268]$
$= -632\,kJ\,mol^{-1}$

Question73

For complete combustion of methanol
$CH_3OH(I) + \frac{3}{2}O_2(g) \rightarrow CO_2(g) + 2H_2O(I)$

the amount of heat produced as measured by bomb calorimeter is 726 kJ mol^{-1} at 27°C. The enthalpy of combustion for the reaction is −x kJ mol^{-1}, where x is____ (Nearest integer)
(Given : R = 8.3$JK^{-1}mol^{-1}$)
[26-Jun-2022-Shift-1]

Answer: 727

Solution:

Solution:

$$CH_3OH(l) + \frac{3}{2}O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$$

$$\Delta H = \Delta U + \Delta n_g RT$$

$$= -726\,kJ + \left(\frac{-1}{2}\right) \times 8.3 \times 300$$

$$\simeq -727\,kJ\,mol^{-1}$$

Question74

CNG is an important transportation fuel. When 100g CNG is mixed with 208g oxygen in vehicles, it leads to the formation of CO 2 and H_2O and produces large quantity of heat during this combustion, then the amount of carbon dioxide, produced in grams is ____ [nearest integer]
[Assume CNG to be methane]
[26-Jun-2022-Shift-2]

Answer: 143

Solution:

Solution:

$$\begin{array}{lccccc} & CH_4 & + 2O_2 & \rightarrow & CO_2 & + 2H_2O \\ \text{Initially:} & 6.25 & 6.5 & & 0 & 0 \\ \text{finally:} & 6.25 - \frac{6.5}{2} & 0 & & \frac{6.5}{2} & 6.5 \end{array}$$

Initially,
Mass of $CNG(CH_4) = 100\,gm$

$\therefore$ Moles of $CH_4 = \frac{100}{16} = 6.25$ moles

Mass of $O_2 = 208\,gm$

$\therefore$ Moles of $O_2 = \frac{208}{32} = 6.5$ moles

From reaction you can see,
1 mole of CH_4 react with 2 mole of O_2

$\therefore 6.25$ mole of CH_4 react with $6.25 \times 2 = 12.5$ moles of O_2

But here only 6.5 moles of O_2 present

So, O_2 will act as limiting reagent.

$\therefore$ Produced CO_2 will depends on moles of O_2.

From 2 moles of $O_2$1 mole of CO_2

∴ From 6.5 moles of O_2 $\frac{6.5}{2}$ moles of CO_2 produced.

∴ Weight of $CO_2 = \frac{6.5}{2} \times 44 = 143$ gm

Question75

A fish swimming in water body when taken out from the water body is covered with a film of water of weight 36g. When it is subjected to cooking at 100°C, then the internal energy for vaporization in kJ mol^{-1} is _____[nearest integer]
[Assume steam to be an ideal gas. Given $\Delta_{vap}H^{\Theta}$ for water at 373K and 1 bar is 41.1 kJ mol^{-1}; R = 8.31$JK^{-1}mol^{-1}$]
[26-Jun-2022-Shift-2]

Answer: 38

Solution:

Solution:
$H_2O(l) \rightarrow H_2O(g)$
$n = \frac{36}{18} = 2\,mol$
$\Delta U = \Delta H - \Delta n_g RT$
$= 41.1 - \frac{1 \times 8.31 \times 373}{1000}\,kJ/mol$
$= 38\,kJ/mol$

Question76

40% of HI undergoes decomposition to H_2 and I_2 at 300K. $\Delta G^{\ominus}$ for this decomposition reaction at one atmosphere pressure is ____ $Jmol^{-1}$. [nearest integer]
(Use .R = 8.31$JK^{-1}mol^{-1}$; log 2 = 0.3010, ln 10 = 2.3, log 3 = 0.477)
[26-Jun-2022-Shift-2]

Answer: 2735

Solution:

Solution:

$$\underset{\text{At } t=0:}{\underset{1}{HI}} \rightleftharpoons \underset{0}{\frac{1}{2}H_2} + \underset{0}{\frac{1}{2}I_2}$$

$$\text{At } t = t_{eq}: \underset{1-\alpha}{} \quad \underset{\frac{\alpha}{2}}{} \quad \underset{\frac{\alpha}{2}}{}$$

$$\therefore K = \frac{\left(\frac{\alpha}{2}\right)^{1/2} \times \left(\frac{\alpha}{2}\right)^{1/2}}{(1-\alpha)}$$

Given $\alpha = \frac{40}{100} = 0.4$

$$\therefore K = \frac{\left(\frac{0.4}{2}\right)^{1/2} \times \left(\frac{0.4}{2}\right)^{1/2}}{(1-0.4)}$$

$$= \frac{1}{3}$$

We know,

$$\Delta G^{\circ} = -RT \ln K$$

$$= -RT \ln\left(\frac{1}{3}\right)$$

$$= +RT \ln 3$$

$$= +8.314 \times 300 \times \ln 3$$

$$= 2735 J/mol$$

Question77

Choose the correct answer from the options given below:

	List - I		List - II
(A)	Spontaneous process	(I)	$\Delta H < 0$
(B)	Process with $\Delta P = 0, \Delta T = 0$	(II)	$\Delta G_{T,P} < 0$
(C)	$\Delta H_{\text{reaction}}$	(III)	Isothermal and isobaric process
(D)	Exothermic Process	(IV)	[Bond energies of molecules in reactants]-[Bond energies of product molecules

[27-Jun-2022-Shift-1]

Options:

A. (A) - (III), (B) - (II), (C) - (IV), (D) - (I)

B. (A) - (II), (B) - (III), (C) - (IV), (D) - (I)

C. (A) - (II), (B) - (III), (C) - (I), (D) - (IV)

D. (A) - (II), (B) - (I), (C) - (III), (D) - (IV)

Answer: B

Solution:

Solution:

(A) For a spontaneous process $\Delta G_{T,P} < 0$

(B) $\Delta P = 0 \rightarrow$ Isobaric process, $\Delta T = 0 \rightarrow$ Isothermal process

(C) $\Delta H_{\text{reaction}} = (\Sigma \text{ Bond energies of reactants}) - (\Sigma \text{ bond energies of products})$

(D) $\Delta H < 0$ is for exothermic reaction

Question78

When 5 moles of He gas expand isothermally and reversibly at 300K from 10 litre to 20 litre, the magnitude of the maximum work obtained is____ J. [nearest integer] (Given : $R = 8.3 JK^{-1}mol^{-1}$ and $\log 2 = 0.3010$)

[27-Jun-2022-Shift-2]

Answer: 8630

Solution:

Solution:

$W_{rev} = -2.303\,nRT\log_{10}\left(\frac{V_2}{V_1}\right)$

$= -2.303 \times 5 \times 8.3 \times 300 \times \log_{10}\left(\frac{20}{10}\right)$

$\simeq -8630J$

Question79

4.0L of an ideal gas is allowed to expand isothermally into vacuum until the total volume is 2.0L. The amount of heat absorbed in this expansion is _____L atm.
[28-Jun-2022-Shift-1]

Answer: 0

Solution:

Solution:
Work done $= -P_{ext}\ \Delta v$
$\because P_{ext} = 0$ (vacuum)
$\therefore w = 0, \quad \Delta U = 0$ (as the process is isothermal)
So, $q = 0$

Question80

For combustion of one mole of magnesium in an open container at 300K and 1 bar pressure, $\Delta_C H^{\Theta} = -601.70\ kJ\ mol^{-1}$, the magnitude of change in internal energy for the reaction is _____ kJ. (Nearest integer)
(Given : $R = 8.3JK^{-1}mol^{-1}$)
[28-Jun-2022-Shift-2]

Answer: 600

Solution:

Solution:
$Mg(s) + \frac{1}{2}O_2(g) \longrightarrow MgO(s)$
$\Delta H = \Delta U + \Delta ngRT$
$\Delta ng = -\frac{1}{2}$

$-601.70 = \Delta U - \frac{1}{2}(8.3)(300) \times 10^{-3}$

$\Delta U = -601.70 + 1.245$

$\Delta U \simeq -600\,kJ$

The magnitude of change in internal energy is $600\,kJ$.

Question81

17.0g of NH_3 completely vapourises at $-33.42^{\circ}C$ and 1 bar pressure and the enthalpy change in the process is 23.4 kJ mol^{-1}. The enthalpy change for the vapourisation of 85g of NH_3 under the same conditions is ____kJ.
[29-Jun-2022-Shift-1]

Answer: 117

Solution:

Solution:
Given data is for 1 moles and asked for 5 moles so
value is 23.4 × 5 = 117 kJ

Question82

2.2g of nitrous oxide (N_2O) gas is cooled at a constant pressure of 1 atm from 310K to 270K causing the compression of the gas from 217.1 mL to 167.75 mL. The change in internal energy of the process, ΔU is $' -x' J$. The value of ' x ' is _____[nearest integer]
(Given : atomic mass of N = 14gmol^{-1} and of O = 16gmol^{-1}. Molar heat capacity of N_2O is 100JK^{-1}mol^{-1})
[29-Jun-2022-Shift-2]

Answer: 195

Solution:

Solution:

$\Delta T = -40K$

$\Delta U = q + w$

$= \frac{100 \times 2.2}{44}(-40) - (-49.39) \times 10^{-3} \times 101.325$

$= -200 + 5$

$= -195J$

$x = 195$

Question83

The enthalpy of combustion of propane, graphite and dihydrogen at 298K are $-2220.0 kJ\ mol^{-1}$, $-393.5 kJ\ mol^{-1}$ and $-285.8 kJ\ mol^{-1}$ respectively. The magnitude of enthalpy of formation of propane (C_3H_8) is kJ mol^{-1}. (Nearest integer)
[25-Jul-2022-Shift-1]

Answer: 104

Solution:

Solution:
Enthalpy of combustion of propane, graphite and H_2 at 298K are

$$C_3H_8(g) + 5O_2(g) \rightarrow 3CO_2(g) + 4H_2O(I),\ \Delta H_1 = -2220\ kJ\ mol^{-1}$$

$$C(\ \text{graphite}\ ) + O_2(g) \rightarrow CO_2(g),\quad \Delta H_2 = -393.5\ kJ\ mol^{-1}$$

$$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(I),\quad \Delta H_3 = -285.8\ kJ\ mol^{-1}$$

The desired reaction is

$$3C\ (\text{graphite}) + 4H_2(g) \rightarrow C_3H_8(g)$$

$$\Delta H_f = 3\Delta H_2 + 4\Delta H_3 - \Delta H_1$$
$$= 3(-393.5) + 4(-285.8) - (-2220)$$
$$= -103.7\ kJ\ mol^{-1}$$
$$|\Delta H_f| \simeq 104\ kJ\ mol^{-1}$$

Question84

While performing a thermodynamics experiment, a student made the following observations.
$HCl + NaOH \rightarrow NaCl + H_2O\ \Delta H = -57.3 kJ\ mol^{-1}$
$CH_3COOH + NaOH \rightarrow CH_3COONa + H_2O\ \Delta H = -55.3 kJ\ mol^{-1}$
The enthalpy of ionization of CH_3COOH as calculated by the student is _____ kJ mol^{-1}. (nearest integer)
[25-Jul-2022-Shift-2]

Answer: 2

Solution:

Solution:
(I) $HCl + NaOH \rightarrow NaCl + H_2O$

$\Delta H_1 = -57.3 KJmol^{-1}$

(II) $CH_3COOH + NaOH \rightarrow CH_3COONa + H_2O$

$\Delta H_2 = -55.3 KJmol^{-1}$

Reaction (I) can be written as
(III) $NaCl + H_2O \rightarrow HCl + NaOH$

$\Delta H_3 = 57.3 KJmol^{-1}$

By adding (II) and (III)

$$CH_3COOH + NaCl \rightarrow CH_3COONa + HCl\quad \Delta H_r$$
$$\Delta H_r = \Delta H_3 + \Delta H_2 = 57.3 - 55.3$$
$$= 2\ kJ\ mol^{-1}$$

Question85

2.4g coal is burnt in a bomb calorimeter in excess of oxygen at 298K and 1 atm pressure. The temperature of the calorimeter rises from 298K to 300K. The enthalpy change during the combustion of coal is $-x\ kJ\ mol^{-1}$. The value of x is _____. (Nearest Integer)
(Given : Heat capacity of bomb calorimeter $20.0\ kJ\ K^{-1}$. Assume coal to be pure carbon)
[26-Jul-2022-Shift-1]

Answer: 200

Solution:

Solution:

$$Q(\text{ Heat evolved }) = -\frac{C_{system}\,\Delta T}{n}$$

$$n_{coal} = \frac{2.4}{12}$$

$$Q = \frac{-20(300-298)}{0.2}$$

$$Q = -200\ kJ/mol$$

$$x = 200$$

Question86

For the reaction
$H_2F_2(g) \rightarrow H_2(g) + F_2(g)$
$\Delta U = -59.6\ kJ\ mol^{-1}$ at 27°C
The enthalpy change for the above reaction is (−) _____ $kJ\ mol^{-1}$ [nearest integer]
Given: $R = 8.314 JK^{-1}mol^{-1}$
[26-Jul-2022-Shift-2]

Answer: 57

Solution:

Solution:

$$H_2F_2(g) \longrightarrow H_2(g) + F_2(g)$$

$$\Delta U = -59.6\ kJ\ mol^{-1} \text{ at } 27^{\circ}C$$

$$\Delta H = \Delta U + \Delta n_g RT$$

$$= -59.6 + \frac{1 \times 8.314 \times 300}{1000}$$

$$= -57.10\ kJ\ mol^{-1}$$

Question87

The molar heat capacity for an ideal gas at constant pressure is $20.785 JK^{-1}mol^{-1}$. The change in internal energy is 5000J upon heating it from 300K to 500K. The number of moles of the gas at constant volume is _____. [[Nearest integer] (Given: R = $8.314 JK^{-1}mol^{-1}$)
[27-Jul-2022-Shift-1]

Answer: 2

Solution:

Solution:
$C_p = 20.785 JK^{-1}mol^{-1}$
and $\Delta U = nCv\ \Delta\ T$
$\therefore\ nC_v = \frac{5000}{200} = 25$
and we know that
$C_p - C_v = R$
$20.785 - \frac{25}{n} = 8.314$
$n = \frac{25}{(20.785 - 8.314)} = 2$

Question88

A gas (Molar mass = $280 gmol^{-1}$) was burnt in excess O_2 in a constant volume calorimeter and during combustion the temperature of calorimeter increased from 298.0K to 298.45K. If the heat capacity of calorimeter is 2.5 kJ K^{-1} and enthalpy of combustion of gas is 9 kJ mol^{-1} then amount of gas burnt is _____ g. (Nearest Integer)
[27-Jul-2022-Shift-2]

Answer: 35

Solution:

Solution:
Let xg is burnt
$\text{moles} = \frac{x}{280}$
heat released by $\frac{x}{280}$ mole $= 2.5 \times 0.45$ kJ
heat released by 1 mole $= \frac{2.5 \times 0.45 \times 280}{x}$ kJ
$\Delta H = \Delta U + \Delta\ ngRT$
$\Delta H \simeq \Delta U$
$9 = \frac{2.5 \times 280 \times 0.45}{x}$
$x = 35g$

Question89

Which of the following relation is not correct?
[28-Jul-2022-Shift-1]

Options:

A. $\Delta H = \Delta U - P\,\Delta V$

B. $\Delta U - q + W$

C. $\Delta S_{sys} + \Delta\, S_{surr} \geq$ sl ant0

D. $\Delta G = \Delta H - T\,\Delta\, S$

Answer: A

Solution:

Solution:
If $U + Pv$ (By definition)
$\Delta 14 = \Delta U + \Delta$ (Pr) at constant pressure
$\Delta H = \Delta U + P\,\Delta\, V$

Question90

Given below are two statements: One is labelled as Assertion AA and the other is labelled as Reason RR
Assertion A: The reduction of a metal oxide is easier if the metal formed is in liquid state than solid state.
Reason RR : The value of $\Delta G^{\ominus}$ Δ G becomes more on negative side as entropy is higher in liquid state than solid state.
In the light of the above statements, choose the most appropriate answer from the options given below
[28-Jul-2022-Shift-2]

Options:

A. Both A and R are correct and R is the correct explanation of A

B. Both A and R are correct but R is NOT the correct explanation of A

C. A is correct but R is not correct

D. A is not correct but R is correct

Answer: A

Solution:

Solution:
Reduction of a metal oxide is easier if the metal is formed in a liquid state at the temperature of reduction because the entropy is higher if the metal is in a liquid state.

Question91

Among the following the number of state variables is _____.
Internal energy (U)
Volume (V)
Heat (q)
Enthalpy (H)
[28-Jul-2022-Shift-2]

Answer: 3

Solution:

Solution:
State variables are internal energy (U), Volume (V) and Enthalpy (H).

Question92

When 600 mL of 0.2MHNO_3 is mixed with 400 mL of 0.1M NaOH solution in a flask, the rise in temperature of the flask is _____ $\times 10^{-2}$°C
(Enthalpy of neutralisation $= 57\ kJ\ mol^{-1}$ and Specific heat of water $= 4.2JK^{-1}g^{-1}$) (Neglect heat capacity of flask)
[29-Jul-2022-Shift-1]

Answer: 54

Solution:

Solution:
HNO_3
$600\ mL \times 0.2M = 120m\ mol$
NaOH
$400\ mL \times 0.1M = 40m\ mol$

	$HNO_3 + NaOH \rightarrow NaNO_3 + H_2O$		
Bef.	120	40	
Aft.	80	0	40m mol

Heat liberated from reaction
$= 40 \times 10^{-3} \times 57 \times 10^3$JHeat gained by solution $= mC\ \Delta\ T$
m = mass of solution $= V \times d = 1000 \times 1$
$= 1000g$
Heat gained by solution $= 1000 \times 4.2 \times \Delta\ T \ldots$ (2)
From (1) and (2)
Heat liberated = Heat gained
$40 \times 10^{-3} \times 57 \times 10^3 = 1000 \times 4.2 \times \Delta\ T$
$\Delta T = 54 \times 10^{-2}$°C
(Rounded off to the nearest integer)

Question93

Five moles of an ideal gas at 293K is expanded isothermally from an initial pressure of 2.1M Pa to 1.3M Pa against at constant external pressure 4.3M Pa. The heat transferred in this process is ..... kJ mol^{-1}. (Rounded off to the nearest integer) [R = 8.314J $mol^{-1}K^{-1}$]
[25 Feb 2021 Shift 2]

Answer: 15

Solution:

Solution:
The gas performs isothermal irreversible work (W).
where, $\Delta U = 0$ (change in internal energy)
From, 1st law of thermodynamics,

$\Rightarrow \Delta U = \Delta Q + W$

$\Rightarrow 0 = \Delta Q + W$

$\Rightarrow \Delta Q = -W$

Now, $W = -p_{ext}(V_2 - V_1)$

$= -p_{ext}\left(\frac{nRT}{p_2} - \frac{nRT}{p_1}\right) = -p_{ext} \times nRT\left(\frac{1}{p_2} - \frac{1}{p_1}\right)$

Given, $p_{ext} = 4.3M\,Pa$, $p_1 = 2.1M\,Pa$, $p_2 = 1.3M\,Pa$,

$n = 5mol$, $T = 293K$ and $R = 8.314J\,mol^{-1}K^{-1}$

$= -4.3 \times 5 \times 8.314 \times 293\left(\frac{1}{1.3} - \frac{1}{2.1}\right)$

$= -15347.70J\,mol^{-1}$

$= -15.347kJ\,mol^{-1} \sim eq - 15kJ\,mol^{-1}$

$\Rightarrow \Delta Q = 15kJ\,mol^{-1}$

Question94

The ionisation enthalpy of Na^+formation from Na(g) is $495.8kJ\,mol^{-1}$, while the electron gain enthalpy of Br is $-325.0kJ\,mol^{-1}$. Given, the lattice enthalpy of NaBr is $-728.4kJ\,mol^{-1}$. The energy for the formation of NaBr ionic solid is (−)..... $\times 10^{-1}kJ\,mol^{-1}$.
[25 Feb 2021 Shift 1]

Answer: 5576

Solution:

Solution:

$Na(g) \xrightarrow{IE_1} Na^+(g)[IE_1 = 495.8kJ\,mol^{-1}]$

$Br(g) \xrightarrow[\text{gain enthalpy}]{\text{Electron}} (EG)Br^-(g)[\text{Gain enthalpy} = -325kJ\,mol^{-1}]$

$Na^+ + Br^- \xrightarrow[\text{energy}]{\text{Lattice}} (LE\,NaBr(s)[\text{Lattice energy} = -728.4kJ\,mol^{-1}]$

$\Delta H_{\text{Formation}} = IE_1 + \text{Gain enthalpy} + \text{Lattice energy}\ \Delta H = 495.8 + (-325) + (-728.4)$

$= -557.6kJ/mol$

$= -5576 \times 10^{-1}kJ$

Question95

The average $S-F$ bond energy in $kJ\,mol^{-1}$ of SF_6 is ..... (Rounded off to the nearest integer) [Given, the values of standard enthalpy of formation of SF_6(g), S(g) and F (g) are -1100, 275 and $80kJ\,mol^{-1}$ respectively.]
[26 Feb 2021 Shift 2]

Answer: 309

Solution:

Solution:
So, $\Delta_f H^\circ[S, g] + 6\times \Delta_f H^\circ[F, g] = \Delta_f H^\circ[SF_6, g] + 6 \times E_{S-F}$
$[\therefore E_{S-F} =$ Average $S-F$ bond energy in $SF_6]$
$275 + 6\times 80 = -1100 + 6 \times E_{S-F}$
$\Rightarrow E_{S-F} = \frac{275 + 6 \times 80 + 1100}{6}$
$= 309.16 kJ\ mol^{-1} = 309 kJ\ mol^{-1}$

Question96

At 25°C, 50g of iron reacts with H Cl to form $FeCl_2$. The evolved hydrogen gas expands against a constant pressure of 1 bar. The work done by the gas during this expansion is .....J. (Round off to the nearest integer) [Given, R = 8.314J $mol^{-1}K^{-1}$. Assume, hydrogen is an ideal gas]
[Atomic mass off Fe is 55.85u]
[16 Mar 2021 Shift 2]

Answer: 2218

Solution:

Solution:
$Fe(s) + HCl(aq) \longrightarrow FeCl_2(aq) + H_2(g)$
$\underset{50g}{}$
(Moles of Fe = Moles of H_2)
moles $= \frac{50}{55.85}$ $0.8952mol$ $0.8952mol$
$= 0.8952mol$
Work done can be given by
$W = -p_{ext}\ \Delta V$
$W = -\Delta n_g RT$
where, Δn_g = { gaseous moles on product side } - {gaseous mole on reactant side}
$= 0.8952 - 0$
$W = -0.8952 \times 8.314 \times 298$
$= -2217.92J$
Closest integer $= 2218$

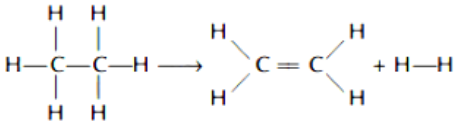
Question97

For the reaction,
$C_2H_6 \longrightarrow C_2H_4 + H_2$ (Round off to the nearest integer). [Given : Bond enthalpies in kJ $mol\ m^{-1}$: C − C = 347, C = C = 611; C − H = 414; H − H = 436
[18 Mar 2021 Shift 1]

Answer: 128

Solution:

Solution:
The equation representing various substances involved in chemical reaction is



Given bond enthalpies of various bonds in $kJ\ mol^{-1}$ i.e.
$C-C=347, C=C=611, C-H=414, H-H=436$
This reaction involves the breaking of $2C-H$ bonds and formation, of $1C=C$ bond and $1H-H$ bond.
$\Delta_r H$ = [Sum of bond enthalpies of reactants] - [Sum of bond enthalpies of products]

$= [1\,\Delta_{C-C}H + 6\,\Delta_{C-H}H] - [1\Delta_{C}=CH + 4\,\Delta_{C-H}H + 1\,\Delta_{H-H}H]$
$= (357 + 6\times 414) - [611 + 4\times 414 + 436]$
$= 2831 - 2703$
$= 128 kJ/mol$

Question98

The standard enthalpies of formation of Al_2O_3 and CaO are $-1675 kJ\ mol^{-1}$ and $-625 kJ\ mol^{-1}$ respectively. For the reaction, $3CaO + 2Al \longrightarrow 3Ca + Al_2O_3$ the standard reaction enthalpy $\Delta_t H^\circ$ ..... kJ . (Round off to the nearest integer).
[17 Mar 2021 Shift 1]

Answer: 230

Solution:

Solution:
Given, $\Delta_f H_{Al_2O_3}^{o} = -1675 kJ/mol$
$\Delta_f H_{CaO}^{\circ} = -625 kJ/mol$
To find $\Delta_r H^\circ$ for the reaction
$3CaO + 2Al \longrightarrow 3Ca + Al_2O_3;\ \Delta_r H^\circ = ?$

$\Delta_r H^\circ = \sum_{\Delta_f} H_{product}^{\circ} - \sum_{\Delta_f} H_{Reactant}^{\circ}$
$= \left\{\Delta_f H_{Al_2O_3}^{\circ} + 3\,\Delta_f H_{Ca}^{\circ}\right\} - \{3\,\Delta_f H_{CaO}^{\circ} + 2\,\Delta_f H_{Al}^{\circ}\}$
$= \Delta H_f^{\circ}(Al_2O_3) - 3\times \Delta H_f^{a}(CaO)$
We know, $\Delta_f H^\circ$ for elemental state $= 0$
$\Rightarrow \Delta_f H_{Ca}^{o} = 0, \Delta_f H_{Al}^{o} = 0$
Putting the value,
$\Delta_t H^\circ = \{-1675 + 0\} - \{3(-625) + 0\}$
$= -1675 + 1905$
$= 230 kJ$

Question99

During which of the following processes, does entropy decrease?
A. Freezing of water to ice at $0^\circ C$.
B. Freezing of water to ice at $-10^\circ C$.

C. $N_2(g) + 3H_2(g) \longrightarrow 2NH_3(g)$
D. Adsorption of CO(g) and lead surface.
E. Dissolution of NaCl in water.
[17 Mar 2021 Shift 2]

Options:

A. A, B, C and D

B. B and C

C. A and E

D. A, C and E

Answer: A

Solution:

Solution:
Entropy will decrease in A, B, C and D processes.
A, B→ Freezing of water will decrease entropy as particles will move closer and forces of attraction will increase. This leads to a decrease in randomness. So, entropy decreases.
(A) Water $\xrightarrow{0^\circ C}$ ice; $\Delta S = -$ ve
(B) Water $\xrightarrow{-10^\circ C}$ ice; $\Delta S = -$ ve
(C) $N_2(g) + 3H_2(g) \longrightarrow 2NH_3(g)$; $\Delta S = -$ ve
Number of moles are decreasing
$n = 2 - (3 + 1)$
$n = -2$
So, entropy decreases.
(D) Adsorption; $\Delta S = -$ ve
Adsorption will lead to a decrease in the randomness of gaseous particles.
So, entropy decreases.
(E) $NaCl(s) \longrightarrow Na^+(aq) + Cl^-(aq)$; $\Delta S > 0$
The number of species on product side is more than the number of species on reactant side. So, entropy increases on dissolution of NaCl in water.

Question100

When 400mL of 0.2M H_2SO_4 solution is mixed with 600mL of 0.1M NaOH solution, the increase in temperature of the final solution is -10^{-2}K. (Round off to the nearest integer).
[Use : $H^+(aq) + OH^-(aq) \rightarrow H_2O$:
$\Delta_\gamma H = -57.1 kJ\ mol^{-1}$]
Specific heat of $H_2O = 4.18 J\ K^{-1} g^{-1}$
density of $H_2O = 1.0 gcm^{-3}$
Assume no change in volume of solution on mixing.
[27 Jul 2021 Shift 2]

Answer: 82

Solution:

Solution:

$n_{H^+} = \frac{400 \times 0.2}{1000} \times 2 = 0.16$

$n_{OH^-} = \frac{600 \times 0.1}{1000} = 0.06(L.R)$

Now, heat liberated from reaction = heat gained by solutions

or, $0.06 \times 57.1 \times 10^3$

$= (1000 \times 1.0) \times 4.18 \times \Delta T$

$\therefore \Delta T = 0.8196K$

$= 81.96 \times 10^{-2}K \approx 82 \times 10^{-2}K$

Question101

For water at 100°C and 1 bar,
$\Delta_{vap}H - \Delta_{vap}U = \text{——} \times 10^2 J\ mol^{-1}$.
(Round off to the Nearest Integer)
[Use : R = 8.31J $mol^{-1}K^{-1}$]
[Assume volume of $H_2O(l)$ is much smaller than volume of $H_2O(g)$. Assume $H_2O(g)$ treated as an ideal gas]
[27 Jul 2021 Shift 1]

Answer: 31

Solution:

Solution:

$H_2O_{(1)} \rightleftharpoons H_2O_{(v)}$

$\Delta H = \Delta U + \Delta n_g RT$

for 1 mole waters ;

$\Delta n_g = 1$

$\therefore \Delta n_g RT = 1mol \times 8.31J/mol - k \times 373K$

$= 3099.63J \cong 31 \times 10^2 J$

Question102

A system does 200 J of work and at the same time absorbs 150 J of heat. The magnitude of the change in internal energy is _____ J. (Nearest integer)
[25 Jul 2021 Shift 2]

Answer: 50

Solution:

Solution:

$w = -200J$, $q = +150$: $\Delta U = q + w$

$\Delta U = 150 - 200 = -50J$: magnitude $= 50J = |\Delta U|$

Question103

At 298K , the enthalpy of fusion of a solid (X) is 2.8~kJ mol $^{-1}$ and the enthalpy of vaporisation of the liquid (X) is 98.2kJ mol $^{-1}$. The enthalpy of sublimation of the substance (X) in kJ mol $^{-1}$ is _____ . (in nearest integer)
[25 Jul 2021 Shift 1]

Answer: 101

Solution:

Solution:
$\Delta H_{sub} = \Delta H_{fivs.} + \Delta H_{vap.}$
$= 2.8 + 98.2$
$= 101 kJ / mol$

Question104

At 298.2K the relationship between enthalpy of bond dissociation (in kJ mol $^{-1}$) for hydrogen (E_H) and its isotope, deuterium (E_D), is best described by:
[25 Jul 2021 Shift 1]

Options:

A. $E_H = \frac{1}{2}E_D$

B. $E_H = E_D$

C. $E_H \simeq E_D - 7.5$

D. $E_H = 2E_D$

Answer: C

Solution:

Solution:
Enthalpy of bond dissociation (kJ/mole) at 298.2K
For , hydrogen = 435.88
For , Deuterium = 443.35
$\therefore E_H \simeq E_D - 7.5$

Question105

If the standard molar enthalpy change for combustion of graphite powder is -2.48×10^2kJ mol $^{-1}$,the amount of heat generated on combustion of 1g. of graphite powder is _____ kJ . (Nearest integer)
[22 Jul 2021 Shift 2]

Answer: 21

Solution:

Solution:
$1\text{mol graphite} = 12\text{gmC}$
Ans. $= \frac{248}{12} = 20.67\text{kJ} / \text{gm}$ heat evolved

---

Question106

For a given chemical reaction A → B at 300K the free energy change is −49.4kJ mol $^{-1}$ and the enthalpy of reaction is 51.4kJ mol $^{-1}$. The entropy change of the reaction is J K $^{-1}$mol $^{-1}$.
[20 Jul 2021 Shift 2]

Answer: 336

Solution:

Solution:
Given chemical reaction:
$A \xrightarrow[T\ 300K]{} B \quad [\Delta G]_{P,T} = -49.4\ \text{kJ/mol}$
$\Delta H_{rxn} = 51.4\ \text{kJ/mol}$
$\Delta S_{rxn} = ?$
⇒ From the relation $[\Delta G]_{P,T} = \Delta H - T\Delta S$
$\Rightarrow \Delta S_{rxn} = \frac{\Delta H_{rn} - [\Delta G]_{P,T}}{T}$
$= \frac{[51.4 - (-49.4)] \times 1000}{300} \frac{J}{\text{mol K}}$
$\Rightarrow \Delta S_{rxn} = 336 \frac{J}{\text{mol K}}$

Question107

For water $\Delta_{vap}H = 41\ \text{kJ mol}^{-1}$ at 373K and 1 bar pressure. Assuming that water vapour is an ideal gas that occupies a much larger volume than liquid water, the internal energy change during evaporation of water is ..... kJ mol $^{-1}$.
[Use R = 8.3Jmol^{-1}K^{-1}]
[26 Aug 2021 Shift 2]

Answer: 38

Solution:

Solution:
$H_2O(l) \rightarrow H_2O(g)$
$\because \Delta H = \Delta U + \Delta n_g RT$
ΔH = enthalpy of vaporisation
ΔU = change in internal energy

Δn_g = number of moles of water vapour
$R = 8.3 J mol^{-1} K^{-1}$
T = 373 K
$\therefore 41\ kJ = \Delta U + RT$
$\Delta U = 41 - 8.3 \times 373 \times 10^{-3} = 41 - 3.095$
$= 37.90\ kJ\ mol^{-1} = 38\ kJ\ mol^{-1}$

Question108

200 mL of 0.2M HCl is mixed with 300 mL of 0.1M NaOH. The molar heat of neutralisation of this reaction is −57.1 kJ. The increase in temperature in °C of the system on mixing is $x \times 10^{-2}$. The value ofx is ..... (Nearest integer)
[Given, specific heat of water = $4.18 J g^{-1} K^{-1}$ Density of water = $1.00 g cm^{-3}$]
(Assume no volume change on mixing)
[27 Aug 2021 Shift 1]

Answer: 82

Solution:

Solution:
Millimoles of HCl = 200 × 0.2 = 40
Millimoles of NaOH = 300 × 0.1 = 30
Heat released (q) = n× molar heat
$= \frac{30}{1000} \times 57.1 \times 1000$
= 1713 J
Mass of solution = × 500 × 1= 500 g
We know that,
$\Delta T = \frac{q}{mc} = \frac{1713 J}{500 g \times 4.18 J/g - K}$
$= 0.8196 K = 81.96 \times 10^{-2} K$
x ≈ 82

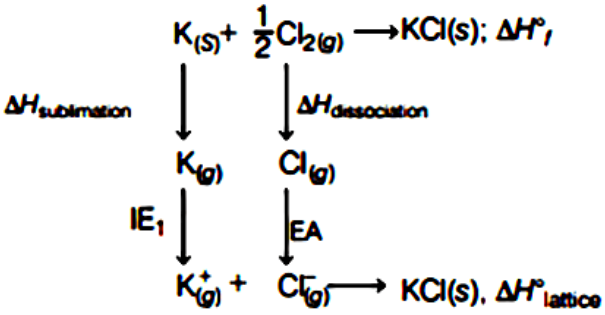
Question109

The Born-Haber cycle for KCl is evaluated with the following data :
$\Delta_f H°$ for KCl = $-436.7\ kJ\ mol^{-1}$
$\Delta_{sub} H°$ for K = $89.2\ kJ\ mol^{-1}$,
$\Delta_{ionisation} H°$ for K = $419.0\ kJ\ mol^{-1}$;
$\Delta_{electron\ gain} H^{\cup}$ for Cl(g) = $-348.6\ kJ\ mol^{-1}$,
$\Delta_{bond} H°$ for Cl_2 = $243.0\ kJ\ mol^{-1}$
The magnitude of lattice enthalpy of KCl in kJ mol^{-1} is . (Nearest integer)
[26 Aug 2021 Shift 1]

Answer: 718

Solution:

Solution:
Born-Haber cycle for KCl is as follows



$\therefore \Delta H_f^\circ = \Delta H_{sublimation} + \frac{1}{2}\Delta H_{dissociation} + IE_1 + EA + \Delta H_{lattice}$

$-436.7\,kJ/mol = 89.2\,kJ/mol + \frac{1}{2} \times (243\,kJ/mol) + 419\,kJ/mol + (-348.6\,kJ/mol) + \Delta H^\circ_{lattice}$

$\Rightarrow \Delta H^\circ_{lattice} = -717.8\,kJ/mol$

∴ Magnitude of lattice enthalpy of KCl in kJ/mol is 718 (nearest).

Question110

The incorrect expression among the following is [31 Aug 2021 Shift 2]

Options:

A. $\frac{\Delta G_{System}}{\Delta S_{Total}} = -T$ (at constant p)

B. $\ln k = \frac{\Delta H^\circ - T\Delta S^\circ}{RT}$

C. $k = e^{-\frac{\Delta G^\circ}{RT}}$

D. For isothermal process, $W_{reversible} = -nRT \ln \frac{V_f}{V_i}$

Answer: B

Solution:

Solution:
All the expression of thermodynamics are correct except
$\ln k = \frac{\Delta H^\circ - T\Delta S^\circ}{RT}$
As we know,
$\Delta G = \Delta H - T\Delta S^\circ$ ...(i)
Also $\Delta G = -RT \ln K$ ...(ii)
∴ From (i) and (ii)
$-RT \ln K = \Delta H - T\Delta S^\circ$
$\therefore \ln K = \frac{-\Delta H}{R} + \frac{\Delta S^\circ}{R}$
The corect expression is
$\ln K = \frac{-\Delta H^\circ}{RT} + \frac{\Delta S^\circ}{R} = \frac{\Delta H^\circ - T\Delta S^\circ}{RT}$

Question111

Data given for the following reaction is as follows.
$FeO(s) + C_{(graphite)} \rightarrow Fe(s) + CO(g)$

Substance	$\Delta H°(kJ\ mol^{-1})$	$\Delta S°(J\ mol^{-1}K^{-1})$
FeO(s)	-266.3	57.49
$C_{(graphite)}$	0	5.74
Fe(s)	0	27.28
CO(g)	-110.5	197.6

The minimum temperature in K at which the reaction becomes spontaneous is..... (Integer answer)
[27 Aug 2021 Shift 2]

Answer: 964

Solution:

Solution:
$FeO(s) + C_{(graphite)} \rightarrow Fe(s) + CO(g)$
$\Delta H°_{reaction} = \Delta H°_{f(product)} - \Delta H°_{f(reactants)}$
$= [\Delta H°_{f(Fe)} + \Delta H°_{f(CO)}] - [\Delta H°_{f(FeO)} - \Delta H°_{f(C)}]$
$= [0 + (-110.5)] - [-266.3 - 0]$
$= 156 kJ^{-1} mol^{-1}$
$\Delta S°_{reaction} = \Delta S°_{product} - \Delta S°_{reactant}$
$= [\Delta S°_{(Fe)} + \Delta S°_{(CO)}] - [\Delta S°_{(FeO)} - \Delta S°_{(O)}]$
$= [27.28 + 197.6] - [57.49 + 5.79]$
$= 161 JK^{-1} mol^{-1}$
According to Gibb's equation,
$\Delta G° = \Delta H° - T\Delta S°$
The reaction becomes spontaneous when $\Delta G°$ is atleast zero or negative.
$0 = \Delta H° - T\Delta S°$
$T\Delta S° = \Delta H°$
$\Rightarrow T = \frac{\Delta H°}{\Delta S°} = \frac{156\ kJ\ mol^{-1}}{161 JK^{-1} mol^{-1}}$
$= \frac{156000 mol^{-1}}{161 JK^{-1} mol^{-1}} = 964 K$
The temperature at which reaction becomes spontaneous is 964 K.

Question112

For the reaction, $2NO_2(g) \rightleftharpoons N_2O_4(g)$, when $\Delta S = -176.0 \times JK^{-1}$ and $\Delta H = -57.8\ kJ\ mol^{-1}$, the magnitude of ΔG at 298K for the reaction is .....kJ mol^{-1}. (Nearest integer)
[1 Sep 2021 Shift 2]

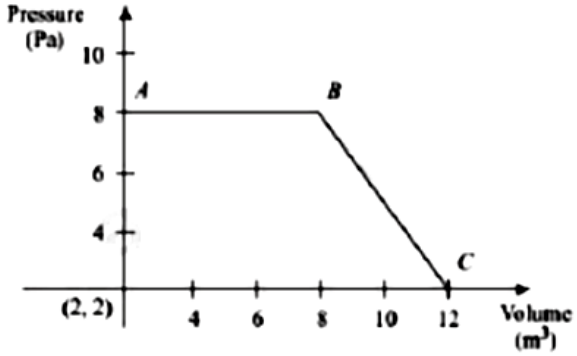
Answer: 5

Solution:

Solution:
Given, $\Delta H = -57.8\,kJ\,mol^{-1}$
$\Delta S = -176JK^{-1}mol^{-1}$
$T = 298K$
Using Gibb's free energy relation
$\Delta G = \Delta H - T\Delta S$
where, ΔG = change in Gibb's free energy
ΔH = change in enthalpy
T = temperature
ΔS = change in entropy
$\Delta G = 57.8\,kJ/mol - [298K \times (-176Jk^{-1}\,mol^{-1})]$
$= 57.8\,kJ/mol - \left(298 \times \frac{-176}{1000}kJ\right)$ [∴1kJ = 1000 J]
= − 5.352 kJ/mol
| ΔG | = 5.352
Hence, answer is 5.

Question113

The magnitude of work done by a gas that undergoes a reversible expansion along the path ABC shown in the figure is _____.



[NV, Jan. 08,2020(I)]

Answer: 48

Solution:

Solution:
Work done is given by the area under the trapezium.
$\therefore |w| = \frac{1}{2}(6+10) \times 6 = 48J$

Question114

At constant volume, 4 mol of an ideal gas when heated from 300K to 500K changes its internal energy by 5000J . The molar heat capacity at constant volume is
[NV, Jan. 08, 2020 (II)]

Answer: 6.25

Solution:

Solution:
$\Delta U = nC_v \Delta T$
$5000 = 4 \times C_v(500 - 300)$
$C_v = 6.25 J K^{-1} mol^{-1}$

Question115

If enthalpy of atomisation for $Br_2(\ell)$ is xkJ / mol and bond enthalpy for Br_2 is ykJ / mol, the relation between them:
[Jan. 09,2020 (I)]

Options:

A. is x = y

B. does not exist

C. is x > y

D. is x < y

Answer: C

Solution:

Solution:
$\Delta H_{atomisation} = \Delta H_{vap} +$
Bond energy Hence x > y

Question116

For the reaction ; A(l) → 2B(g)
ΔU = 2.1kcal , ΔS = 20cal K^{-1} at 300K
Hence Δ G in kcal is.
[NV, Jan. 07, 2020 (I)]

Answer: -2.7

Solution:

Solution:
$\Delta U = 2.1 kcal = 2.1 \times 10^3 cal$
$\Delta n_g = 2$
$\Delta H = \Delta U + \Delta n_g RT$
$= 2.1 \times 10^3 + 2 \times 2 \times 300$
$= 2100 + 1200$
$= 3300 cal$
$\Delta G = \Delta H - T \Delta S$
$= 3300 - 300 \times 20$
$= 3300 - 6000$
$= -2700 cal s$
$= -2.7 kcal$

Question117

The standard heat of formation ($\Delta_f H_{298}^{\circ}$) of ethane (in kJ/mol), if the heat of combustion of ethane, hydrogen and graphite are −1560, −393.5 and −286kJ/mol, respectively is _____. [NV, Jan. 07,2020 (II)]

Answer: -192.5

Solution:

Solution:

$C(s) + O_2(g) \rightarrow CO_2(g)$

$\Delta_C H^0[\text{Cgraphite}] = -286.0\text{kJ/mol} \quad \ldots$ (i)

$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(g)$

$\Delta_C H^0[H_2(g)] = -393.5\text{kJ/mol} \quad \ldots$ (ii)

$C_2H_6(g) + \frac{7}{2}O_2(g) \rightarrow 2CO_2(g) + 3H_2O(g)$

$\Delta_C H^0[C_2H_6(g)] = -1560\text{kJ/mol} \quad \ldots$ (iii)

The reaction of formation of ethane is

$2C(s) + 3H_2(g) \rightarrow C_2H_6(g)$

$\therefore \Delta_f H^0 \text{ of } C_2H_6(g) =$

$2\times \Delta_C{}^0H[C_{\text{gaphite}}] + 3\times \Delta_c H^0[H_2(g)] - \Delta_C H^0[C_2H_6(g)]$

$= 2\times(-286.0) + 3(-393.5) - (-1560)$

$= -192.5\text{kJ/mol}$

Question118

The true statement amongst the following is : [Jan. 09,2020 (II)]

Options:

A. Both ΔS and S are functions of temperature.

B. Both S and ΔS are not functions of temperature.

C. S is not a function of temperature but ΔS is a function of temperature.

D. S is a function of temperature but ΔS is not a function of temperature.

Answer: A

Solution:

Solution:

A system at higher temperature has greater entropy (randomness). S and Δ S are related with T as:

$S_T = \int_0^T \frac{nC \cdot dT}{T} \quad \Delta S = \int \frac{dq}{T}$

Thus both S and ΔS are function of temperature.

Question119

Five moles of an ideal gas at 1 bar and 298K is expanded into vacuum to double the volume. The work done is:
[Sep. 04,2020 (II)]

Options:

A. $C_v(T_2 - T_1)$

B. $-RT\,(V_2 - V_1)$

C. $-RT\,\ln V_1 / V_1$

D. zero

Answer: D

Solution:

Solution:
In expansion against vacuum
$P_{ext} = 0$
$W = -P_{ext}\,\Delta V = 0$
$W = 0$

Question120

Lattice enthalpy and enthalpy of solution of NaCl are 788kJ mol^{-1} and 4kJ mol^{-1}, respectively. The hydration enthalpy of NaCl is :
[Sep. 05, 2020(II)]

Options:

A. -780kJ mol^{-1}

B. 780kJ mol^{-1}

C. -784kJ mol^{-1}

D. 784kJ mol^{-1}

Answer: C

Solution:

Solution:
$\Delta_{sol.}\,H^\circ = \Delta_{lattice}\,H^\circ + \Delta_{Hyd.}\,H^\circ$
$4 = 788 + \Delta_{Hyd}\,H^\circ$
$\Delta_{Hyd.}H^\circ = -784\text{kJ mol}^{-1}$

Question121

For one mole of an ideal gas, which of these statements must be true?
(1) U and H each depends only on temperature (2) Compressibility factor z is not equal to 1
(3) $C_{P,m} - C_{V,m} = R$ (4) $dU = C_y dT$ for any process
[Sep. 04, 2020 (I)]

Options:

A. (1) and (3)

B. (2), (3) and (4)

C. (3) and (4)

D. (1), (3) and (4)

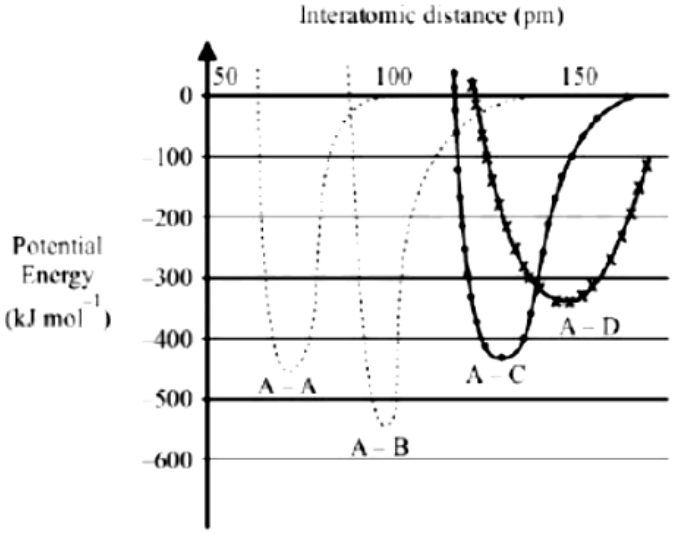
Answer: D

Solution:

Solution:
(1) For ideal gas U and H are function of temperature.
(2) Compressibility factor for an ideal gas is 1 .
(3) $C_P - C_V = R$
(4) ΔU = CvdT for all processes.

Question122

The intermolecular potential energy for the molecules A, B, C and D given below suggests that :



[Sep. 04,2020 (1)]

Options:

A. A-D has the shortest bond length

B. A-A has the largest bond enthalpy

C. D is more electronegative than other atoms

D. A-B has the stiffest bond

Answer: D

Solution:

Solution:
A-B bond has highest intermolecular potential energy among the given molecules. Hence, it is strongest bond and has maximum bond enthalpy.

Question123

The internal energy change (in J) when 90g of water undergoes complete evaporation at $100^{\circ}C$ is _____.
(Given : ΔH_{vap} for water at 373K $= 41kJ/mol$, $R = 8.314J\,K^{1}mol^{-1}$)
[NV, Sep. 02,2020(I)]

Answer: 189494

Solution:

Solution:
$\Delta H = \Delta U + \Delta n_g RT$
$n = \frac{90}{18} = 5mol$
$H_2O(l) \rightleftharpoons H_2O(g)$
$\Delta n = 1$
$41000 = \Delta U + 1 \times 8.314 \times 373$
$\Rightarrow \Delta U = 37898.875J$
For 5 moles, $\Delta U = 37898.87 \times 5 = 189494J$

Question124

The heat of combustion of ethanol into carbon dioxids and water is −327 kcal at constant pressure. The heat evolved (in cal) at constant volume and $27^{\circ}C$ (if all gases behave ideally) is ($R = 2cal\,mol^{-1}K^{-1}$)_____.
[NV, Sep. 02, 2020 (II)]

Answer: -326400

Solution:

Solution:
$C_2H_5OH(l) + O_2(g) \longrightarrow 2O_2(g) + 3H_2O(I)$
$\Delta H_C = -327K\,cal$
$\Delta H = \Delta U + \Delta n_g RT$
$\Rightarrow -327 \times 10^3 = \Delta U + (-1) \times 2 \times 300$
$\Rightarrow \Delta U = -327 \times 10^3 + 600$
$\therefore \Delta U = -326400cal$

Question125

For a dimerization reaction, $2A(g) \rightarrow A_2(g)$,
at 298K , $\Delta U^{\Theta} = -20kJ\,mol^{-1}$, $\Delta S^{\Theta} = -30J\,K^{-1}mol^{-1}$,
then the ΔG^{Θ} will be _____ J.
[NV, Sep. 05,2020 (II)]

Answer: -13538

Solution:

Solution:

From $\Delta H^\circ = \Delta U^\circ + \Delta n_g RT$

$\Delta H^\circ = -20 \times 1000 - 1 \times 8.314 J/mol.K \times 298K$

$= -22477.57J$

$\Delta G^\circ = \Delta H^\circ - T\ \Delta S^\circ = -22477.57 - (298 \times -30)$

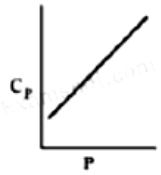
$= -13538J$

Question126

For a diatomic ideal gas in a closed system, which of the following plots does not correctly describe the relation between various thermodynamic quantities?
[Jan. 12, 2019 (I)]

Options:

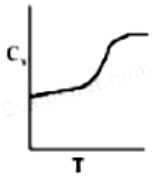
A.



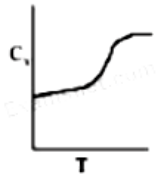
B.



C.



D.



Answer: A

Solution:

Solution:
For ideal gas, C_p and C_v are dependent on temperaturc only.

$C_p = \frac{7}{2}R$ (Independent of P)

$C_v = \frac{5}{2}R$ (Independent of V)

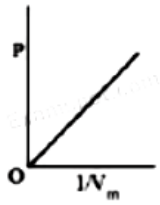
Thus, C_p will not change with pressure.

Question127

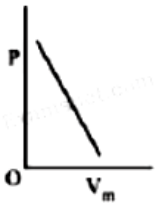
The combination of plots which does not represent isothermal expansion of an ideal gas is: [Jan. 12, 2019 (II)]

Options:

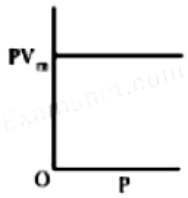
A.



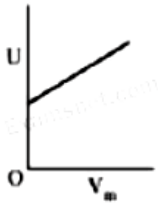
B.



C.



D.

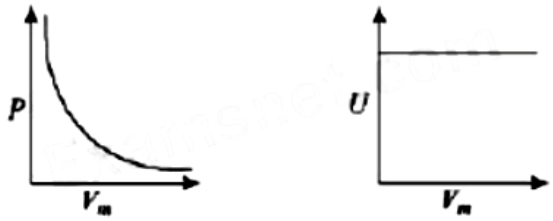


Answer: 0

Solution:

Solution:
Isothermal expansion
$PV_m = K$ (graph - C)

$P = \frac{K}{V_m}$(graph − A)



Therefore (B) and (D) are not correct representation.

Question128

An ideal gas undergoes isothermal compression from $5m^3$ to $1m^3$ against a constant external pressure of $4N\ m^{-2}$. Heat released in this process is used to increase the temperature of 1 mole of Al . If molar heat capacity of Al is $24J\ mol^{-1}K^{-1}$, the temperature of Al increases by: [Jan. 10, 2019 (II)]

Options:

A. $\frac{3}{2}K$

B. 2K

C. $\frac{2}{3}K$

D. 1K

Answer: C

Solution:

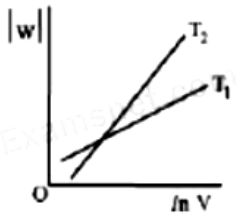
Solution:
We know that,
$w = -P_{ext}(V_f - V_i)$
$w = -4N\ m^{-2}(1-5)m^3$
$w = 16N\ m \Rightarrow 16J$
For isothermal compression,
$\Delta U = q + w$
$\Rightarrow q = -w = -16J\ (\because \Delta U = 0$ for isothermal process)
From calorimetry,
Heat given $= nC\Delta T$
So, $16 = \frac{1 \times 24J \times \Delta T}{mol\ K}$
$\therefore$ Change in temperature, $\Delta T = \frac{2}{3}K$

Question129

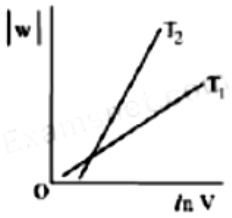
Consider the reversible isothermal expansion of an ideal gas in a closed system at two different temperatures, T_1 and $T_2(T_1 < T_2)$. The correct graphical depiction of the dependence of work done (w) on the final volume (V) is: [Jan. 9, 2019 (I)]

Options:

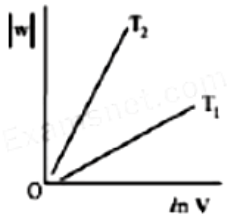
A.



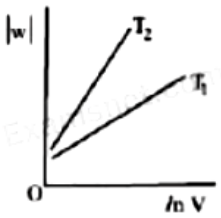
B.



C.



D.



Answer: B

Solution:

Solution:
For reversible isothermal expansion,

$$w = -nRT \ln \frac{V_2}{V_1}$$

$$\Rightarrow |w| = nRT \ln \frac{V_2}{V_1}$$

$$|w| = nRT (\ln V_2 - \ln V_1)$$

$$|w| = nRT \ln V_2 - nRT V_1$$

$$y = mx + c$$

So, slope of curve 2 is more than curve 1 and intercept of curve 2 is more negative than curve 1 .

Question130

(i) C (graphite) $+O_2(g) \rightarrow CO_2(g)$; $\Delta_r H^0 = xkJ\ mol^{-1}$

(ii) C(graphite) $+ \frac{1}{2}O_2(g) \rightarrow CO(g)$; $\Delta_r H^0 = ykJ\ mol^{-1}$

(iii) $CO(g) + \frac{1}{2}O_2(g) \rightarrow CO_2(g)$; $\Delta_r H^0 = zkJ\ mol^{-1}$

Based on the above thermochemical equations, find out which one of the following algebraic relationships is correct?
[Jan. 12, 2019 (II)]

Options:

A. $x = y + z$

B. $z = x + y$

C. $y = 2z - x$

D. $x = y - z$

Answer: A

Solution:

Solution:
Equation (i) can be obtained by adding equations (ii) and equation (iii)

$C(\text{ graphite }) + \frac{1}{2}O_2(g) \rightarrow CO(g);\ \Delta H_1^\circ = y\text{kJ mol}^{-1}\ldots$ (ii)

$CO(g) + \frac{1}{2}O_2(g) \rightarrow CO_2(g);\ \Delta H_2^\circ = z\text{kJ mol}^{-1}\ldots$ (iii)

$C\text{ (graphite)} + O_2(g) \rightarrow CO_2(g);\ \Delta H_3^\circ = x\text{kJ mol}^{-1}\ldots$ (i)

$\because\ \Delta H_3^\circ = \Delta H_1^\circ + \Delta H_2^\circ$

$x = y + z$

Question131

Two blocks of the same metal having same mass and at temperature T_1, and T_2, respectively, are brought in contact with each other and allowed to attain thermal equilibrium at constant pressure. The change in entropy, ΔS, for this process is :
[Jan. 11, 2019 (I)]

Options:

A. $C_P \ln\left[\frac{(T_1 + T_2)^2}{4T_1T_2}\right]$

B. $2C_p \ln\left[\frac{(T_1 + T_2)^{\frac{1}{2}}}{T_1T_2}\right]$

C. $2C_P \ln\left(\frac{T_1 + T_2}{4T_1T_2}\right)$

D. $2C_P \ln\left[\frac{T_1 + T_2}{2T_1T_2}\right]$

Answer: A

Solution:

Solution:

Final temperature $= \frac{T_1 + T_2}{2}$, let $T_2 > T_1$

$\therefore dS = \frac{dq}{T} = \frac{C_p dT}{T}$

$\therefore \Delta S = C_p \ln\left(\frac{T_f}{T_i}\right)$

$\therefore \Delta S_{Total} = C_p \ln\left(\frac{T_1 + T_2}{2T_1}\right) + C_p \ln\left(\frac{T_1 + T_2}{2T_2}\right)$

$= C_p \ln\left[\frac{(T_1 + T_2)^2}{4T_1 T_2}\right]$

Question132

The process with negative entropy change is:
[Jan. 10, 2019 (II)]

Options:

A. Dissociation of $CaSO_4(s)$ to $CaO(s)$ and $SO_3(g)$

B. Sublimation of dry ice

C. Dissolution of iodine in water

D. Synthesis of ammonia from N_2 and H_2

Answer: D

Solution:

Solution:
In the process of synthesis of ammonia from N_2 and H_2, number of moles decreases which implies that the change in entropy will be negative.
$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

Question133

The entropy change associated with the conversion of 1 kg of ice at 273K to water vapours at 383K is:
(Specific heat of water liquid and water vapour are 4.2kJ $K^{-1}kg^{-1}$ and 2.0kJ $K^{-1}kg^{-1}$; heat of liquid fusion and vapourisation of water are 334kJ kg^{-1} and 2491kJ kg^{-1}, respectively).
(log 273 = 2.436, log 373 = 2.572, log 383 = 2.583)
[Jan. 9, 2019 (II)]

Options:

A. 7.90kJ $kg^{-1}K^{-1}$

B. 2.64kJ $kg^{-1}K^{-1}$

C. 8.49kJ $kg^{-1}K^{-1}$

D. 9.26kJ $kg^{-1}K^{-1}$

Answer: D

Solution:

Solution:
∴ Total entropy change
$\Delta S = 1.28 + 6.68 + 1.31 + 0.05$
$= 9.26 kJ\, kg^{-1} K^{-1}$

Question134

For the chemical reaction $X \rightleftharpoons Y$, the standard reaction Gibbs energy depends on temperature T (in K) as $\Delta_r G^\circ$ (in kJ mol^{-1}) = $120 - \frac{3}{8}T$
The major component of the reaction mixture at T is :
[Jan. 11, 2019 (I)]

Options:

A. Y if T = 300K

B. Y if T = 280K

C. X if T = 350K

D. X if T = 315K

Answer: D

Solution:

Solution:
At 315K ; $\Delta G^\circ = 120 - \frac{3}{8}T$
$\Delta G^\circ = 120 - 118.125 =$ positive
Since ΔG° is positive then $K_{eq} < 1$.
So $\frac{[Y]}{[X]} < 1$.
$\therefore [X] > [Y]$

Question135

The reaction
MgO(s) + C(s) → Mg(s) + CO(g), for which $\Delta H^\circ = +491.1 kJ\, mol^{-1}$ and $\Delta S^\circ = 198.0 J\, K^{-1} mol^{-1}$ is not feasible at 298K. Temperature above which reaction will be feasible is
[Jan. 11, 2019 (II)]

Options:

A. 2040.5K

B. 1890.0K

C. 2480.3K

D. 2380.5K

Answer: C

Solution:

Solution:
$MgO(s) + C(s) \rightarrow Mg(s) + CO(g)$
For a reaction to be spontaneous
$\Delta G < 0$
$\Delta H^\circ - T\ \Delta S^\circ < 0$
$\Rightarrow T > \frac{\Delta H^\circ}{\Delta S^\circ}$
$T > \frac{491.1 \times 1000}{198}$
$T > 2480.3K$

Question136

The standard reaction Gibbs energy for a chemical reaction at an absolute temperature T is given by $\Delta G^\circ = A - BT$ where A and B are non-zero constants. Which of the following is true about this reaction?
[Jan. 11, 2019 (II)]

Options:

A. Exothermic if $B < 0$

B. Endothermic if $A > 0$

C. Endothermic if $A < 0$ and $B > 0$

D. Exothermic if $A > 0$ and $B < 0$

Answer: B

Solution:

Solution:
$\Delta G^\circ = A - BT$
A and B are non-zero constants
$\therefore \Delta G^\circ = \Delta H^\circ - T\ \Delta S^\circ = A - BT$
$\therefore$ Reaction will be endothermic if $\Delta G^0 > 0$.
Hence, $A > 0$ and $B < 0$.

Question137

A process has $\Delta H = 200J\ mol^{-1}$ and $\Delta S = 40J\ K^{-1}mol^{-1}$ Out of the values given below, choose the minimum temperature above which the process will be spontaneous:
[Jan. 10, 2019 (I)]

Options:

A. 20K

B. 12K

C. 5K

D. 4K

Answer: C

Solution:

Solution:
$\Delta H = 200 J\,mol^{-1}$
$\Delta S = 40 J\,K^{-1} mol^{-1}$
For spontaneous reaction. $\Delta G < 0$
$\Delta H - T\,\Delta S < 0; \Delta H < T\,\Delta S$
$\frac{\Delta H}{\Delta S} < T;\ \frac{200}{40} < T$
$5 < T$
So, minimum temperature is $5K$

Question138

An ideal gas is allowed to expand from 1L to 10L against a constant external pressure of 1 bar. The work done in kJ is:
[April 12, 2019(I)]

Options:

A. -9.0

B. $+10.0$

C. -0.9

D. -2.0

Answer: C

Solution:

Solution:
$W = -P\,\Delta V$
$= -(1\ bar) \times (9L)$
$= -(10^5 Pa) \times (9 \times 10^{-3}) m^3$
$= -9 \times 10^2 N \cdot m\ \ [\because 1Pa = 1N/m^2]$
$= -900J = -0.9kJ$

Question139

Among the following, the set of parameters that represents path functions, is:
(A) q + w
(B) q
(C) w
(D) H − T S
[April 9, 2019 (I)]

Options:

A. (B) and (C)

B. (B), (C) and (D)

C. (A) and (D)

D. (A), (B) and (C)

Answer: A

Solution:

Solution:
We know that heat and work are not state functions but $q + w = \Delta U$ is a state function. $H - TS$(i.e. G) is also a state function.

Question140

During compression of a spring the work done is 10kJ and 2kJ escaped to the surroundings as heat. The change in internal energy, "U (in kJ) is:
[April 9, 2019 (II)]

Options:

A. -12

B. -8

C. 8

D. 12

Answer: C

Solution:

Solution:
$w = 10kJ$
$q = -2kJ$
$\Delta U = q + w = -2 + 10 = 8kJ$

Question141

Which one of the following equations does not correctly represent the first law of thermodynamics for the given processes involving an ideal gas ? (Assume non-expansion work is zero)
[April 8, 2019 (I)]

Options:

A. Cyclic process : $q = -w$

B. Adiabatic process : $\Delta U = -w$

C. Isochoric process: $\Delta U = q$

D. Isothermal process: $q = -w$

Answer: B

Solution:

Solution:
From first law of thermodynamics, $\Delta U = q + w$
For adiabatic process, $q = 0$
$\therefore \Delta U = w$
For isothermal process, $\Delta U = 0 \Rightarrow q = -w$
For cyclic process, $\Delta U = 0 \Rightarrow q = -w$
For isochoric process, $w = 0 \Rightarrow \Delta U = q$

Question142

5 moles of an ideal gas at 100K are allowed to undergo reversible compression till its temperature becomes 200 K. If $C_V = 28J\,K^{-1}mol^{-1}$, calculate ΔU and ΔpV for this process. ($R = 8.0J\,K^{-1}mol^{-1}$)
[April 8, 2019 (II)]

Options:

A. $\Delta U = 14kJ$; $\Delta(pV) = 18kJ$

B. $\Delta U = 14kJ$; $\Delta(pV) = 0.8kJ$

C. $\Delta U = 14kJ$; $\Delta(pV) = 4kJ$

D. $\Delta U = 14kJ$; $\Delta(pV) = 8.0kJ$

Answer: C

Solution:

Solution:
$\Delta U = nC_v \Delta T = 5 \times 28 \times 100 = 14kJ$
$\Delta(PV) = nR(T_2 - T_1) = 5 \times 8 \times 100 = 4kJ$

Question143

Enthalpy of sublimation of iodine is 24 cal g^{-1} at 200°C. If specific heat of I_2(s) and I_2 (vap) are 0.055 and 0.031 cal $g^{1}K^{-1}$ respectively, then enthalpy of sublimation of iodine at 250°C in cal g^{-1} is:
[April 12, 2019(I)]

Options:

A. 2.85

B. 5.7

C. 22.8

D. 11.4

Answer: C

Solution:

Solution:
$I_2(s) \longrightarrow I_2(g)$
Heat of reaction depend upon temperature i.e., it varies with temperature, as given by Kirchoff's equation,
$\Delta H_{T_2} = \Delta H_{T_1} + \int_{T_1}^{T_2} \Delta C_p dT$
where $\Delta C_p = C_p$ of product $- C_p$ of reactant
$\therefore \quad \Delta C_p = 0.031 - 0.055 = -0.024 cal/g$
Now, $\Delta H_{T_2} - \Delta H_{T_1} = \Delta C_p(T_2 - T_1)$
$\Delta H_{(250)} - \Delta H_{(200)} = -0.024(523 - 473)$
$\Delta H_{(250)} = 24 - 50 \times 0.024 = 22.8 cal/g$

Question144

The difference between ΔH and ΔU ($\Delta H - \Delta U$), when the combustion of one mole of heptane (I) is carried out at a temperature T, is equal to:
[April 10, 2019 (II)]

Options:

A. $-4RT$

B. $-3RT$

C. $4RT$

D. $3RT$

Answer: A

Solution:

Solution:

$$C_7H_{16}(l) + 11O_2(g) \xrightarrow{\Delta} 7CO_2(g) + 8H_2O(I)$$

$$\Delta H - \Delta U = \Delta n_g RT$$

$\Delta n_g =$ no. of moles of product in gaseous state - no. of moles of reactant in gaseous state.

$$\because \Delta n_g = -4$$

$$\therefore \Delta H - \Delta U = -4RT$$

Question145

For silver, $C_p(J\,K^{-1}mol^{-1}) = 23 + 0.01T$. If the temperature (T) of 3 moles of silver is raised from 300K to 1000K at 1 atm pressure, the value of ΔH will be close to:
[April 8, 2019 (I)]

Answer: 62

Solution:

Solution:
Given: $n = 3$

$$T_1 = 300;\ T_2 = 1000$$

$$C_p = 23 + 0.01T$$

The relation between ΔH and $mathcal\,C_p$ is

$$\Delta H = \int_{T_1}^{T_2} nC_p dT \quad \ldots (i)$$

After putting all variable values in eq. (i), we get

$$\Delta H = n\int_{300}^{1000} (23 + 0.01T)dT$$

$$= 3\left[23T + \frac{0.01T^2}{2}\right]_{300}^{1000}$$

$$= 3\left[23\left(1000 - 300\right[+ \frac{0.01}{2}(1000^2 - 300^2)\right]$$

$$= 3[16100 + 4550]$$

$$= 3 \times 20650 = 61950J$$

$= 61.95kJ$
$\approx 62kJ$

Question146

A process will be spontaneous at all temperatures if: [April 10, 2019 (I)]

Options:

A. $\Delta H < 0$ and $\Delta S < 0$

B. $\Delta H > 0$ and $\Delta S < 0$

C. $\Delta H < 0$ and $\Delta S > 0$

D. $\Delta H > 0$ and $\Delta S > 0$

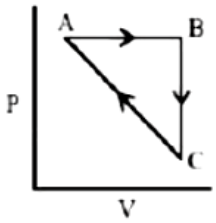
Answer: C

Solution:

Solution:
A reaction is spontaneous if ΔG_{sys} is negative.
$\Delta G_{sys} = \Delta H_{sys} - T\ \Delta S_{sys}$
A reaction will be spontaneous at all temperatures if
ΔH_{sys} is negative and ΔS_{sys} is positive.

Question147

An ideal gas undergoes a cyclic process as shown in Figure.



$\Delta U_{BC} = -5kJ\ mol^{-1}$, $q_{48} = 2kJ\ mol^{-1}$

$W_{AB} = -5kJ\ mol^{-1}$, $W_{CA} = 3kJ\ mol^{-1}$

Heat absorbed by the system during process CA is: [Online April 15,2018 (I)]

Options:

A. $-5kJ\ mol^{-1}$

B. $+5kJ\ mol^{-1}$

C. $18kJ\ mol^{-1}$

D. $-18kJ\ mol^{-1}$

Answer: B

Solution:

Solution:
$\Delta U_{AB} = q_{AB} + W_{AB} = 2 + (-5) = -3kJ/mol$
$\Delta U_{BC} = -5kJ/mol$
For cyclic process, $\Delta U = 0$
$\Delta U_{AB} + \Delta U_{BC} + \Delta U_{CA} = 0$
$\Delta U_{CA} = -\Delta U_{AB} - \Delta U_{BC}$
$\Delta U_{CA} = -(-3) - (-5) = 8kJ/mol$
$\Delta U_{CA} = q_{CA} + W_{CA}$
$8 = q_{CA} + 3$
$q_{CA} = +5kJ/mol$
Heat absorbed has positive sign.

Question148

The combustion of benzene (I) gives $CO_2(g)$ and $H_2O(l)$. Given that heat of combustion of benzene at constant volume is $-3263.9kJ\ mol^{-1}$ at $25^{\circ}C$; heat of combustion (in $kJ\ mol^{-1}$) of benzene at constant pressure will be:
($R = 8.314J\ K^{-1}mol^{-1}$)
[2018]

Options:

A. 4152.6

B. −452.46

C. 3260

D. −3267.6

Answer: D

Solution:

Solution:
$C_6H_6(l) + \frac{15}{2}O_2(g) \rightarrow {}_6CO_2(g) + 3H_2O(l)$
$\Delta n_g = 6 - \frac{15}{2} = -\frac{3}{2}$ $\Delta H = \Delta U + \Delta n_g RT$
$= -3263.9 + \left(-\frac{3}{2}\right) \times 8.314 \times 10^{-3} \times 298$
$= -3263.9 + (-3.71) = -3267.6kJ\ mol^{-1}$

Question149

For which of the following reactions, ΔH is equal to ΔU ?
[Online April 15, 2018 (I)]

Options:

A. $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$

B. $2HI(g) \rightarrow H_2(g) + I_2(g)$

C. $2SO_2(g) + O_2(g) \rightarrow 2SO_3(g)$

D. $2NO_2(g) \rightarrow N_2O_4(g)$

Answer: B

Solution:

Solution:
$\Delta H = \Delta U + \Delta n_g RT$
$2HI(g) \rightarrow H_2(g) + I_2(g); \Delta n_g = (1+1) - 2 = 0$
$\therefore \Delta H = \Delta U + 0$

Question150

For which of the following processes, ΔS is negative? [Online April 16, 2018]

Options:

A. C (diamond) $\rightarrow$C (graphite)

B. $N_2(g, 1atm) \rightarrow N_2(g, 5atm)$

C. $N_2(g, 273K) \rightarrow N_2(g, 300K)$

D. $H_2(g) \rightarrow 2H(g)$

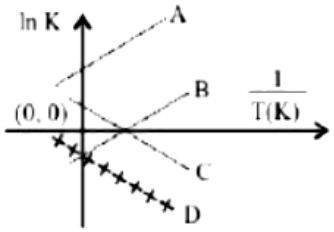
Answer: B

Solution:

Solution:
When diamond is converted into graphite (it is heated to $1500^{\circ}C$) entropy is increased, $\Delta S > 0$
(b) When pressure increases then molecules of gas will come closer and intermolecular distance decreases, so cntropy will also decrease, $\Delta S < 0$
(c) When we increase the temperature of a gas then randomness is increased as the kinetic energy gained by molecules. So, $\Delta S > 0$
(d) H_2 molecule is converted into atoms, the no. of particles increases. Thus entropy will increase, $\Delta S > 0$

Question151

Which of the following lines correctly show the temperature dependence of equilibrium constant, K , for an exothermic reaction?



[2018]

Options:

A. A and B

B. B and C

C. C and D

D. A and D

Answer: A

Solution:

Solution:
From thermodynamics relation.
$\Delta G^{\circ} = -RT\,\ell nK \;\; \Delta H^{\circ} - T\;\Delta S^{\circ} = RT\,\ell nK$
$-\frac{\Delta H^{\circ}}{RT} + \frac{T\;\Delta S^{\circ}}{RT} = \ln K$
or $\ln K = -\frac{\Delta H^{\circ}}{RT} + \frac{\Delta S^{\circ}}{R}$
For exothermic reaction, $\Delta H^{\circ} = -ve$
slope $= -\frac{\Delta H^{\circ}}{R} = +ve$
So from graph, lines should be A& B.

Question152

At 320K , a gas A_2 is 20% dissociated to A(g). The standard free energy change at 320K and 1atm in J mol $^{-1}$ is approximately:
(R = 8.314J K $^{-1}$mol $^{-1}$; ln 2 = 0.693; ln 3 = 1.098)
[Online April 16, 2018]

Options:

A. 1844

B. 2068

C. 4281

D. 4763

Answer: D

Solution:

Solution:
$\Delta G_{rxn}^{o} = \Delta_f G^{\circ}(vapour) - \Delta_f\, G^{\circ}(\text{ liquid })$
$\Delta G_{rxn}^{o} = 103 - 100.7 = 2.3kcal\,/mol\; = 2300cal\,/mol$
$\Delta G_{rxn}^{0} = -RT\;\ln K_p (K_p = P_{vap})$
$2300cal\,/mol\; = -2cal\,/mol\,/K\; \times 500K\; \times \ln K_p$
$\ln K_p = -2.3\;\log_{10} K_p = -1$
$K_p =$ Antilog $-1 = 0.1atm$
∴ Vapour pressure of liquid ' S ' at $500K$ is approximately equal to $0.1atm$.

Question153

$\Delta_f G^{\circ}$ at 500K for substance 'S' in liquid state and gaseous state are +100.7kcal mol $^{-1}$ and +103kcal mol $^{-1}$, respectively.
Vapour pressure of liquid 'S' at 500K is approximately equal to: (R = 2cal K $^{-1}$mol $^{-1}$).
[Online April 15, 2018(II)]

Options:

A. 100atm

B. 1atm

C. 10atm

D. 0.1atm

Answer: C

Solution:

Solution:
As we know that, at equilibrium,
$\Delta G^{\circ} = -2.303 \ RT \ \log K_{p} \ldots\ldots (1)$
$S_{(l)} \rightleftharpoons S_{(g)}$
From the above reaction,
$\Delta G^{\circ} = \Delta G_{f_{(product)}} - \Delta G_{(reactant)}^{\circ}$
$\Rightarrow \Delta G^{\circ} = 103 - 100.7 = 2.3 \, kcal/mol = 2.3 \times 10^{3} \, kcal/mol$
Given:-
$T = 500K$
$R = 2 \, cal/mol - K$
From eq^{n} (1), we have
$2.3 \times 10^{3} = -2.303 \times 2 \times 500 \times \log K_{p}$
$\Rightarrow \log K_{p} = -1$
$\Rightarrow K_{p} = 10^{-1} \, atm$
Now, from the above reaction,
$K_{p} = \frac{1}{P_{S_1}}$
$\Rightarrow P_{S_1} = \frac{1}{10^{-1}} = 10 \, atm$
Hence the vapou pressure of liquid S is approximately equal to $10 \, atm$.

Question154

Given
(i) $2Fe_2O_3(s) \rightarrow 4Fe(s) + 3O_2(g)$;
$\Delta_r G^{\circ} = +1487.0 kJ \, mol^{-1}$
(ii) $2CO(g) + O_2(g) \rightarrow 2CO_2(g)$;
$\Delta_r G^{\circ} = -514.4 kJ \, mol^{-1}$
Free energy change, $\Delta_r G^{\circ}$ for the reaction
$2Fe_2O_3(s) + 6CO(g) \rightarrow 4Fe(s) + 6CO_2(g)$ will be:
[Online April 15, 2018 (II)]

Options:

A. $-112.4 kJ \, mol^{-1}$

B. $-56.2 kJ \, mol^{-1}$

C. $-208.0 kJ \, mol^{-1}$

D. $-168.2 kJ \, mol^{-1}$

Answer: B

Solution:

Solution:
(i) $2Fe_2O_3(s) \rightarrow 4Fe(s) + 3O_2(g)$; $\Delta_r G^0 = +1487.0\,kJ\,mol^{-1}$
(ii) $2CO(g) + O_2(g) \rightarrow 2CO_2(g)$; $\Delta_r G^0 = -514.4\,kJ\,mol^{-1}$
Multiply above reaction with 3
(iii) $6CO(g) + 3O_2(g) \rightarrow 6CO_2(g)$; $\Delta_r G^0 = 3 \times -514.4 = -1543.2\,kJ\,mol^{-1}$
When we add reaction (i) and reaction (iii), we get reaction (iv)
(iv) $2Fe_2O_3(s) + 6CO(g) \rightarrow 4Fe(s) + 6CO_2(g)$
Free energy change, $\Delta_r G^0$ for the reaction will be, $1487.0 - 1543.2 = -56.2\,kJ\,mol^{-1}$

Question155

ΔU is equal to
[2017]

Options:

A. Isochoric work

B. Isobaric work

C. Adiabatic work

D. Isothermal work

Answer: C

Solution:

Solution:
From 1st law of thermodynamics
$\Delta U = q + w$
For adiabatic process :$q = 0$
$\therefore \Delta U = w$

Question156

A gas undergoes change from state A to state B. In this process, the heat absorbed and work done by the gas is 5 J and 8J , respectively. Now gas is brought back to A by another process during which 3J of heat is evolved. In this reverse process of B to A:
[Online April 9,2017]

Options:

A. 10J of the work will be done by the gas.

B. 6J of the work will be done by the gas.

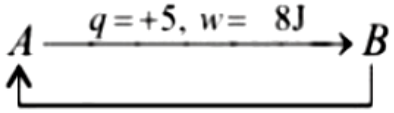
C. 10J of the work will be done by the surrounding on gas.

D. 6J of the work will be done by the surrounding on gas.

Answer: D

Solution:

Solution:



$\Delta U_{AB} = q + w = +5 + (-8) = -3$
$q = -3, \Delta U_{BA} = +3$
$\Delta U_{BA} = q + w$
$\Rightarrow 3 = -3 + w \Rightarrow w = +6$ (work done on the system).

Question157

Given

$C_{(\text{graphite})} + O_2(g) \rightarrow CO_2(g); \Delta_r H^\circ = -393.5 kJ\ mol^{-1}$

$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(l); \Delta_, H^\circ = -285.8 kJ\ mol^{-1}$

$CO_2(g) + 2H_2O(l) \rightarrow CH_4(g) + 2O_2(g);$

$\Delta_r H^\circ = +890.3 kJ\ mol^{-1}$

Based on the above thermochemical equations, the value of $\Delta_r H^\circ$ at 298K for the reaction $C_{(\text{gaphite})} + 2H_2(g) \rightarrow CH_4(g)$ will be
[2017]

Options:

A. $+74.8 kJ\ mol^{-1}$

B. $+144.0 kJ\ mol^{-1}$

C. $-74.8 kJ\ mol^{-1}$

D. $-144.0 kJ\ mol^{-1}$

Answer: C

Solution:

Solution:
$C_{(\text{graphite})} + O_2(g) \rightarrow CO_2(g);$
$\Delta H^\circ = -393.5 kJ/mol^{-1} \ldots$ (i)
$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(I)$
$\Delta_r H^\circ = -285.8 kJ/mol^{-1} \ldots$ (ii)
$CO_2(g) + 2H_2O(I) \rightarrow CH_4(g) + 2O_2(g);$
$\Delta H_r^\circ = +890.3 kJ/mol^{-1} \ldots$ (iii)
$C_{(\text{graphite})} + 2H_2(g) \rightarrow CH_4(g)$
$\Delta H = ? \ldots$ (iv)
[Eq. (i) + Eq. (iii)] + [2 × Eq. (ii)] = Eq(iv)
$\therefore [\Delta H_1 + \Delta H_3] + [2 \times \Delta H_2] = \Delta H_4$
$[(-393.5) + (890.3)] + [2(-285.8)]$
$= -74.8 kJ/mol$

Question158

For a reaction, $A(g) \rightarrow A(l)$; $\Delta H = -3RT$. The correct statement for the reaction is: [Online April 8,2017]

Options:

A. $\Delta H = \Delta U \neq O$

B. $\Delta H = \Delta U = O$

C. $|\Delta H| < |\Delta U|$

D. $|\Delta H| > |\Delta U|$

Answer: D

Solution:

Solution:
$A(g) \longrightarrow A(I)$
$\Delta H = \Delta U + \Delta n_g RT$
Given, $\Delta H = -3RT$
Here
$\Delta n_g = n_p - n_r = 0 - 1 = -1$
$\Delta H = \Delta U + \Delta RT$
$\Rightarrow -3RT = \Delta U - RT$
$\Rightarrow -3RT + RT = \Delta U$
$\Rightarrow -2RT = \Delta U$
$|\Delta H| > |\Delta U|$

Question159

The enthalpy change on freezing of 1mol of water at $5^{\circ}C$ to ice at $-5^{\circ}C$ is : (Given $\Delta_{fis} H = 6kJ\ mol^{-1}$ at $0^{\circ}C$, $C_p(H_2O, l) = 75.3J\ mol^{-1}K^{-1}$ $C_p(H_2O, s) = 36.8J\ mol^{-1}K^{-1}$) [Online April 8, 2017]

Options:

A. $5.44kJ\ mol^{-1}$

B. $5.81kJ\ mol^{-1}$

C. $6.56kJ\ mol^{-1}$

D. $6.00kJ\ mol^{-1}$

Answer: B

Solution:

Solution:
In order to calculate the enthalpy change for H_2O at $5^{\circ}C$ to ice at $-5^{\circ}C$, we need to calculate the enthalpy change of all the transformation involved in the process.
(a) Energy change of $1mol, H_2O(l)$, at $5^{\circ}C$
$\rightarrow 1mol, H_2O(I), 0^{\circ}C$
(b) Energy change of $1mol, H_2O(l)$, at $0^{\circ}C$
$\rightarrow 1mol, H_2O(s)(ice), 0^{\circ}C$

(c) Energy change of 1mol, ice (s), at 0°C
$\rightarrow 1\text{mol}$, ice (s), -5°C
Total ΔH
$= C_p[H_2O(l)]\,\Delta T + \Delta H_{\text{freezing}} + C_p[H_2O(s)]\,\Delta T$
$= (75.3\text{J mol}^{-1}\text{K}^{-1})(0-5)\text{K} + (-6\times 10^3\text{J mol}^{-1})$
$+(36.8\text{J mol}^{-1}\text{K}^{-1})(-5-0)\text{K}$
$\Delta H = -6.56\text{kJ mol}^{-1}$ (cxothermic process)
So, $\Delta H = 6.56\text{kJ mol}^{-1}$.

Question160

An ideal gas undergoes isothermal expansion at constant pressure. During the process: [Online April9, 2017]

Options:

A. enthalpy increases but entropy decreases.

B. enthalpy remains constant but entropy increases.

C. enthalpy decreases but entropy increases.

D. Both enthalpy and entropy remain constant.

Answer: B

Solution:

Solution:
$\Delta H = nC_p\,\Delta T = 0$ means ΔH constant.

$\Delta S = nR\ln\left(\frac{V_f}{V_i}\right) \geq 0$ ΔS increases

---

Question161

If 100 mole of H_2O_2 decomposes at l bar and 300K , the work done (kJ) by one mole of $O_2(g)$ as it expands against 1 bar pressure is : $2H_2O_2(l) \rightleftharpoons H_2O(l) + O_2(g)$ $(R = 83\text{J K}^{-1}\text{mol}^{-1})$ [Online April 10, 2016]

Options:

A. 124.50

B. 249.00

C. 498.00

D. 62.25

Answer: A

Solution:

Solution:

$2H_2O_2(1) \rightleftharpoons 2H_2O(1) + O_2(g)$
$w = -P_{ext}(\Delta V) = -n_{O_2}RT$
$\because 100 mol\ eH_2O_2$ on decomposition give $50 mol\ eO_2$.
$\therefore W = -(50)(8.3)(300) = -124500J = -124.5kJ$.

Question162

The heats of combustion of carbon and carbon monoxide are −393.5 and −283.5kJ mol^{-1}, respectively. The heat of formation (in kJ) of carbon monoxide per mole is : [2016]

Options:

A. −676.5

B. −110.5

C. 110.5

D. 676.5

Answer: B

Solution:

Solution:
Given
$C(s) + O_2(g) \rightarrow CO_2(g); \Delta H = -393.5kJ\ mol^{-1}$ . . . (i)
$CO(g) + \frac{1}{2}O_2(g) \rightarrow CO_2(g); \Delta H = -283.5kJ\ mol^{-1}$ . . . (ii)
$\therefore$ Heat of formation of $CO = eq^n(i) - eq^n(ii)$
$= -393.5 - (-283.5)$
$= -110kJ$

Question163

A reaction at 1 bar is non-spontaneous at low temperature but becomes spontaneous at high temperature. Identify the correct statement about the reaction among the following : [Online April 9, 2016]

Options:

A. ΔH is negative while ΔS is positive

B. Both ΔH and ΔS are negative

C. ΔH is positive while ΔS is negative

D. Both ΔH and ΔS are positive.

Answer: D

Solution:

Solution:
$\Delta G = \Delta H - T\Delta S$
At low temperature,
$T\Delta S < \Delta H$
$\Delta G > 0$ and the reaction is non spontaneous.

At high temperature,
$T\Delta S > \Delta H$
$\Delta G < 0$ and the reaction is spontaneous.

Question164

The heat of atomization of methane and ethane are 360kJ/mol and 620kJ /mol , respectively. The longest wavelength of light capable of breaking the C − C bond is (Avogadro number $= 6.02 \times 10^{23}$, h = 6.62×10^{-34}J s):
[Online April 10,2015]

Options:

A. 2.48×10^{4}nm

B. 1.49×10^{3}nm

C. 2.48×10^{3}nm

D. 1.49×10^{4}nm

Answer: B

Solution:

Solution:
In CH_4, $4 \times BE_{(C-H)} = 360kJ/mol$
$\therefore BE_{(C-H)} = 90kJ/mol$
In C_2H_6, $BE_{(C-C)} + 6 \times BE_{(C-H)} = 620kJ/mol$
$\therefore BE_{(C-C)} = 80kJ/mol$
$\therefore BE_{(C-C)} = \frac{80 \times 10^3}{6.023 \times 10^{23} J/mol}$
Now, $E = \frac{hc}{\lambda}$
$\therefore \lambda = \frac{6.626 \times 10^{-34} \times 3 \times 10^8 \times 6.023 \times 10^{23}}{80 \times 10^3}$
$\lambda = 1.49 \times 10^{-6}m \ (\because 1nm = 10^{-9}m)$
$\therefore \lambda = 1.49 \times 10^{3}nm$

Question165

For complete combustion of ethanol,
$C_2H_5OH(l) + 3O_2(g) \longrightarrow 2CO_2(g) + 3H_2O(l)$, the amount of heat produced as measured in bomb calorimeter, is 1364.47kJ mol^{-1} at 25°C. Assuming ideality the enthalpy of combustion, $\Delta_c H$, for the reaction will be:
(R = 8.314kJ mol^{-1})
[2014]

Options:

A. -1366.95kJ mol^{-1}

B. -1361.95kJ mol^{-1}

C. -1460.95kJ mol^{-1}

D. $-1350.50 kJ\ mol^{-1}$

Answer: A

Solution:

Solution:
$C_2H_5OH(l) + 3O_2(g) \longrightarrow 2CO_2(g) + 3H_2O(l)$
Bomb calorimeter gives ΔU of the reaction
Given, $\Delta U = -1364.47 kJ\ mol^{-1}$
$\Delta n_g = -1$
$\Delta H = \Delta U + \Delta n_g RT$
$= -1364.47 - \frac{I \times 8.314 \times 298}{1000}$
$= -1366.95 kJ\ mol^{-1}$

Question166

The standard enthalpy of formation of NH_3 is $-46.0 kJ/mol$. If the enthalpy of formation of H_2 from its atoms is $436 kJ/mol$ and that of N_2 is $-712 kJ/mol$, the average bond enthalpy of $N-H$ bond in NH_3 is:
[Online April 9, 2014]

Options:

A. $-1102 kJ/mol$

B. $-964 kJ/mol$

C. $+352 kJ/mol$

D. $+1056 kJ/mol$

Answer: C

Solution:

Solution:
Given
$\frac{1}{2}N_2 + \frac{3}{2}H_2 \rightleftharpoons NH_3$
$\Delta_f H = -46.0 kJ/mol$
$H + H \rightleftharpoons H_2; \Delta_f H = -436 kJ/mol$
$N + N \rightleftharpoons N_2; \Delta_f H = -712 kJ/mol$
$\Delta_f H(NH_3) = \frac{1}{2}\Delta H_{N-N} + \frac{3}{2}\Delta H_{H-H} - 3\Delta H_{N-H}$
$-46 = \frac{1}{2}(712) + \frac{3}{2}(436) - 3\Delta H_{N-H}$
$-3\Delta H_{N-H} = -1056$
$\Delta H_{N-H} = 352 kJ/mol$

Question167

The standard enthalpy of formation ($\Delta_f H_{298}^{\circ}$) for methane, CH_4 is $-74.9 kJ\ mol^{-1}$. In order to calculate the average energy given out in the formation of a $C-H$ bond from this it is necessary to know which one of the following?
[Online April 12, 2014]

Options:

A. The dissociation energy of the hydrogen molecule, H_2.

B. The first four ionisation energies of carbon.

C. The dissociation energy of H_2 and enthalpy of sublimation of carbon (graphite).

D. The first four ionisation energies of carbon and electron affinity of hydrogen.

Answer: C

Solution:

Solution:
To calculate average enthalpy of $C-H$ bond in methane following informations are needed
(i) dissociation energy of H_2 i.e.

$\frac{1}{2}H_2(g) \longrightarrow H(g); \Delta H = x(\text{ suppose })$

(ii) Sublimation energy of C(graphite) to $C(g)$

$C_{(graphite)} \longrightarrow C(g); \Delta H = y(\text{ Suppose })$

Given

$C_{(graphite)} + 2H_2(g) \longrightarrow CH_4(g)$

$\Delta H\, g = -74.9 kJ\, mol^{-1}$

Question168

The molar heat capacity (C_p) of CD_2O is 10 cals at 1000K . The change in entropy associated with cooling of 32g of CD_2O vapour from 1000K to 100K at constant pressure will be: (D = deuterium, atomic mass = 2u) [Online April 11, 2014]

Options:

A. $23.03\ cal\ deg^{-1}$

B. $-23.03\ cal\ deg^{-1}$

C. $2.303\ cal\ deg^{-1}$

D. $-2.303\ cal\ deg^{-1}$

Answer: B

Solution:

Solution:
(b) Given, $C_p = 10cal$ at $1000K$
$T_1 = 1000K$. $T_2 = 100K$
$m = 32g$
$\Delta S = ?$
at constant pressure

$$\Delta S = C_p \ln \frac{T_2}{T_1}$$
$$= 2.303 \times C_p \log \frac{T_2}{T_1}$$
$$= 2.303 \times 10 \log \frac{100}{1000}$$
$$= -23.03 cal\, deg^{-1}$$

Question169

The (S°) of the following substances are:
$CH_4(g) 186.2 J\,K^{-1}mol^{-1}$ $O_2(g) 205.2 J\,K^{-1}mol^{-1}$
$CO_2(g) 213.6 J\,K^{-1}mol^{-1}$
$H_2O(l) 69.9 J\,K^{-1}mol^{-1}$
The entropy change (ΔS°) for the reaction
$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$ is:
[Online April 12, 2014]

Options:

A. $-312.5 J\,K^{-1}mol^{-1}$

B. $-242.8 J\,K^{-1}mol^{-1}$

C. $-108.1 J\,K^{-1}mol^{-1}$

D. $-37.6 J\,K^{-1}mol^{-1}$

Answer: B

Solution:

Solution:
$\Delta S^\circ = S^\circ CO_2 + 2 \times S^\circ H_2O - \left(S^\circ CH_4 + 2 \times S_{O_2}{}^\circ\right)$
$= (213.6 + 2 \times 69.9) - (186.2 + 2 \times 205.2)$
$= -242.8 J\,K^{-1}mol^{-1}$

Question170

A piston filled with 0.04 mol of an ideal gas expands reversibly from 50.0mL to 375mL at a constant temperature of 37.0°C. As it does so, it absorbs 208J of heat. The values of q and w for the process will be:
(R = 8.314J / mol K)(ln 7.5 = 2.01)
[2013]

Options:

A. $q = +208 J$, $w = -208 J$

B. $q = -208 J$, $w = -208 J$

C. $q = -208 J$, $w = +208 J$

D. $q = +208 J$, $w = +208 J$

Answer: A

Solution:

Solution:
Process is isothermal reversible expansion, hence
$\Delta U = 0$, therefore $q = -w$
Since $q = +208 J$, $w = -208 J$

Question171

Which of the following statements\/relationships is not correct in thermodynamic changes? [Online April 23,2013]

Options:

A. $\Delta U = 0$ (isothermal reversible expansion of a gas)

B. $w = -nRT \ln \frac{V_2}{V_1}$ (isothermal reversible expansion of an ideal gas)

C. $w = nRT \ln \frac{V_2}{V_1}$ (isothermal reversible expansion of an ideal gas)

D. For a system of constant volume heat involved directly changes to internal energy.

Answer: C

Solution:

Solution:
For isothermal reversible expansion.
$w = -nRT \ln \frac{V_2}{V_1}$

Question172

Given:
(I) $H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(l)$
$\Delta H_{298K}^{\circ} = -285.9 kJ\ mol^{-1}$
(II) $H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(g);$
$\Delta H_{298K}^{\circ} = -241.8 kJ\ mol^{-1}$
The molar enthalpy of vapourisation of water will be: [Online April 9, 2013]

Options:

A. $241.8 kJ\ mol^{-1}$

B. $22.0 kJ\ mol^{-1}$

C. $44.1 kJ\ mol^{-1}$

D. $527.7 kJ\ mol^{-1}$

Answer: C

Solution:

Solution:
Given
$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(l)$
$\Delta H^{\circ} = -285.9 kJ\ mol^{-1}$ . . . (i)

$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(g)$

$\Delta H^\circ = -241.8 kJ\ mol^{-1}$ . . . (ii)

We have to calculate

$H_2O(l) \rightarrow H_2O(g); \Delta H^\circ = ?$

On substracting eqn. (ii) from eqn. (i) we get

$H_2O(l) \rightarrow H_2O(g); \Delta H^\circ = -241.8 - (-285.9)$

$= 44.1 kJ\ mol^{-1}$

Question173

Reaction	Energy Change (in kJ)
$Li(s) \rightarrow Li(g)$	161
$Li(g) \rightarrow Li^+(g)$	520
$\frac{1}{2}F_2(g) \rightarrow F(g)$	77
$F(g) + e^- \rightarrow F^-(g)$	(Electron gain enthalpy)
$Li^+(g) + F(g) \rightarrow LiF(s)$	-1047
$Li(s) + \frac{1}{2}F_2(g) \rightarrow LiF(s)$	-617

Based on data provided, the value of electron gain enthalpy of fluorine would be : [Online April 22,2013]

Options:

A. $-300 kJ\ mol^{-1}$

B. $-350 kJ\ mol^{-1}$

C. $-328 kJ\ mol^{-1}$

D. $-228 kJ\ mol^{-1}$

Answer: C

Solution:

Solution:
Applying Hess's Law
$\Delta_f H^\circ = \Delta_{sub}\ H + \frac{1}{2}\Delta_{d\,iss} H + I.E. + E.A + \Delta_{lattice}\ H$
$-617 = 161 + 520 + 77 + E.A. + (-1047)$
$E.A. = -617 + 289 = -328 kJ\ mol^{-1}$
$\therefore$ electron affinity of fluorine $= -328 kJ\ mol^{-1}$

Question174

In which of the following exothermic reactions, the heat liberated per mole is the highest? [Online April 25, 2013]

Options:

A. $CaO + H_2O \rightarrow Ca(OH)_2$

B. $SrO + H_2O \rightarrow Sr(OH)_2$

C. $BaO + H_2O \rightarrow Ba(OH)_2$

D. $MgO + H_2O \rightarrow Mg(OH)_2$

Answer: A

Solution:

Solution:

Question175

Given that:
(i) $\Delta_f H^\circ$ of N_2O is $82 kJ\ mol^{-1}$
(ii) Bond energies of $N = N$, $N = N$, $O = O$ and $N = O$ are 946, 418, 498 and $607 kJ\ mol^{-1}$ respectively,
The resonance energy of N_2O is:
[Online April 25, 2013]

Options:

A. $-88 kJ$

B. $-66 kJ$

C. $-62 kJ$

D. $-44 kJ$

Answer: A

Solution:

Solution:

$N_2(g) + \frac{1}{2}O_2 \rightarrow N_2O(g)$

$N \equiv N(g) + \frac{1}{2}(O = O) \rightarrow \overset{+}{N} = N = O(g)$

$\Delta_f H^o = [$
Energy required for breaking of bonds]
$-[$ Energy released for forming of bonds]

$= \left(\Delta H_{N \equiv N} + \frac{1}{2}\Delta H_{O=O} - (\Delta H_{N=N} + \Delta H_{N=O}\right)$

$= \left(946 + \frac{1}{2} \times 498\right) - (418 + 607) = 170 kJ\ mol^{-1}$

Resonance energy = observed $\Delta_f H^\circ$ − calculated $\Delta_f H^\circ$

$82 - 170 = -88 kJ\ mol^{-1}$

Question176

The difference between the reaction enthalpy change (ΔH) and reaction internal energy change ($\Delta_f U$) for the reaction:

$2C_6H_6(I) + 15O_2(g) \longrightarrow 12CO_2(g) + 6H_2O(I)$

at 300K is ($R = 8.314 J\ mol^{-1}K^{-1}$)
[Online May 12, 2012]

Options:

A. $0 J\ mol^{-1}$

B. $2490 J\ mol^{-1}$

C. $-2490 J\ mol^{-1}$

D. $-7482 J\ mol^{-1}$

Answer: D

Solution:

Solution:
$\Delta H = \Delta U + \Delta n_g RT$
For the reaction $\Delta n_g = 12 - 15 = -3$
$\Delta H - \Delta U = -3 \times 8.314 \times 300$
$= -7482.6 J\ mol^{-1}$

Question177

The enthalpy of neutralisation of NH_4OH with HCl is $-51.46 kJ\ mol^{-1}$ and the enthalpy of neutralisation of NaOH with HCl is $-55.90 kJ\ mol^{-1}$. The enthalpy of ionisation of NH_4OH is
[Online May 19, 2012]

Options:

A. $-107.36 kJ\ mol^{-1}$

B. $-4.44 kJ\ mol^{-1}$

C. $+107.36 kJ\ mol^{-1}$

D. $+4.44 kJ\ mol^{-1}$

Answer: D

Solution:

Solution:

$\underset{\text{Strong acid (Complete ionisation)}}{HCl} \longrightarrow H^+ + Cl^-$ ...(i)

$\underset{\text{Weak base}}{NH_4OH} \rightleftharpoons NH_4^+ + OH^- \ldots$ (ii) $\Delta H = x kJ\ mol^{-1}$

$H^+ + OH^- \longrightarrow H_2O \ldots$ (iii)

$\Delta H = -55.90 kJ\ mol^{-1}$
(from neutralisation of strong acid and strong base)
From equation (i), (ii) and (iii)

$NH_4OH + HCl \longrightarrow NH_4^+ + Cl^- + H_2O$

$\Delta H = -51.46 kJ\ mol^{-1}$

$\therefore\ x + (-55.90) = -51.46$

$x = -51.46 + 55.90$

$= 4.44 kJ\ mol^{-1}$

$\therefore$ Enthalpy of ionisation of

$NH_4OH = 4.44 kJ\ mol^{-1}$

Question178

The entropy of a sample of a certain substance increases by $0.836 J\ K^{-1}$ on adding reversibly 0.3344J of heat at constant temperature. The temperature of the sample is:
[Online May 7, 2012]

Options:

A. 2.5K

B. 0.3K

C. 0.016K

D. 0.4K

Answer: D

Solution:

Solution:

$\Delta S = \frac{\Delta H}{T}; T = \frac{\Delta H}{\Delta S} = \frac{0.3344}{0.836} = 0.4K$

Question179

One mole of an ideal gas is expanded isothermally and reversibly to half of its initial pressure. ΔS for the process in $J\ K^{-1} mol^{-1}$ is [ln 2 = 0.693 and R = 8.314, J / (mol K)]
[Online May 26, 2012]

Options:

A. 6.76

B. 5.76

C. 10.76

D. 8.03

Answer: B

Solution:

Solution:
For isothermal process
$(\Delta T = 0)$

$$\Delta S = R \ln \frac{P_1}{P_2} = 8.314 \ln 2$$
$$= 8.314 \times 0.693 = 5.76$$

Question180

The incorrect expression among the following is: [2012]

Options:

A. $\frac{\Delta G_{system}}{\Delta S_{total}} = -T$

B. In isothermal process, $w_{reversible} = -nRT \ln \frac{V_f}{V_i}$

C. $\ln K = \frac{\Delta H^\circ - T \Delta S^\circ}{RT}$

D. $K = e^{-\Delta G^\prime / RT}$

Answer: C

Solution:

Solution:
$\Delta G^\circ = \Delta H^\circ - T \Delta S^\circ$
$-RT \ln K = \Delta H^\circ - T \Delta S^\circ$
$\ln K = -\frac{\Delta H^\circ - T \Delta S^\circ}{RT}$

Question181

The value of enthalpy change (ΔH) for the reaction $C_2H_5OH(l) + 3O_2(g) \rightarrow 2CO_2(g) + 3H_2O(l)$ at $27^\circ C$ is $-1366.5 kJ\ mol^{-1}$. The value of internal energy change for the above reaction at this temperature will be : [2011 RS]

Options:

A. −1369.0kJ

B. −1364.0kJ

C. −1361.5kJ

D. −1371.5kJ

Answer: B

Solution:

Solution:

$C_2H_5OH(l) + 3O_2(g) \longrightarrow 2CO_2(g) + 3H_2O(l)$

$\Delta n_g = 2 - 3 = -1$

$\Delta U = \Delta H - \Delta n_g RT$

$= -1366.5 - [(-1) \times 8.314 \times 300]$

$= -1366.5 + \left[(1) \times \frac{8.314}{10^3} \times 300\right]$

$= -1366.5 + 0.8314 \times 3 = -1364kJ$

Question182

Consider the reaction:
$4NO_2(g) + O_2(g) \rightarrow 2N_2O_5(g)$ $\Delta H = -111kJ$ If $N_2O_5(s)$ is formed instead of $N_2O_5(g)$ in the above reaction, the $\Delta_r H$ value will be :
(given, ΔH of sublimation for N_2O_5 is 54kJ mol^{-1})
[2011 RS]

Options:

A. +54kJ

B. +219kJ

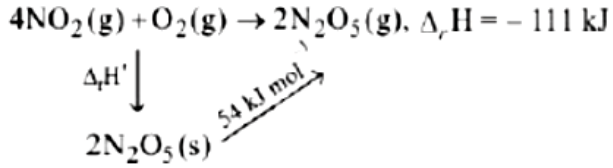
C. −219J

D. −165kJ

Answer: D

Solution:

Solution:



$\Delta_r H' = \Delta_r H - 54$

$= -111 - 54$

$= -165\ kJ$

Question183

The standard enthalpy of formation of NH_3 is −46.0kJ mol^{-1}. If the enthalpy of formation of H_2 from its atoms is −436kJ mol^{-1} and that of N_2 is −712kJ mol^{-1}, the average bond

enthalpy of $N-H$ bond in NH_3 is
[2010]

Options:

A. $-964 kJ\ mol^{-1}$

B. $+352 kJ\ mol^{-1}$

C. $+1056 kJ\ mol^{-1}$

D. $-1102 kJ\ mol^{-1}$

Answer: B

Solution:

Solution:
(b) $N_2 + 3H_2 \longrightarrow 2NH_3 \quad \Delta H = 2\times -46.0 kJ\ mol^{-1}$
Let x be the bond enthalpy of $N-H$ bond then [Note: Enthalpy of formation or bond formation enthalpy is given which is negative but the given reaction involves bond breaking hence values should be taken as positive.] $\Delta H = \Sigma$ Bond energies of reactants $-\Sigma$ Bond energies of products
$2\times -46 = 712 + 3\times(436) - 6x - 92 = 2020 - 6x$
$6x = 2020 + 92$
$6x = 2112$
$x = +352 kJ/mol$

Question184

For a particular reversible reaction at temperature T , ΔH and ΔS were found to be both +ve. If T_e is the temperature at equilibrium, the reaction would be spontancous when
[2010]

Options:

A. $T_c > T$

B. $T > T_c$

C. T_e is 5 times T

D. $T = T_e$

Answer: B

Solution:

Solution:
At equilibrium $\Delta G = 0$
Hence, $\Delta G = \Delta H - T\ \Delta S = 0$
$\therefore \Delta H = T_e \Delta S$ or $T_e = \frac{\Delta H}{\Delta S}$
For a spontaneous reaction
ΔG must be negative which is possible only if $\Delta H < T\ \Delta S$
or $T > \frac{\Delta H}{\Delta S}; T_e < T$

Question185

On the basis of the following thermochemical data :

$(\Delta_f G^\circ H_{(aq)}^+ = 0)$

$H_2O(l) \rightarrow H^+(aq) + OH^-(aq); \Delta H = 57.32kJ$

$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(l); \Delta H = -286.20kJ$

The value of enthalpy of formation of OH^- ion at $25^\circ C$ is:
[2009]

Options:

A. $-228.88kJ$

B. $+228.88kJ$

C. $-343.52kJ$

D. $-22.88kJ$

Answer: A

Solution:

Solution:
$H_2O(l) \rightarrow H^+(aq.) + OH^-(aq.)$
$\Delta H = 57.32kJ$ . . . (i)
$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(l)$
$\Delta H = -286.20kJ$ . . . (ii)
By adding equation (i) & (ii) we get,
$H_2(g) + \frac{1}{2}(O_2) \rightarrow H^+(aq) + OH^-(aq)$
$\Delta H = 57.32 - 286.2 = -228.88kJ$
Here $\Delta_f H^\circ$ of $H^+(aq.) = 0$
$\therefore \Delta_f H^\circ$ of $OH^- = -228.88kJ$

Question186

Oxidising power of chlorine in aqueous solution can be determined by the parameters indicated below:

$$\frac{1}{2}Cl_2(g) \xrightarrow{\frac{1}{2}\Delta_{diss}H^\circ} Cl(g) \xrightarrow{\Delta_{eg}H^\circ} Cl^-(g) \xrightarrow{\Delta_{hyd}H^\circ} Cl^-(aq)$$

The energy envolved in the conversion of
$\frac{1}{2}Cl_2(g) to Cl^{-1}(aq)$
(using the data,
$\Delta_{diss} H_{Cl_2}^\circ = 240kJ\ mol^{-1}$, $\Delta_{eg} H_{Cl}^o = -349kJ\ mol^{-1}$
$\Delta_{hyd} H_{Cl}^\circ = -381kJ\ mol^{-1}$), will be
[2008]

Options:

A. $+152kJ\ mol^{-1}$

B. $-610kJ\ mol^{-1}$

C. $-850kJ\ mol^{-1}$

D. $+120\text{kJ mol}^{-1}$

Answer: B

Solution:

Solution:
The energy involved in the conversion of
$\frac{1}{2}Cl_2(g)$ to $Cl^-(aq)$ is given by

$$\Delta H = \frac{1}{2}\Delta_{diss}H_{Cl_2}{}^{O} + \Delta_{eg}H_{Cl}{}^{\circ} + \Delta_{hyd}H_{Cl}{}^{\circ}$$

Substituting various values from given data, we get

$$\Delta H = \left(\frac{1}{2}\times 240\right) + (-349) + (-381)$$
$$= (120 - 349 - 381) = -610\text{kJ mol}^{-1}$$

Question187

Standard entropy of X_2, Y_2 and XY_3 are 60,40 and 50J $K^{-1}mol^{-1}$, respectively. For the reaction, $\frac{1}{2}X_2 + \frac{3}{2}Y_2 \rightarrow XY_3$, $\Delta H = -30\text{kJ}$, to be at equilibrium, the temperature will be [2008]

Options:

A. 1250K

B. 500K

C. 750K

D. 1000K

Answer: C

Solution:

Solution:
For a reaction to be at equilibrium $\Delta G^\circ = 0$. Since $\Delta G^\circ = \Delta H^\circ - T\ \Delta S^\circ$ so at equilibrium $\Delta H^\circ - T\ \Delta S^\circ = 0$
or $\Delta H^\circ = T\ \Delta S^\circ$
For the reaction

$$\frac{1}{2}X_2 + \frac{3}{2}Y_2 \longrightarrow XY_3;\ \Delta H^\circ = -30\text{kJ}$$

Calculating ΔS° for the above reaction, we get

$$\Delta S^\circ = 50 - \left[\frac{1}{2}\times 60 + \frac{3}{2}\times 40\right]$$
$$= 50 - (30 + 60) = -40\text{J K}^{-1}$$

At equilibrium, $T\ \Delta S^\circ = \Delta H^\circ$ $[\because \Delta G^\circ = 0]$
$\therefore T \times (-40\text{J K}^{-1}) = -30 \times 1000\text{J}$ $[\because 1\text{kJ} = 1000\text{J}]$
or $T = \frac{-30\times 1000}{-40} = 750\text{K}$

Question188

Assuming that water vapour is an ideal gas, the internal energy change (ΔU) when 1mol of water is vapourised at 1 bar pressure and 100°C, (given : molar enthalpy of vapourisation of

water at 1 bar and 373K = 41kJ mol $^{-1}$ and R = 8.3J mol $^{-1}$K $^{-1}$) will be
[2007]

Options:

A. 41.00kJ mol $^{-1}$

B. 4.100kJ mol $^{-1}$

C. 3.7904kJ mol $^{-1}$

D. 37.904kJ mol $^{-1}$

Answer: D

Solution:

Solution:
Given $\Delta H = 41\text{kJ mol}^{-1} = 41000\text{J mol}^{-1}$
$T = 100^{\circ}C = 273 + 100 = 373K$
$H_2O(I) \xrightarrow{\Delta} H_2O(g)$
$\Delta n_g = 1 - 0 = 1$
$\Delta U = \Delta H - \Delta nRT = 41000 - (1 \times 8.314 \times 373)$
$= 37898.88\text{J mol}^{-1} \sim \text{eq}37.9\text{kJ mol}^{-1}$

Question189

In conversion of lime-stone to lime, $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$ the values of ΔH° and ΔS° are +179.1kJ mol $^{-1}$ and 160.2J / K respectively at 298K and 1 bar. Assuming that ΔH° and ΔS° do not change with temperature, temperature above which conversion of limestone to lime will be spontaneous is
[2007]

Options:

A. 1118K

B. 1008K

C. 1200K

D. 845K .

Answer: A

Solution:

Solution:
$\Delta G^{\circ} = \Delta H^{\circ} - T\ \Delta S^{\circ}$
For a spontaneous reaction $\Delta G^{\circ} < 0$
or $\Delta H^{\circ} - T\ \Delta S^{\circ} < 0 \Rightarrow T > \frac{\Delta H^{\circ}}{\Delta S^{\circ}}$
$\Rightarrow T > \frac{179.3 \times 10^3}{160.2}$
$>1117.9K \approx 1118K$

Question190

Identify the correct statement regarding a spontaneous process:
[2007]

Options:

A. Lowering of energy in the process is the only criterion for spontaneity.

B. For a spontaneous process in an isolated system, the change in entropy is positive.

C. Endothermic processes are never spontaneous.

D. Exothermic processes are always spontaneous.

Answer: B

Solution:

Solution:
Spontaneity of reaction depends on tendency to acquire minimum energy state and maximum randomness. For a spontaneous process in an isolated system the change in entropy is positive.

Question191

An ideal gas is allowed to expand both reversibly and irreversibly in an isolated system. If T_i is the initial temperature and T_f is the final temperature, which of the following statements is correct?
[2006]

Options:

A. $(T_f)_{rev} = (T_f)_{irrev}$

B. $T_f = T_i$ for both reversible and irreversible processes

C. $(T_f)_{imcv} > (T_f)_{rev}$

D. $T_f > T_i$ for reversible process but $T_f = T_i$ for irreversible process

Answer: C

Solution:

Solution:
According to first law of thermodynamics $\Delta Q = \Delta U + \Delta W$
An isolated system is adiabatic. This means $\Delta Q = 0$. The first law in this case yields
$0 = \Delta U + \Delta W \Rightarrow \Delta W = -\Delta U \ldots$ (i)
For expansion, ΔW is positive and hence ΔU is negative.
This means T_f is less than T_i in both the cases.
For the same expansion of volume, the work done in irreversible process is greater than that in reversible one because the system has to work against friction etc. Thus
$\Delta W_{irreversible} > \Delta W_{reversible}$
$\Rightarrow -\Delta U_{irreversible} > -\Delta U_{reversible}$ [from equation (i)]
$\Rightarrow \Delta U_{irreversible} < \Delta U_{reversible}$
$\Rightarrow \Delta T_{irreversible} < \Delta T_{reversible}$
$\Rightarrow T_{f\ irreversible} > T_{f\ reversible}$

Question192

The standard enthalpy of formation ($\Delta_f H^\circ$) at 298K for methane, $CH_4(g)$ is $-74.8 kJ\ mol^{-1}$. The additional information required to determine the average energy for C − H bond formation would be
[2006]

Options:

A. the first four ionization energies of carbon and electron gain enthalpy of hydrogen

B. the dissociation energy of hydrogen molecule, H_2

C. the dissociation energy of H_2 and enthalpy of sublimation of carbon

D. latent heat of vapourization of methane

Answer: C

Solution:

Solution:
The standard enthalpy of formation of CH_4 is given by the equation:
$C(s) + 2H_2(g) \longrightarrow CH_4(g)$
Hence, dissociation energy of hydrogen and enthalpy of sublimation of carbon is required.

Question193

The enthalpy changes for the following processes are listed below:
$Cl_2(g) = 2Cl(g)$, 242.3kJ mol^{-1}
$I_2(g) = 2I(g)$, 151.0kJ mol^{-1}
$ICl(g) = I(g) + Cl(g)$, 211.3kJ mol^{-1}
$I_2(s) = I_2(g)$, 62.76kJ mol^{-1}
Given that the standard states for iodine and chlorine are $I_2(s)$ and $Cl_2(g)$, the standard enthalpy of formation for $ICl(g)$ is :
[2006]

Options:

A. $+16.8 kJ\ mol^{-1}$

B. $+244.8 kJ\ mol^{-1}$

C. $-14.6 kJ\ mol^{-1}$

D. $-16.8 kJ\ mol^{-1}$

Answer: A

Solution:

Solution:
$I_2(s) + Cl_2(g) \longrightarrow 2ICl(g)$
$\Delta H = [\Delta H_{12(s) \rightarrow 12(g)} + \Delta H_{I-I} + \Delta H_{Cl-Cl}] - 2[\Delta H_{I-Cl}]$

$= 62.76 + 151.0 + 242.3 - 2 \times 211.3 = 33.46$

$\Delta H_f^{\circ}(ICl) = \frac{33.46}{2} = 16.73 kJ/mol$

Question194

$(\Delta H - \Delta U)$ for the formation of carbon monoxide (CO) from its elements at 298K is $(R = 8.314 J K^{-1} mol^{-1})$
[2006]

Options:

A. $-2477.57 J\ mol^{-1}$

B. $2477.57 J\ mol^{-1}$

C. $-1238.78 J\ mol^{-1}$

D. $1238.78 J\ mol^{-1}$

Answer: D

Solution:

Solution:
For the reaction,

$C(s) + \frac{1}{2}O_2(g) \rightarrow CO(g)$

$\Delta H = \Delta U + \Delta n_g RT$ or $\Delta H - \Delta U = \Delta n_g RT$

$\Delta n_g = 1 - \frac{1}{2} = \frac{1}{2};$

$\Delta H - \Delta U = \frac{1}{2} \times 8.314 \times 298$

$= 1238.78 J\ mol^{-1}$

Question195

Consider the reaction : $N_2 + 3H_2 \rightarrow 2NH_3$ carried out at constant temperature and pressure. If ΔH and ΔU are the enthalpy and internal energy changes for the reaction, which of the following expressions is true?
[2005]

Options:

A. $\Delta H > \Delta U$

B. $\Delta H < \Delta U$

C. $\Delta H = \Delta U$

D. $\Delta H = 0$

Answer: B

Solution:

Solution:

$\Delta H = \Delta U + \Delta n_g RT$ for
$N_2 + 3H_2 \longrightarrow 2NH_3$
$\Delta n_g = 2 - 4 = -2$
$\therefore \Delta H > \Delta U - 2RT$
or $\Delta U = \Delta H + 2RT \quad \therefore \Delta U > \Delta H$

Question196

If the bond dissociation energies of XY, X_2 and Y_2 (all diatomic molecules) are in the ratio of 1 : 1 : 0.5 and ΔH_f for the formation of XY is $-200 kJ$ mole $^{-1}$. The bond dissociation energy of X_2 will be
[2005]

Options:

A. 400kJ mol $^{-1}$

B. 300kJ mol $^{-1}$

C. 200kJ mol $^{-1}$

D. 100kJ mol $^{-1}$

E. None of above

Answer: E

Solution:

Solution:
(N) $\frac{1}{2}X_2 + \frac{1}{2}Y_2 \longrightarrow XY, \Delta H = (-200)J$
Let x be the bond dissociation energy of X_2. Then
$\Delta H_f = -200 = \frac{1}{2}\Delta H_{x-x} + \frac{1}{2}\Delta H_{y-y} - \Delta H_{x-y}$
$-400 = x + 0.5x - 2x = -0.5x$
or $x = \frac{400}{0.5} = 800 kJ\ mol^{-1}$
(In the question paper, this option was not mentioned. So the answer has been marked 'N')

Question197

An ideal gas expands in volume from 1×10^{-3} to $l \times 10^{-2} m^3$ at 300K against a constant pressure of $1 \times 10^5 N\ m^{-2}$. The work done is
[2004]

Options:

A. 270kJ

B. $-900kJ$

C. $-900J$

D. 900kJ

Answer: C

Solution:

Solution:
$w = -P\Delta V = -10^5[(1\times10^{-2}) - (1\times10^{-3})]$
$= -900J$

Question198

The enthalpies of combustion of carbon and carbon monoxide are −393.5 and −283kJ mol $^{-1}$ respectively. The enthalpy of formation of carbon monoxide per mole is
[2004]

Options:

A. −676.5kJ

B. 676.5kJ

C. 110.5kJ

D. −110.5kJ

Answer: D

Solution:

Solution:
(i) $C + O_2 \rightleftharpoons CO_2, \Delta H = -393.5kJ\,mol^{-1}$
(ii) $CO + \frac{1}{2}O_2 \rightleftharpoons CO_2, \Delta H = -283.0kJ\,mol^{-1}$
Operating (i) - (ii), we have
$C + \frac{1}{2}O_2 \rightarrow CO\ \Delta H = -110.5kJ\,mol^{-1}$

Question199

The internal energy change when a system goes from state A to B is 40kJ/mole. If the system goes from A to B by a reversible path and returns to state A by an irreversible path what would be the net change in internal energy?
[2003]

Options:

A. >40kJ

B. <40kJ

C. Zero

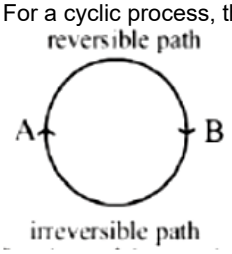
D. 40kJ

Answer: C

Solution:

Solution:

For a cyclic process, the net change in the internal energy is zero because the change in internal energy does not depend on the path.



Question200

If at 298K the bond energies of C − H , C − C, C = C and H − H bonds are respectively 414, 347, 615 and 435kJ mol $^{-1}$, the value of enthalpy change for the reaction $H_2C = CH_2(g) + H_2(g) \rightarrow H_3C - CH_3(g)$ at 298K will be [2003]

Options:

A. −250kJ

B. +125kJ

C. −125kJ

D. +250kJ

Answer: C

Solution:

Solution:
$CH_2 = CH_2(g) + H_2(g) \rightarrow CH_3 - CH_3$
Enthalpy change = Bond energy of reactants − Bond energy of products.
$\Delta H = 1(C = C) + 4(C - H) + 1(H - H) - 1(C - C) - 6(C - H)$
$= 1(C = C) + 1(H - H) - 1(C - C) - 2(C - H)$
$= 615 + 435 - 347 - 2 \times 414 = 1050 - 1175$
$= -125kJ$.

Question201

The enthalpy change for a reaction does not depend upon [2003]

Options:

A. use of different reactants for the same product

B. the nature of intermediate reaction steps

C. the differences in initial or final temperatures of involved substances

D. the physical states of reactants and products

Answer: B

Solution:

Solution:
Enthalpy change for a reaction does not depend upon the nature of intermediate reaction steps.

Question202

In an irreversible process taking place at constant T and P and in which only pressure-volume work is being done, the change in Gibbs free energy (d G) and change in entropy (dS), satisfy the criteria
[2003]

Options:

A. $(d S)_{V, E} > 0, (d G)_{T, P} < 0$

B. $(d S)_{V, E} = 0, (d G)_{T, P} = 0$

C. $(d S)_{V, E} = 0, (d G)_{T, P} > 0$

D. $(d S)_{V, E} < 0, (d G)_{T, P}^{T} < 0$

Answer: A

Solution:

Solution:
For spontaneous reaction, $d S > 0$ and $d G$ should be negative i.e. <0.

Question203

The correct relationship between free energy change in a reaction and the corresponding equilibrium constant K_c is
[2003]

Options:

A. $-\Delta G = RT / nK_c$

B. $\Delta G^{\circ} = RT / nK_c$

C. $-\Delta G^{\circ} = RT \ln K_c$

D. $\Delta G = RT \ln K_c$

Answer: C

Solution:

Solution:
$\Delta G^{\circ} = -RT \ln K_c$ or $-\Delta G^{0} = RT \ln K_c$

Question204

A heat engine absorbs heat Q_1 at temperature T_1 and heat Q_2 at temperature T_2. Work done by the engine is J $(Q_1 + Q_2)$. This data

[2002]

Options:

A. violates 1st law of thermodynamics

B. violates 1st law of themodynamics if Q_1 is − ve

C. violates 1st law of thermodynamics of Q_2 is − ve

D. does not violate 1st law of themodynamics.

Answer: A

Solution:

Solution:
According to first law of thermodynamics energy can neither be created nor destroyed although it can be converted from one form to another.
Note: Carnot cycle is based upon this principle but during the conversion of heat into work some mechanical energy is always converted to other form of energy hence this data violates 1 st law of thermodynamics.

Question205

The heat required to raise the temperature of body by 1K is called
[2002]

Options:

A. specific heat

B. thermal capacity

C. water equivalent

D. none of these.

Answer: B

Solution:

Solution:
The heat required to raise the temperature of body by $1K$ is called thermal capacity or heat capacity.

Question206

For the reactions,
$$2C + O_2 \rightarrow 2CO_2; \quad \Delta H = -393J$$
$$2Zn + O_2 \rightarrow 2ZnO; \quad \Delta H = -412J$$
[2002]

Options:

A. carbon can oxidise Z n

B. oxidation of carbon is not feasible

C. oxidation of Z n is not feasible

D. Z n can oxidise carbon.

Answer: D

Solution:

Solution:
ΔH negative shows that the reaction is spontancous. Higher negative value for $Z n$ shows that the reaction is more feasible.

Question207

If an endothermic reaction is non-spontaneous at freezing point of water and becomes feasible at its boiling point, then
[2002]

Options:

A. ΔH is $-$ve, ΔS is +ve

B. ΔH and ΔS both are +ve

C. ΔH and ΔS both are $-$ve

D. ΔH is +ve, ΔS is $-$ve

Answer: B

Solution:

Solution:
$\Delta G = \Delta H - T \Delta S$
For an endothermic reaction,
ΔH = + ve and at low temperature ΔS = + ve
Hence $\Delta G = (+) \Delta H - T (+) \Delta S$
and if $T \Delta S < \Delta H$ (at low temp)
$\Delta G = +ve$ (non spontancous)
But at high temperature, reaction becomes spontaneous i.e. $\Delta G = -ve$
because at higher temperature $T \Delta S > \Delta H$
